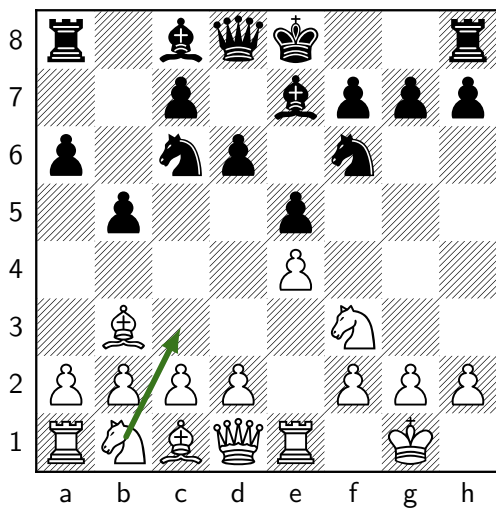


Melanie-BOTS (CCRL engines) — Ruy Lopez, Closed

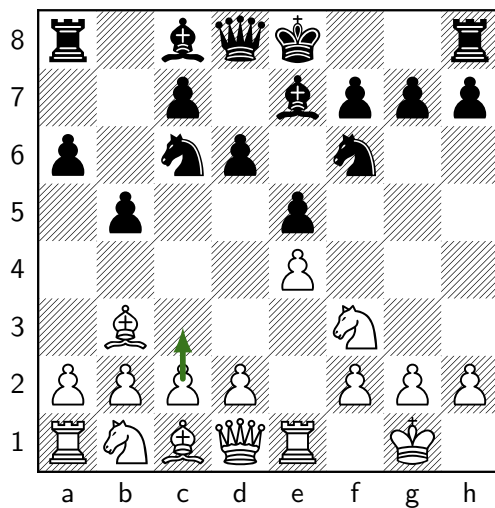
Melanie (Deep_Vision)

Abstract

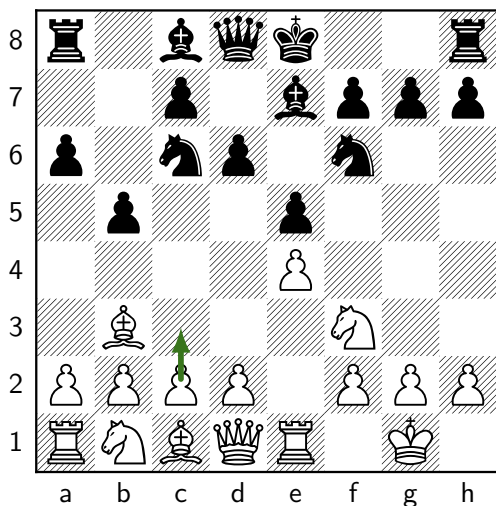
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



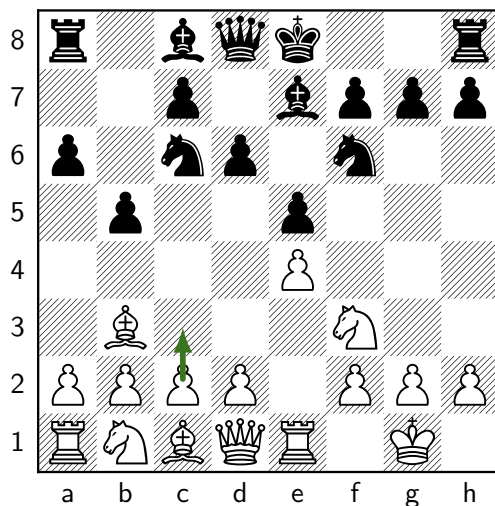
(a)



(b)

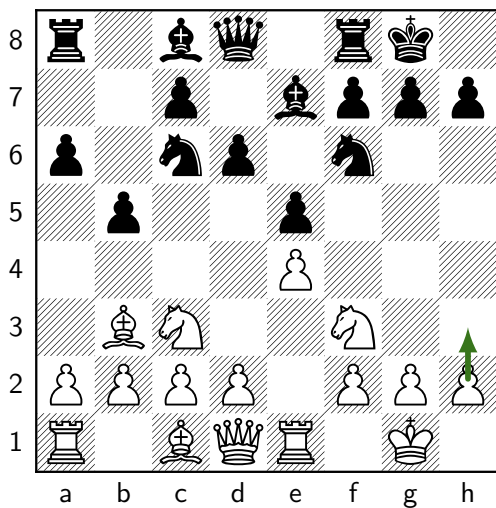


(c)

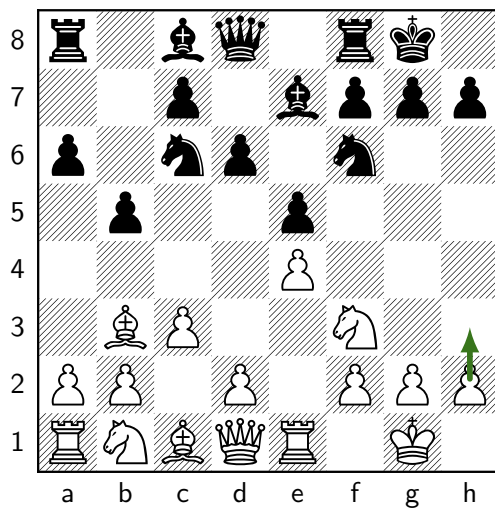


(d)

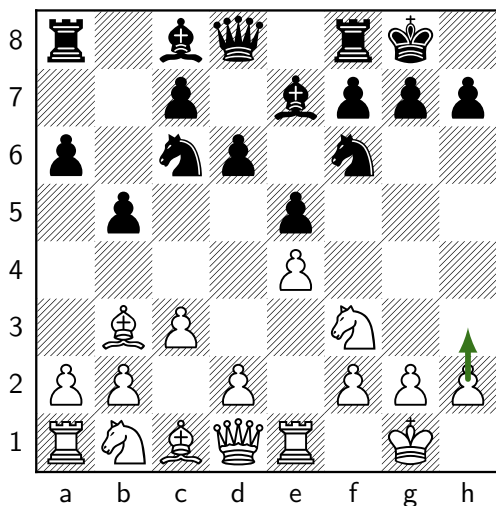
Figure 1: **Move 8, White: Nc3**. (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#152 (ply 14) [structural distance $d=0$, QAP obj=80]: the master played c3; under μ our Nb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Nb1; transferred move b1c3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#153 (ply 14) [structural distance $d=0$, QAP obj=80]: the master played c3; under μ our Nb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Nb1; transferred move b1c3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#168 (ply 14) [structural distance $d=0$, QAP obj=80]: the master played c3; under μ our Nb1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Nb1; transferred move b1c3 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Nc3 (summed QAP objective 1282, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



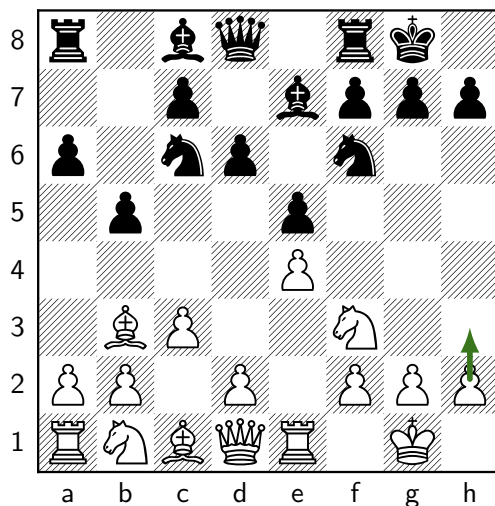
(a)



(b)

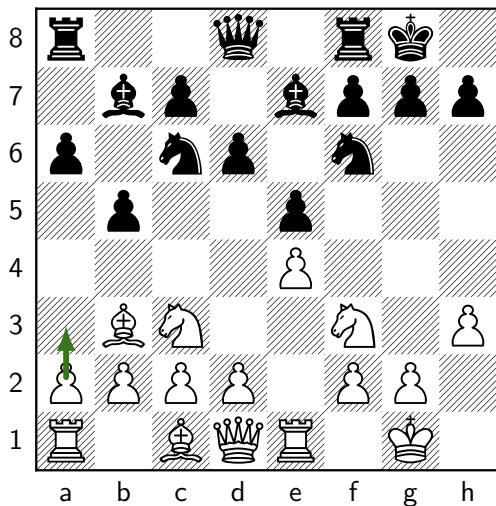


(c)

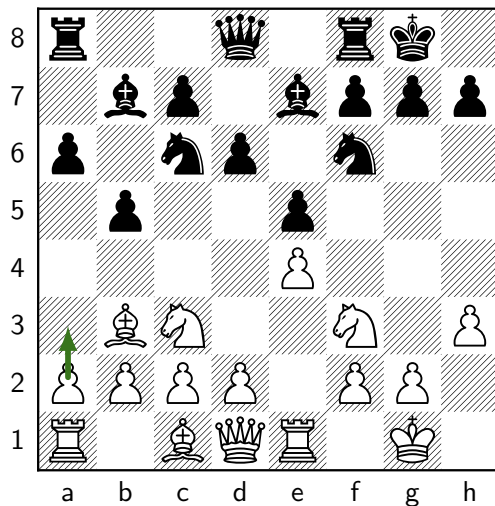


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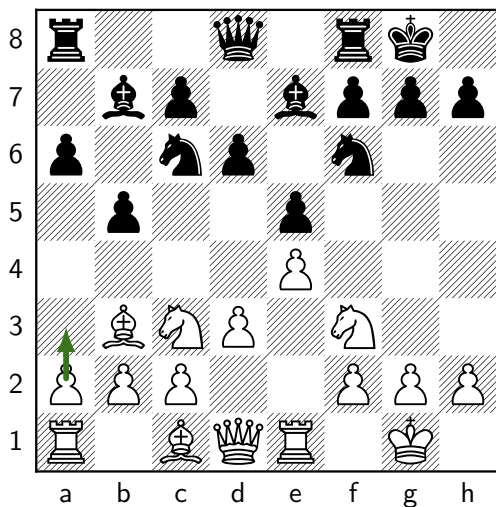
Figure 2: **Move 9, White: h3.** (a) Our position; Melanie plays **h3** (**h2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#152` (ply 16) [structural distance $d=25$, QAP obj=111]: the master played **h3**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *pa6* as their *Nb1*, our *Pa2* as their *Qd1*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#153` (ply 16) [structural distance $d=25$, QAP obj=111]: the master played **h3**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *pa6* as their *Nb1*, our *Pa2* as their *Qd1*; transferred move **h2h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#168` (ply 16) [structural distance $d=25$, QAP obj=111]: the master played **h3**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *pa6* as their *Nb1*, our *Pa2* as their *Qd1*; transferred move **h2h3** (**endorses this move**). Synthesis: 16 retrieved analogs transferred to **h3** (summed QAP objective 1779, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



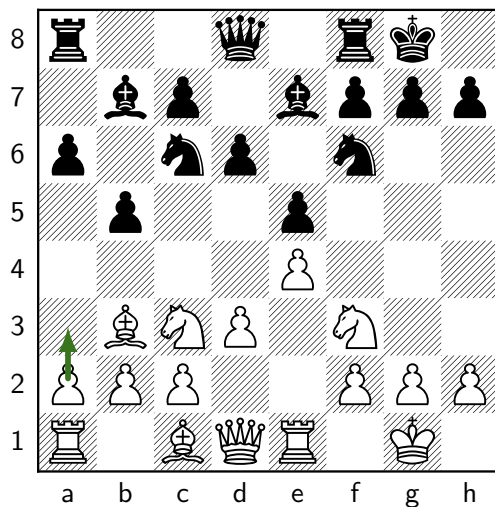
(a)



(b)

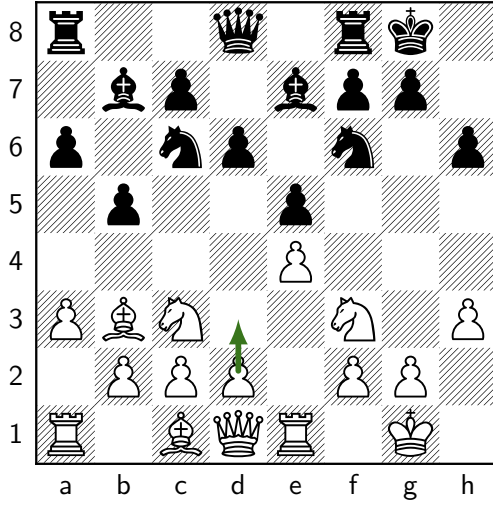


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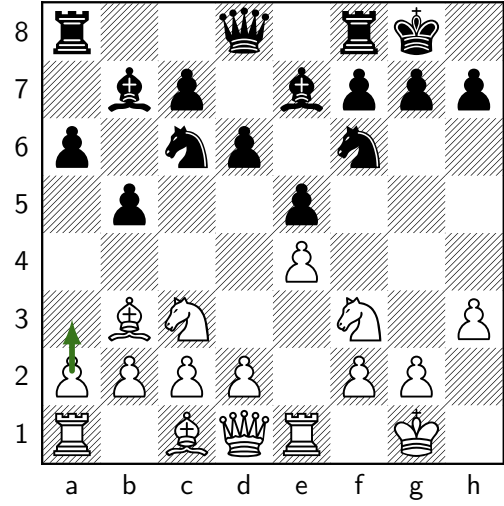


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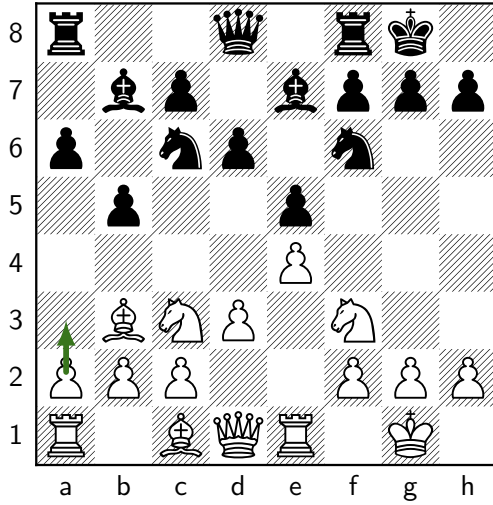
Figure 3: **Move 10, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2081 (ply 18) [structural distance $d=0$, QAP obj=145]: the master played a3; under μ our *Pa2* plays the role of that game’s *Pa2*; transferred move a2a3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2776 (ply 18) [structural distance $d=14$, QAP obj=143]: the master played a3; under μ our *Pa2* plays the role of that game’s *Pa2*; transferred move a2a3 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#181 (ply 18) [structural distance $d=14$, QAP obj=143]: the master played a3; under μ our *Pa2* plays the role of that game’s *Pa2*; transferred move a2a3 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to a3 (summed QAP objective 1864, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



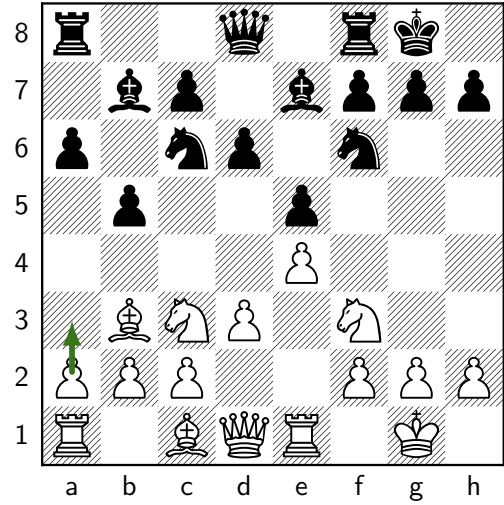
(a)



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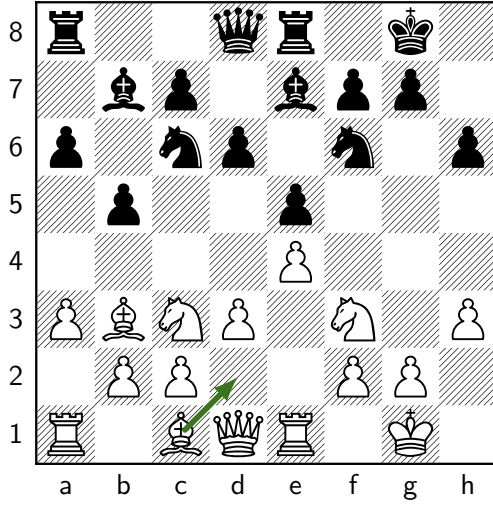


(c)

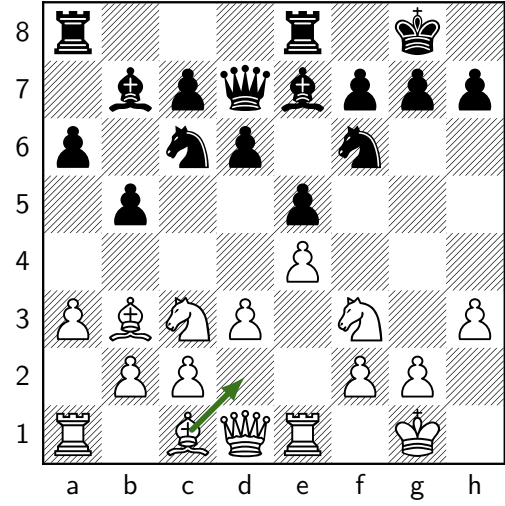


(d)

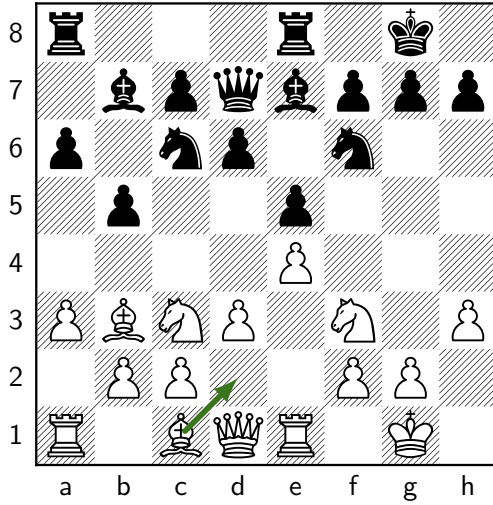
Figure 4: **Move 11, White: d3.** (a) Our position; Melanie plays d3 (d2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2081` (ply 18) [structural distance $d=25$, QAP obj=137]: the master played a3; under μ our *Pd2* plays the role of that game’s *Pa2*; transferred move d2d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2776` (ply 18) [structural distance $d=29$, QAP obj=135]: the master played a3; under μ our *Pd2* plays the role of that game’s *Pa2*; cross-type roles: our *nc6* as their *Ph2*, our *ph6* as their *nc6*; transferred move d2d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#181` (ply 18) [structural distance $d=29$, QAP obj=135]: the master played a3; under μ our *Pd2* plays the role of that game’s *Pa2*; cross-type roles: our *nc6* as their *Ph2*, our *ph6* as their *nc6*; transferred move d2d3 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to d3 (summed QAP objective 1755, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



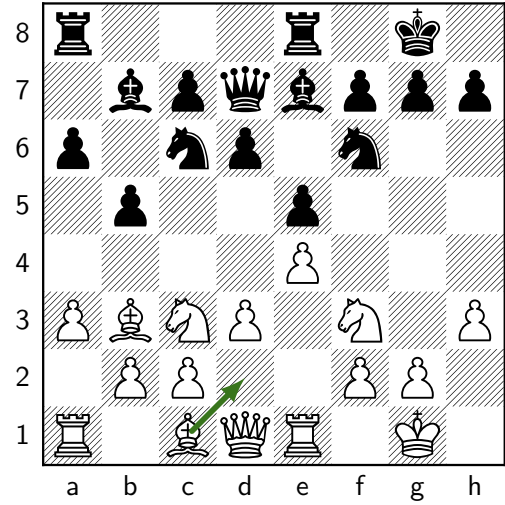
(a)



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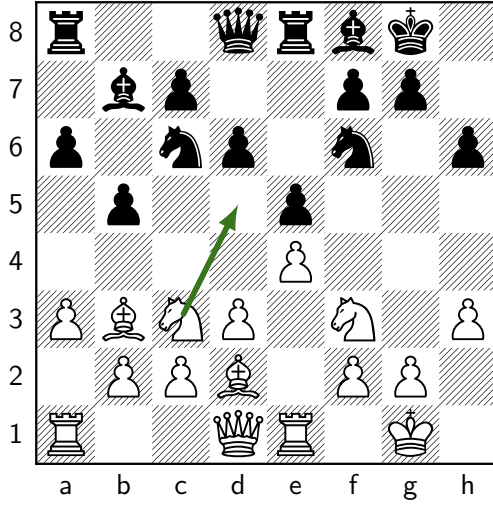


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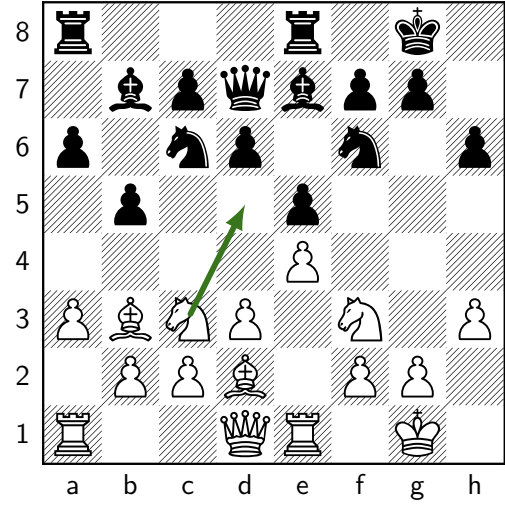


(d)

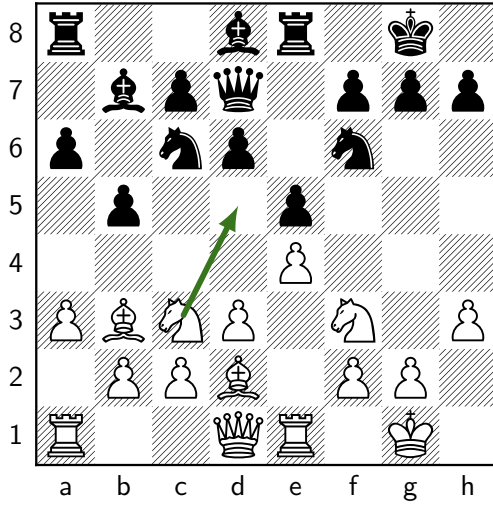
Figure 5: **Move 12, White: Bd2.** (a) Our position; Melanie plays Bd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2776` (ply 22) [structural distance $d=42$, QAP obj=138]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Nf3* as their *qd7*, our *re8* as their *ph7*; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1507` (ply 22) [structural distance $d=42$, QAP obj=138]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Nf3* as their *qd7*, our *re8* as their *ph7*; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2486` (ply 22) [structural distance $d=42$, QAP obj=138]: the master played Bd2; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Nf3* as their *qd7*, our *re8* as their *ph7*; transferred move c1d2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bd2 (summed QAP objective 831, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



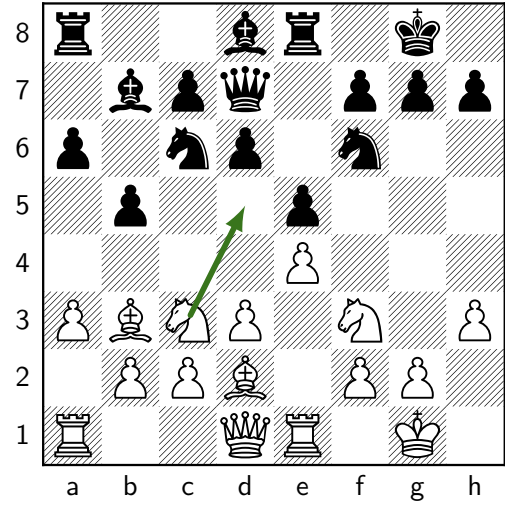
(a)



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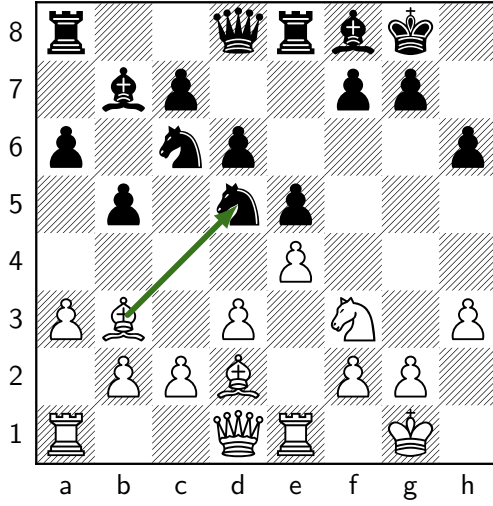


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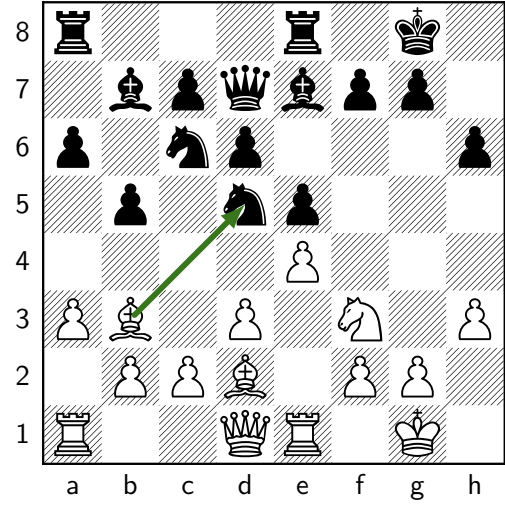


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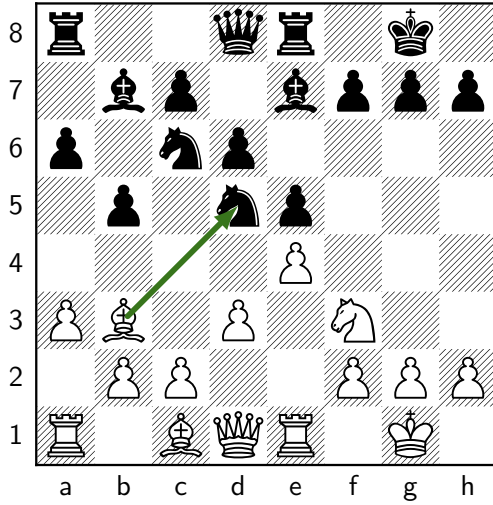
Figure 6: **Move 13, White: Nd5.** (a) Our position; Melanie plays Nd5 (c3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2776` (ply 24) [structural distance $d=34$, QAP obj=145]: the master played Nd5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *bb7* as their *Pb2*, our *re8* as their *nc6*; transferred move c3d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1956` (ply 24) [structural distance $d=47$, QAP obj=141]: the master played Nd5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Bb3* as their *qd7*, our *qd8* as their *bd8*; transferred move c3d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1957` (ply 24) [structural distance $d=47$, QAP obj=141]: the master played Nd5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Bb3* as their *qd7*, our *qd8* as their *bd8*; transferred move c3d5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd5 (summed QAP objective 562, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



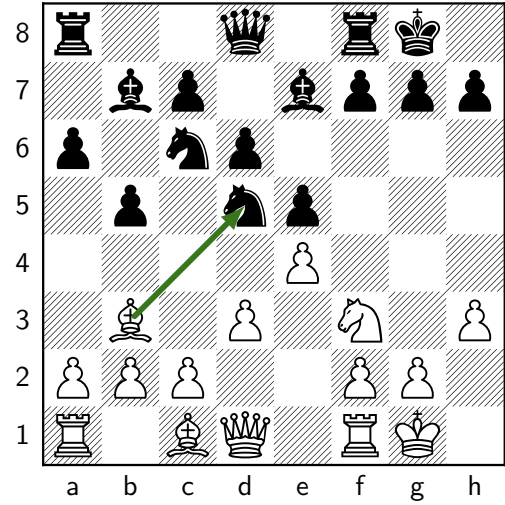
(a)



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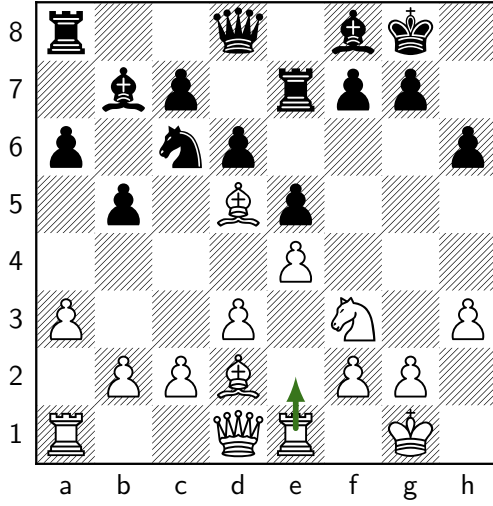


(c)

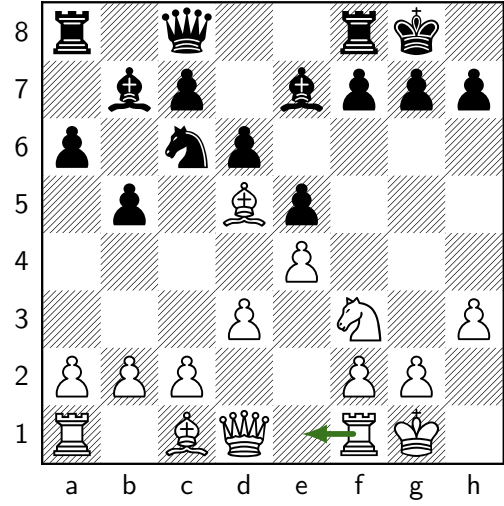


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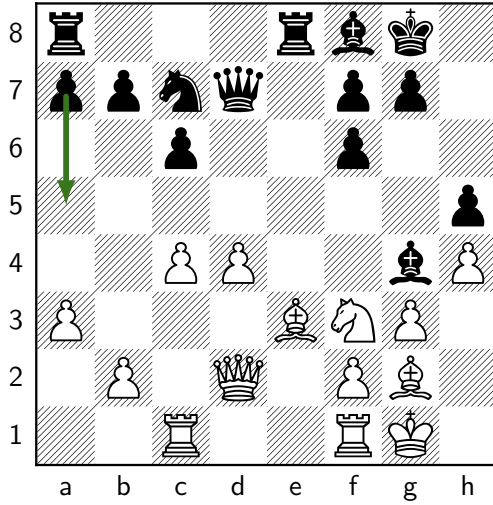
Figure 7: **Move 14, White: Bxd5.** (a) Our position; Melanie plays Bxd5 (b3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2776` (ply 26) [structural distance $d=53$, QAP obj=46]: the master played Bxd5; under μ our *Bb3* plays the role of that game’s *Bb3*; cross-type roles: our *ph6* as their *Ra1*, our *Pa3* as their *Bd2*; transferred move b3d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3015` (ply 22) [structural distance $d=78$, QAP obj=45]: the master played Bxd5; under μ our *Bb3* plays the role of that game’s *Bb3*; cross-type roles: our *Pd3* as their *Ra1*, our *pc7* as their *Qd1*; transferred move b3d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1064` (ply 20) [structural distance $d=88$, QAP obj=43]: the master played Bxd5; under μ our *Bb3* plays the role of that game’s *Bb3*; cross-type roles: our *Ph3* as their *Ra1*, our *Pa3* as their *Bc1*; transferred move b3d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bxd5 (summed QAP objective 205, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



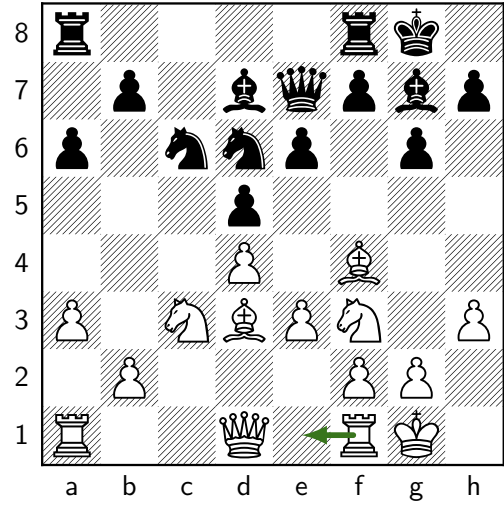
(a)



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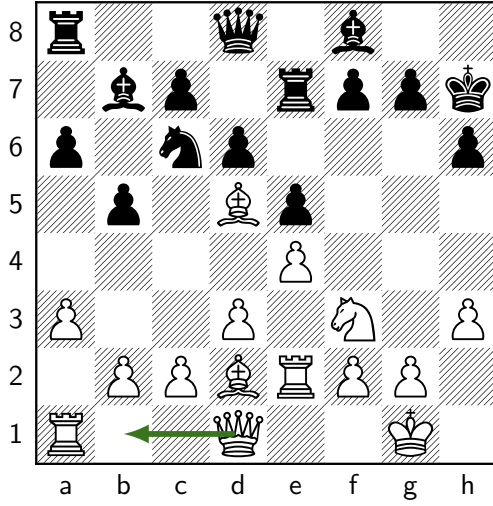


(c)

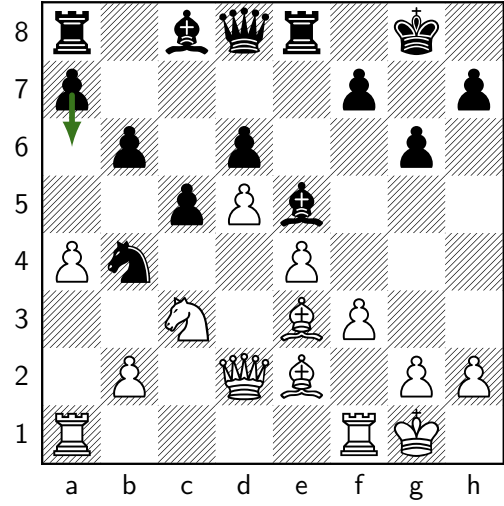


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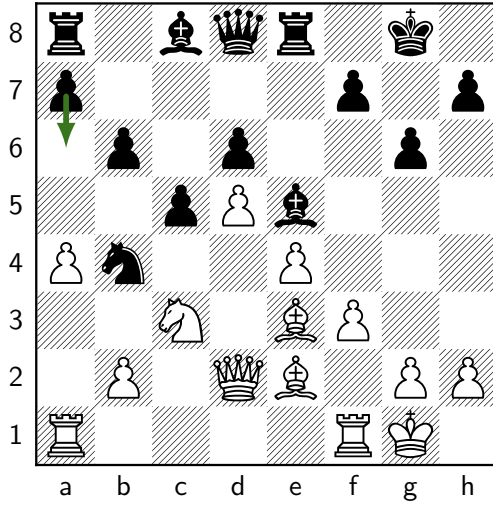
Figure 8: **Move 15, White: Re2.** (a) Our position; Melanie plays Re2 (**e1e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1064` (ply 22) [structural distance $d=82$, QAP obj=78]: the master played **Re1**; under μ our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *b8* as their *Ra1*, our *Nf3* as their *Bc1*; transferred move **e1e2** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#49` (ply 31) [structural distance $d=80$, QAP obj=36]: the master played **a5**; under μ our *Re1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Ph3* as their *Ra8*, our *ph6* as their *Re8*; transferred move **e1e2** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2967` (ply 28) [structural distance $d=68$, QAP obj=27]: the master played **Re1**; under μ our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *ph6* as their *Ra1*, our *pe5* as their *Nc3*; transferred move **e1e2** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re2 (summed QAP objective 140, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



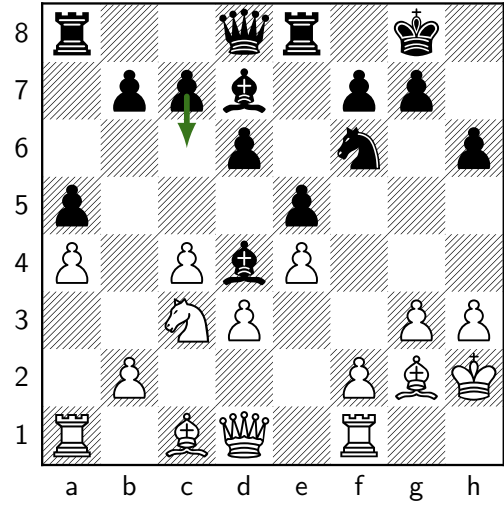
(a)



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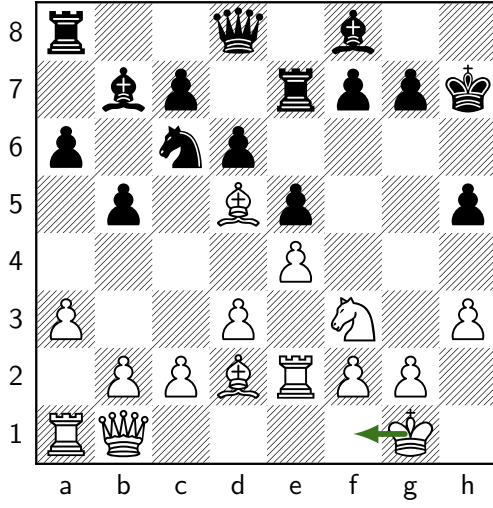


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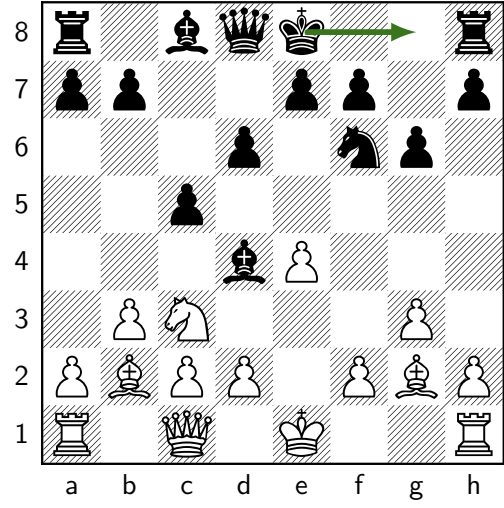


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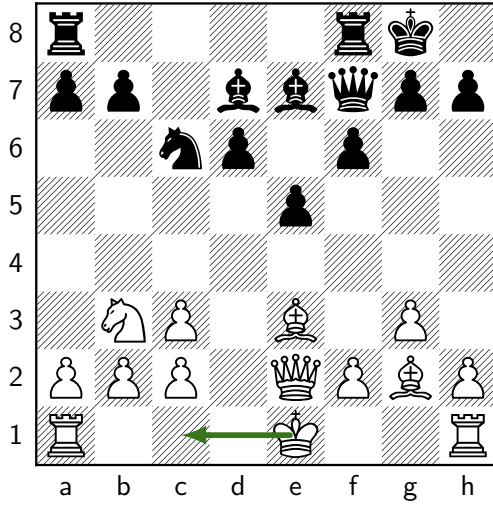
Figure 9: **Move 16, White: Qb1.** (a) Our position; Melanie plays Qb1 (d1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2837` (ply 31) [structural distance $d=91$, QAP obj=21]: the master played a6; under μ our Qd1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our re7 as their Qd8, our Nf3 as their Re8; transferred move d1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3552` (ply 31) [structural distance $d=91$, QAP obj=21]: the master played a6; under μ our Qd1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our re7 as their Qd8, our Nf3 as their Re8; transferred move d1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2306` (ply 27) [structural distance $d=90$, QAP obj=36]: the master played c6; under μ our Re2 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ra8, our qd8 as their Re8; transferred move e2e3. Synthesis: 2 retrieved analogs transferred to Qb1 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



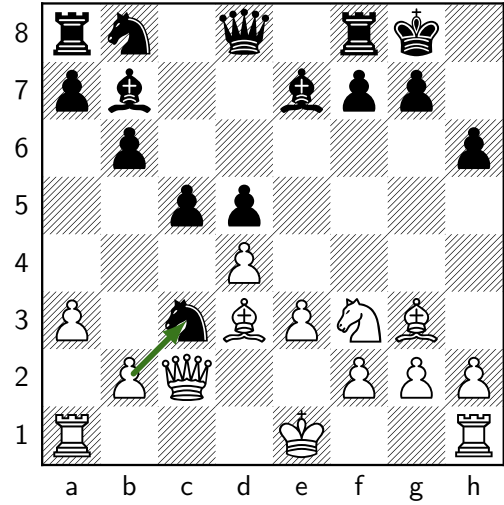
(a)



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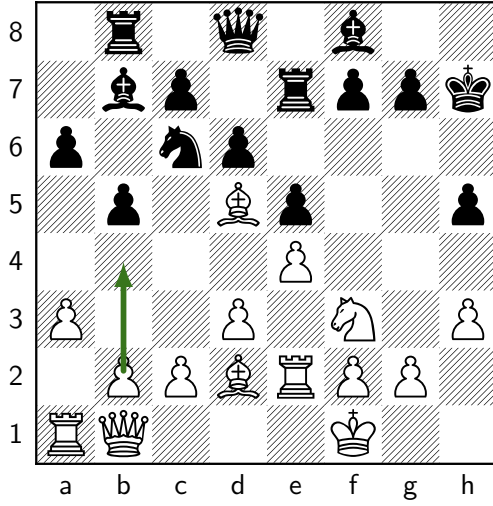


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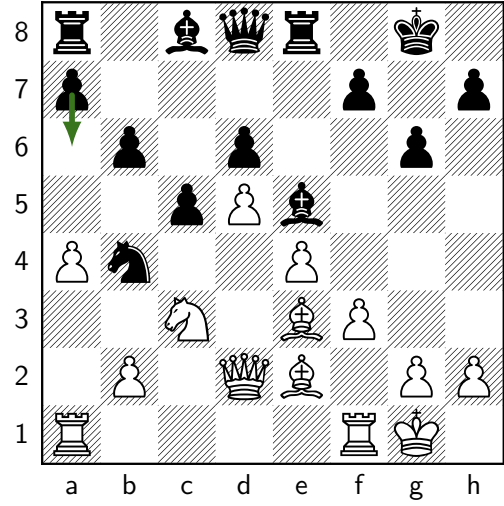


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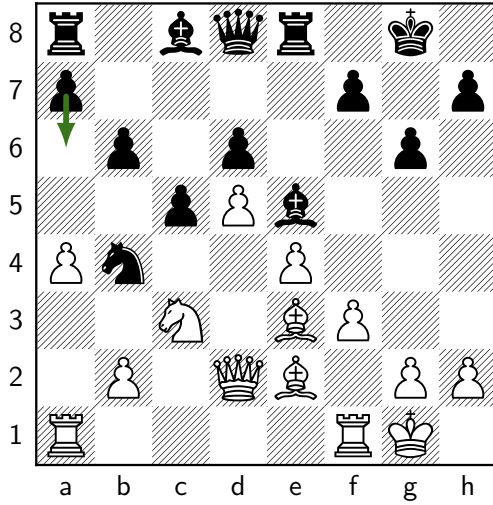
Figure 10: **Move 17, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2542` (ply 17) [structural distance $d=66$, QAP obj=26]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pa6* as their *Ra8*, our *Ph3* as their *Bc8*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#263` (ply 26) [structural distance $d=89$, QAP obj=13]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pe5* as their *Ra1*, our *Qb1* as their *Rh1*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1224` (ply 24) [structural distance $d=79$, QAP obj=38]: the master played bxc3; under μ our *Bd5* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Ra1*, our *Bd5* as their *Qc2*; transferred move d5c4. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



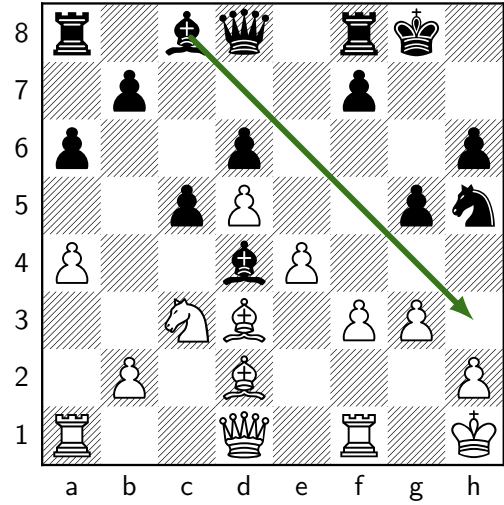
(a)



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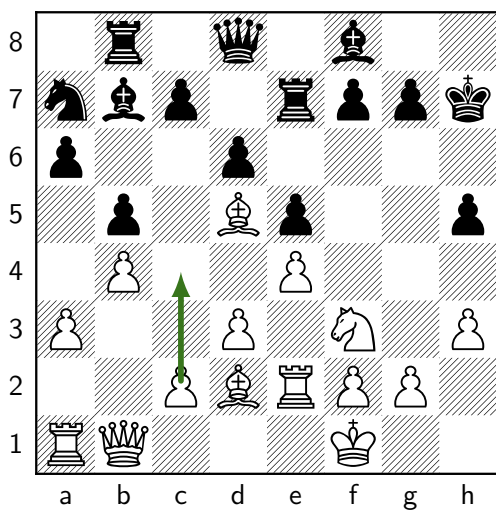


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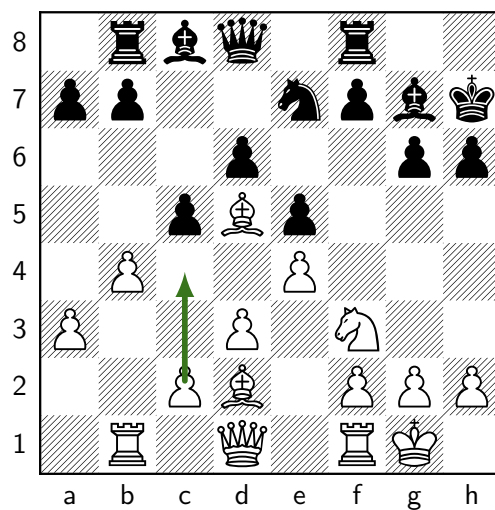


(d)

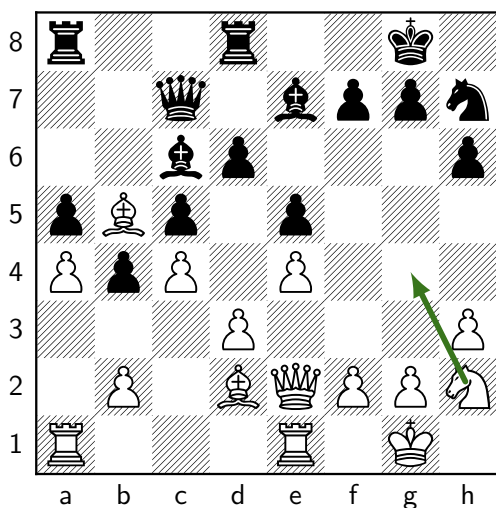
Figure 11: **Move 18, White: b4.** (a) Our position; Melanie plays b4 (b2b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2837` (ply 31) [structural distance $d=75$, QAP obj=23]: the master played a6; under μ our $Pb2$ plays the role of that game’s $Pa2$; cross-type roles: our $bb7$ as their $Ra8$, our $Pc2$ as their $Qd8$; transferred move b2b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3552` (ply 31) [structural distance $d=75$, QAP obj=23]: the master played a6; under μ our $Pb2$ plays the role of that game’s $Pa2$; cross-type roles: our $bb7$ as their $Ra8$, our $Pc2$ as their $Qd8$; transferred move b2b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1115` (ply 35) [structural distance $d=67$, QAP obj=30]: the master played Bh3; under μ our $Pd3$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $Pa3$ as their $Qd8$, our $Qb1$ as their $Rf8$; transferred move d3d4. Synthesis: 2 retrieved analogs transferred to b4 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



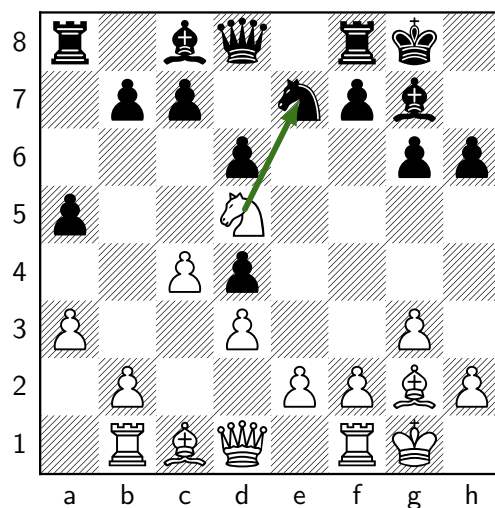
(a)



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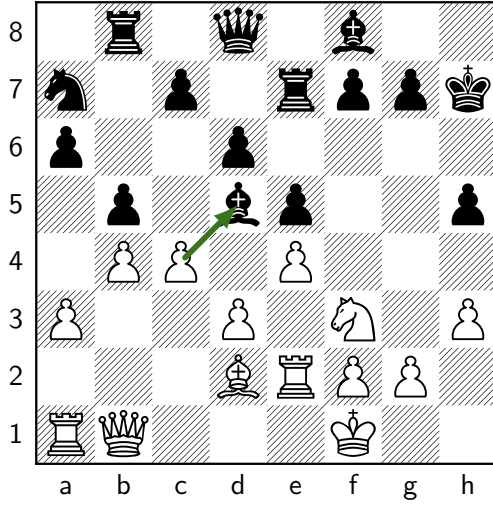


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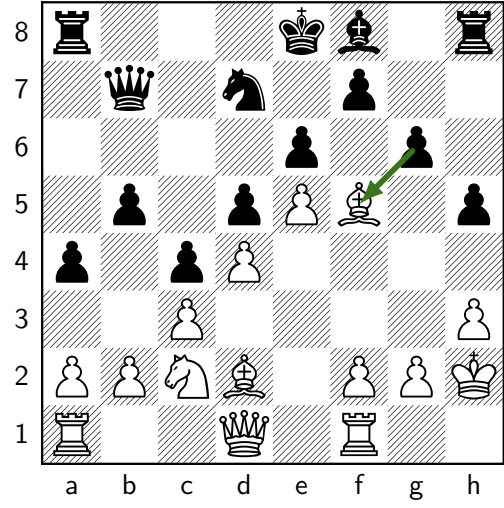


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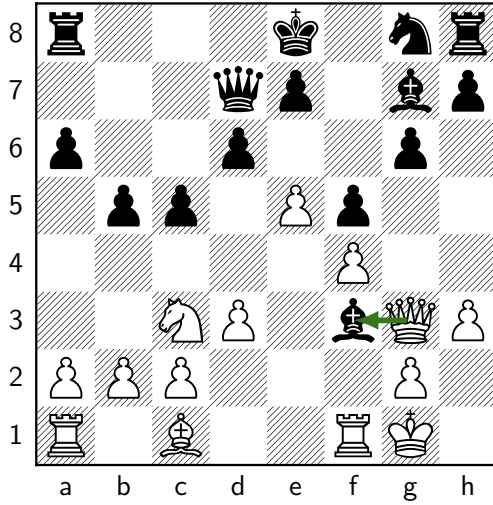
Figure 12: **Move 19, White: c4**. (a) Our position; Melanie plays c4 (c2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3786` (ply 26) [structural distance $d=83$, QAP obj=63]: the master played c4; under μ our *Pc2* plays the role of that game’s *Pc2*; cross-type roles: our *Pf2* as their *Qd1*, our *Qb1* as their *Rf1*; transferred move c2c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1455` (ply 42) [structural distance $d=77$, QAP obj=27]: the master played Ng4; under μ our *Pc2* plays the role of that game’s *Nh2* (a **cross-type** correspondence); cross-type roles: our *pb5* as their *Ra1*, our *Qb1* as their *Re1*; transferred move c2c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3840` (ply 22) [structural distance $d=85$, QAP obj=55]: the master played Nxe7+; under μ our *Bd5* plays the role of that game’s *Nd5* (a **cross-type** correspondence); cross-type roles: our *Ph3* as their *Rb1*, our *Nf3* as their *Bc1*; transferred move d5f7. Synthesis: 2 retrieved analogs transferred to c4 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



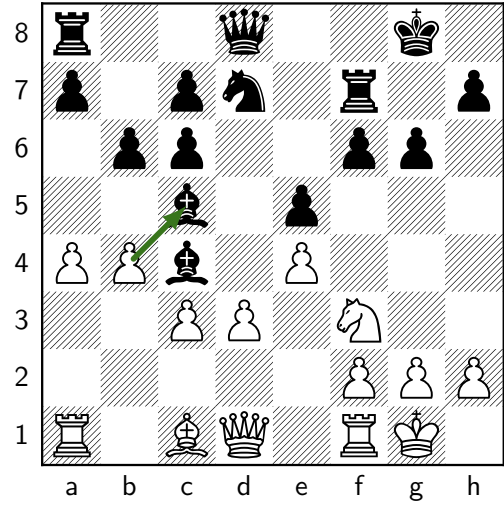
(a)



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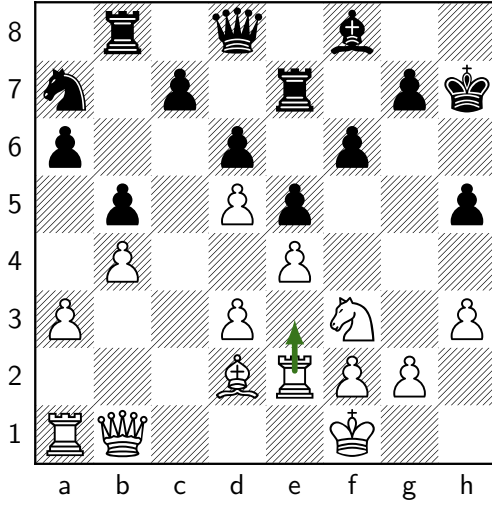


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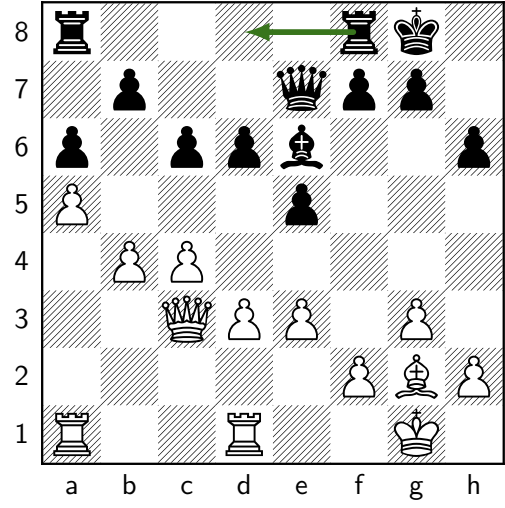


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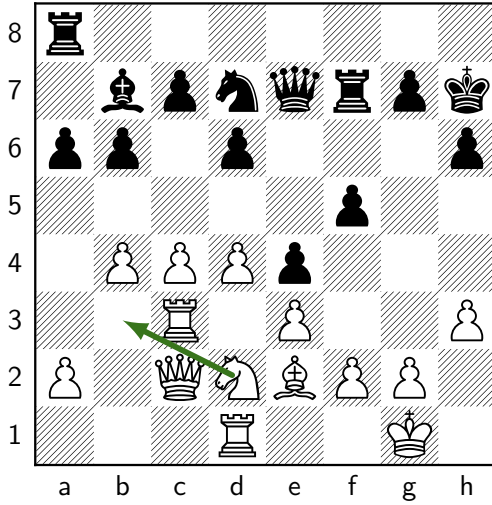
Figure 13: **Move 20, White: cxd5.** (a) Our position; Melanie plays cxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#2584 (ply 47) [structural distance $d=80$, QAP obj=105]: the master played gxf5; under μ our Pc_4 plays the role of that game’s Pg_3 ; cross-type roles: our Qb_1 as their Ra_8 , our pb_5 as their Rh_8 ; transferred move c4d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#544 (ply 24) [structural distance $d=79$, QAP obj=65]: the master played Qxf3; under μ our Pc_4 plays the role of that game’s Qg_3 (a **cross-type** correspondence); cross-type roles: our Bd_2 as their Ra_1 , our Re_2 as their Bc_1 ; transferred move c4d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#959 (ply 26) [structural distance $d=85$, QAP obj=88]: the master played bxc5; under μ our Pc_4 plays the role of that game’s Pb_4 ; cross-type roles: our Pd_3 as their Ra_1 , our Nf_3 as their Bc_1 ; transferred move c4c5. Synthesis: 3 retrieved analogs transferred to cxd5 (summed QAP objective 201, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



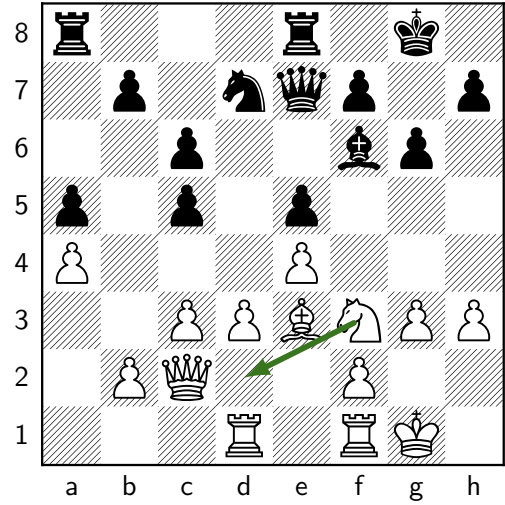
(a)



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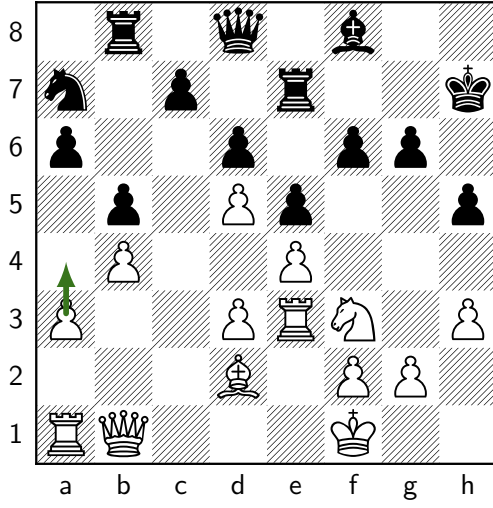


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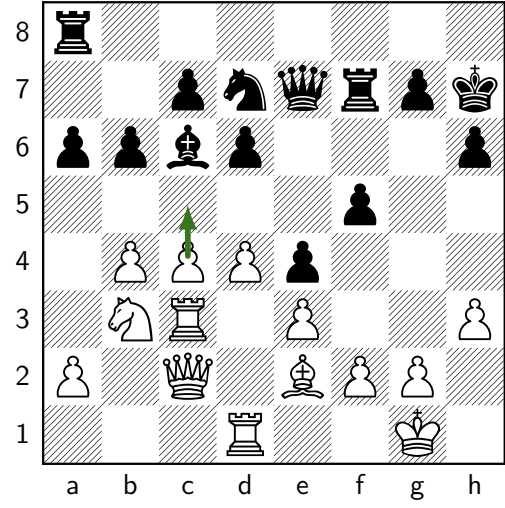


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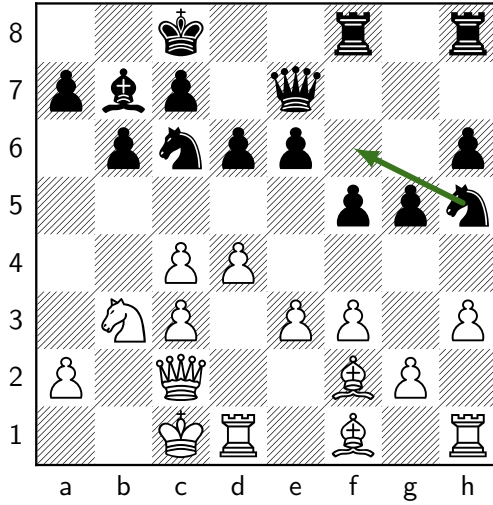
Figure 14: **Move 21, White: Re3.** (a) Our position; Melanie plays Re3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1264` (ply 37) [structural distance $d=85$, QAP obj=34]: the master played Rfd8; under μ our Re2 plays the role of that game’s Rf1; cross-type roles: our Qb1 as their Ra8, our Pe4 as their Qe7; transferred move e2e3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#609` (ply 38) [structural distance $d=57$, QAP obj=42]: the master played Nb3; under μ our Nf3 plays the role of that game’s Nd2; cross-type roles: our Qb1 as their Rd1, our Ra1 as their Pa2; transferred move f3g5. (d) Retrieved from `ccrl_top_p0001#1084` (ply 32) [structural distance $d=78$, QAP obj=36]: the master played Nd2; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pb4 as their Rd1, our Qb1 as their Rf1; transferred move f3g5. Synthesis: 1 retrieved analog transferred to Re3 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



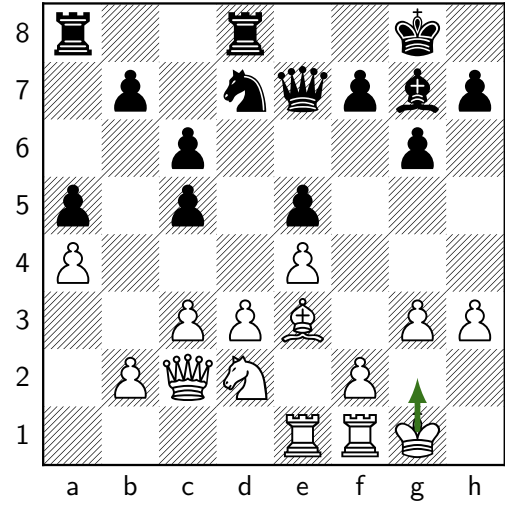
(a)



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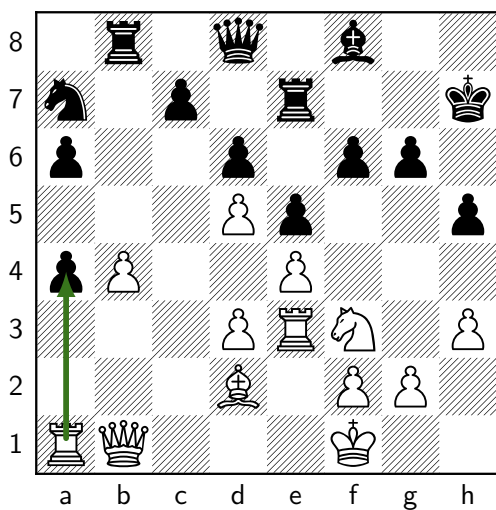


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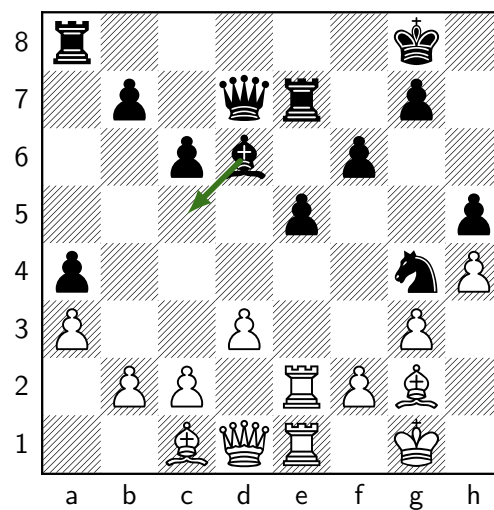


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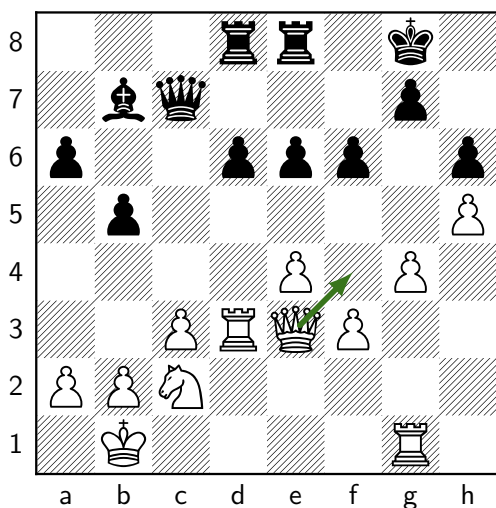
Figure 15: **Move 22, White: a4.** (a) Our position; Melanie plays **a4** (**a3a4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#609` (ply 40) [structural distance $d=66$, QAP obj=27]: the master played **c5**; under μ our *Pa3* plays the role of that game’s *Pc4*; cross-type roles: our *Qb1* as their *Rd1*, our *Bd2* as their *Pa2*; transferred move **a3a4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1955` (ply 31) [structural distance $d=71$, QAP obj=26]: the master played **Nf6**; under μ our *Pa3* plays the role of that game’s *Nh4* (a **cross-type** correspondence); cross-type roles: our *Pe4* as their *Rf8*, our *ph5* as their *Rh8*; transferred move **a3a4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1084` (ply 36) [structural distance $d=74$, QAP obj=26]: the master played **Kg2**; under μ our *Kf1* plays the role of that game’s *Kg1*; cross-type roles: our *Qb1* as their *Rf1*, our *pc7* as their *Qc2*; transferred move **f1e2**. Synthesis: 2 retrieved analogs transferred to **a4** (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



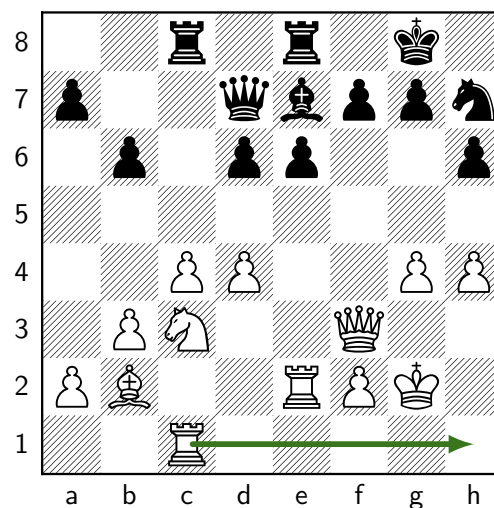
(a)



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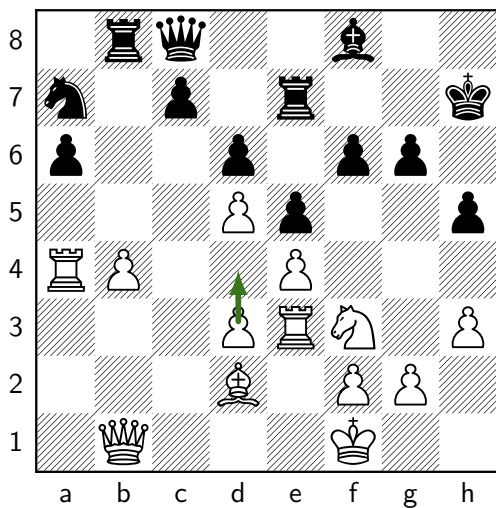


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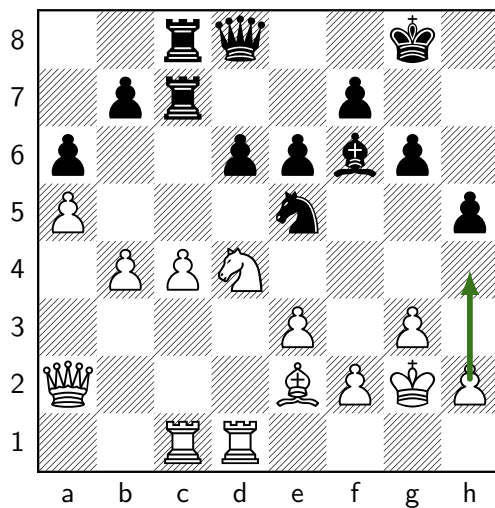


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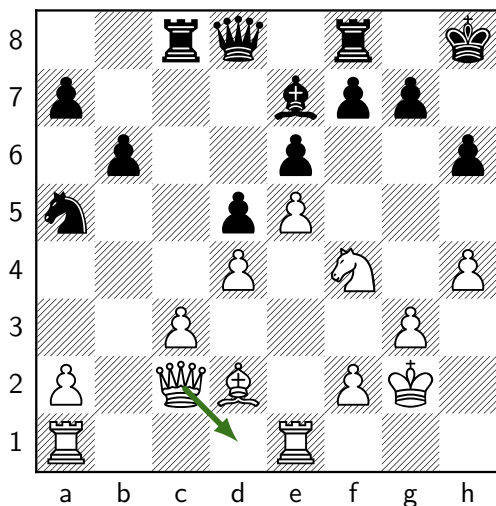
Figure 16: **Move 23, White: Rxa4.** (a) Our position; Melanie plays Rxa4 (a1a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#778` (ply 51) [structural distance $d=59$, QAP obj=66]: the master played Bc5; under μ our *Ra1* plays the role of that game’s *Bd3* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra8*, our *ph5* as their *Qd7*; transferred move a1a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1103` (ply 50) [structural distance $d=68$, QAP obj=61]: the master played Qf4; under μ our *Ra1* plays the role of that game’s *Qe3* (a **cross-type** correspondence); cross-type roles: our *Qb1* as their *Rg1*, our *Pg2* as their *Nc2*; transferred move a1a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#434` (ply 50) [structural distance $d=70$, QAP obj=77]: the master played Rh1; under μ our *Ph3* plays the role of that game’s *Rc1* (a **cross-type** correspondence); cross-type roles: our *Qb1* as their *Bb2*, our *Bd2* as their *Pb3*; transferred move h3h4. Synthesis: 2 retrieved analogs transferred to Rxa4 (summed QAP objective 127, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



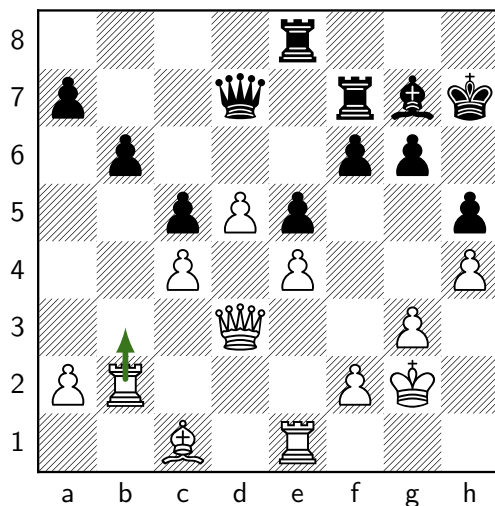
(a)



(b)

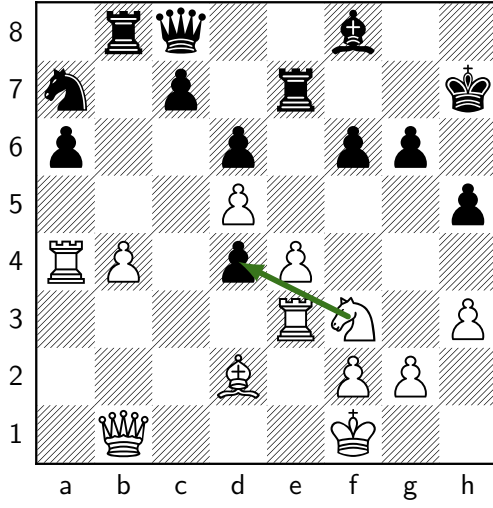


(c)

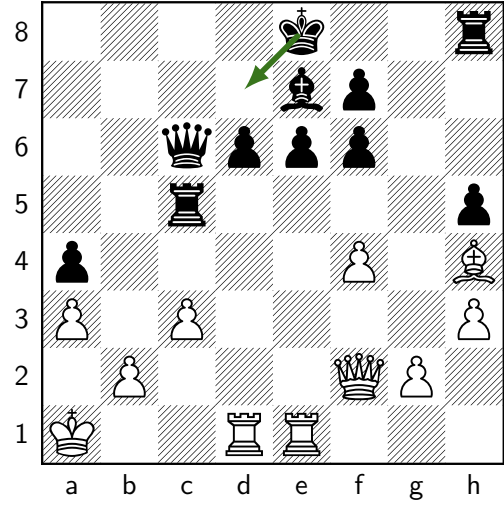


(d)

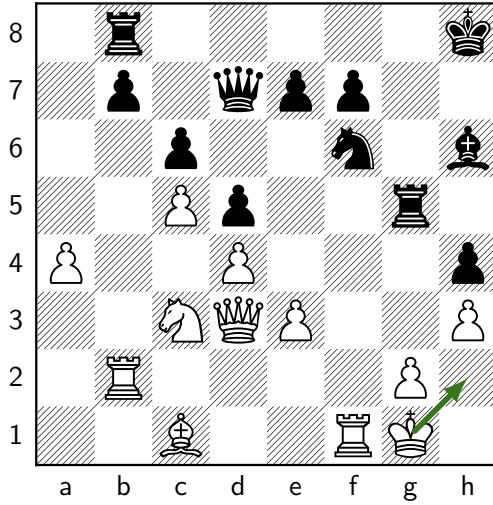
Figure 17: **Move 24, White: d4.** (a) Our position; Melanie plays d4 (d3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#769` (ply 46) [structural distance $d=68$, QAP obj=73]: the master played h4; under μ our $Pd3$ plays the role of that game’s $Ph2$; cross-type roles: our $qc8$ as their $Rc1$, our $Bd2$ as their $Rd1$; transferred move d3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#680` (ply 46) [structural distance $d=46$, QAP obj=71]: the master played Qd1; under μ our $Bd2$ plays the role of that game’s $Qc2$ (a **cross-type** correspondence); cross-type roles: our $Qb1$ as their $Ra1$, our $Pe4$ as their $Re1$; transferred move d2c1. (d) Retrieved from `ccr1_top_p0003#1964` (ply 58) [structural distance $d=73$, QAP obj=55]: the master played Rb3; under μ our $Nf3$ plays the role of that game’s $Rb2$ (a **cross-type** correspondence); cross-type roles: our $Re3$ as their $Bc1$, our $ph5$ as their $Re1$; transferred move f3h2. Synthesis: 1 retrieved analog transferred to d4 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



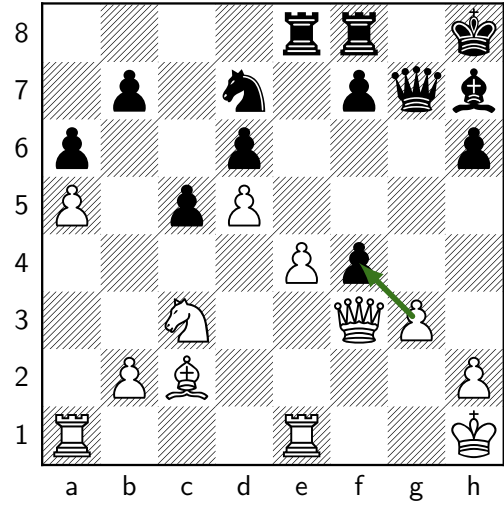
(a)



(b)



(c)



(d)

Figure 18: **Move 25, White: Nxd4.** (a) Our position; Melanie plays Nxd4 (f3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1653` (ply 51) [structural distance $d=76$, QAP obj=37]: the master played Kd7; under μ our Kf1 plays the role of that game’s Ke1; cross-type roles: our Qb1 as their Rh8, our Re3 as their Be7; transferred move f1g1. (c) Retrieved from `ccr1_top_p0000#666` (ply 46) [structural distance $d=72$, QAP obj=34]: the master played Kh2; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our qc8 as their Bc1, our ph5 as their Rf1; transferred move f1g1. (d) Retrieved from `ccr1_top_p0002#1115` (ply 64) [structural distance $d=75$, QAP obj=30]: the master played gxf4; under μ our Bd2 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our pd6 as their Ra1, our pc7 as their Re1; transferred move d2c3. Synthesis: no retrieved analog transferred to Nxd4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

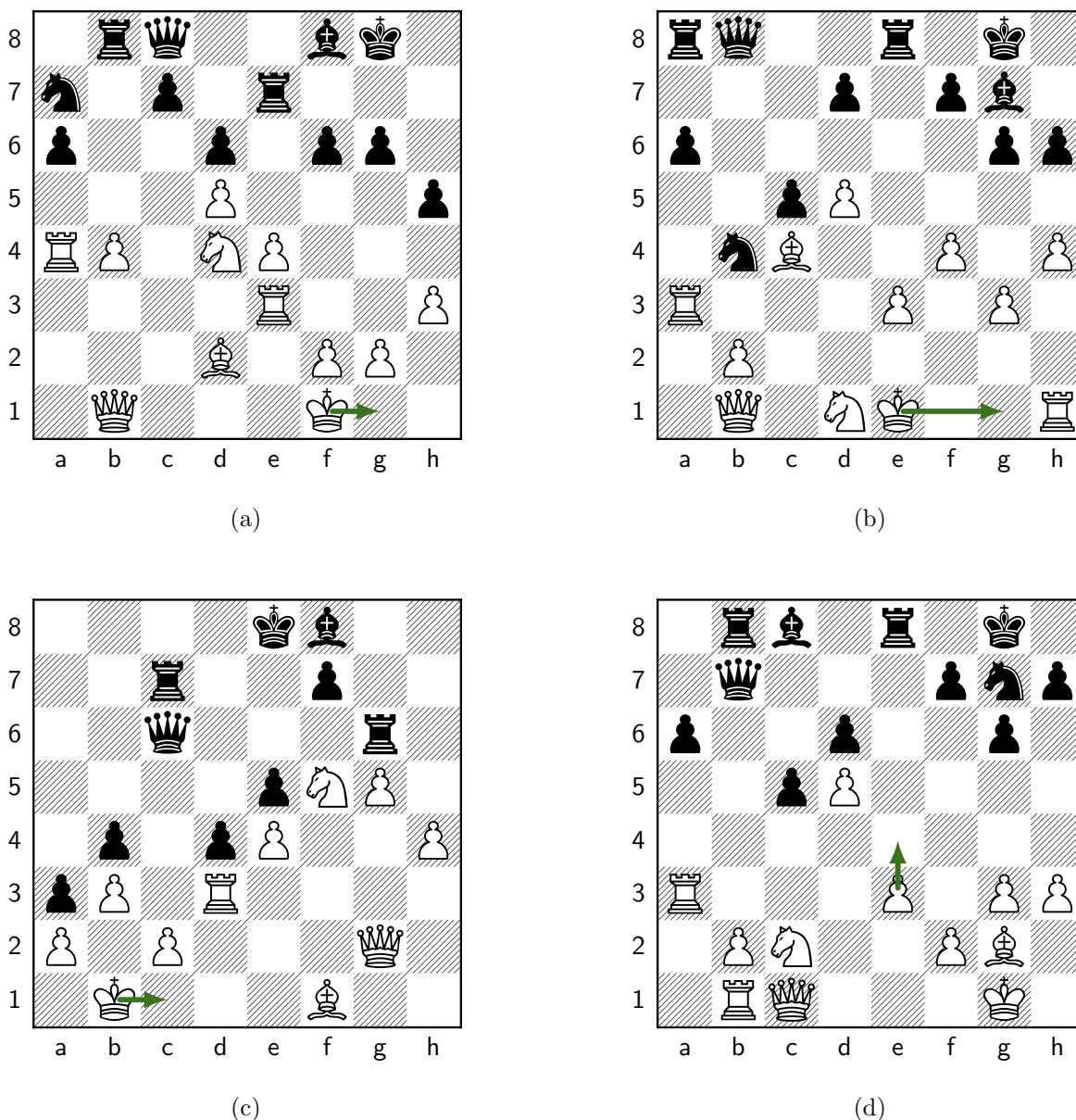
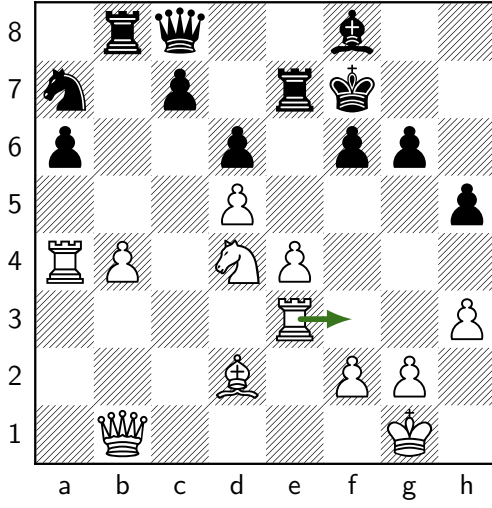
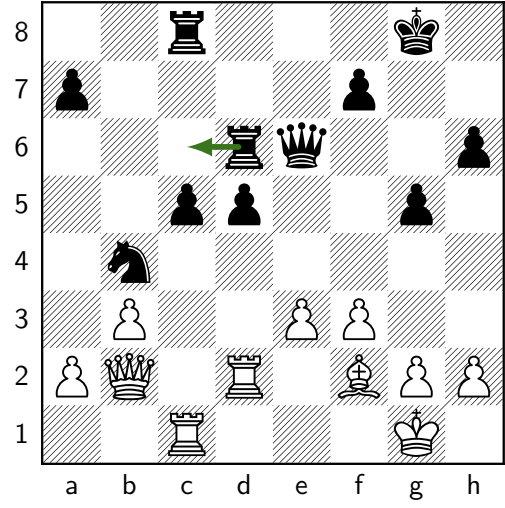


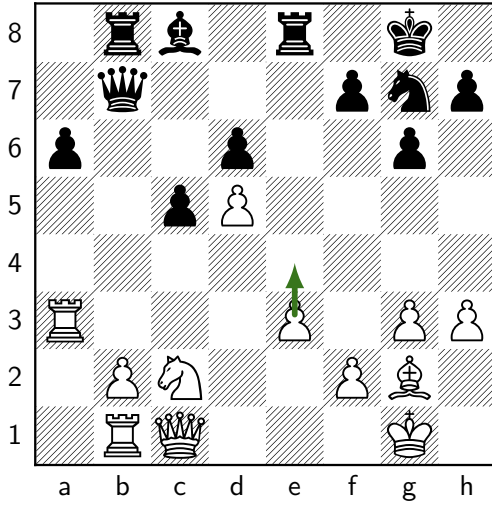
Figure 19: **Move 26, White: Kg1.** (a) Our position; Melanie plays Kg1 (f1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3387 (ply 48) [structural distance $d=86$, QAP obj=82]: the master played 0-0; under μ our Kf1 plays the role of that game’s Ke1; cross-type roles: our Re3 as their Nd1, our Pd5 as their Rh1; transferred move f1g1 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3100 (ply 62) [structural distance $d=80$, QAP obj=47]: the master played Kc1; under μ our Kf1 plays the role of that game’s Kb1; cross-type roles: our Nd4 as their Bf1, our na7 as their pa3; transferred move f1g1 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#523 (ply 58) [structural distance $d=87$, QAP obj=81]: the master played e4; under μ our Pb4 plays the role of that game’s Pe3; cross-type roles: our Pd5 as their Nc2, our Nd4 as their Bg2; transferred move b4b5. Synthesis: 2 retrieved analogs transferred to Kg1 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



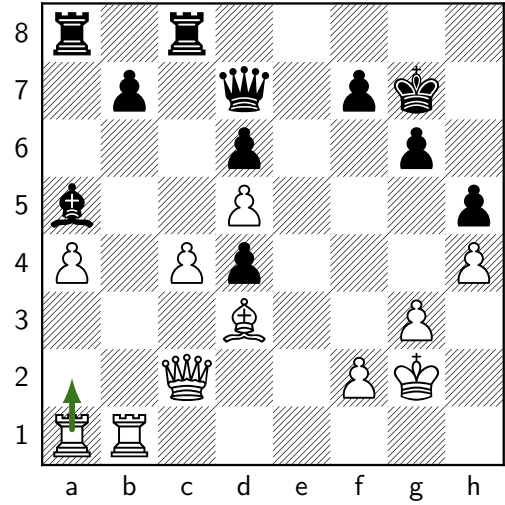
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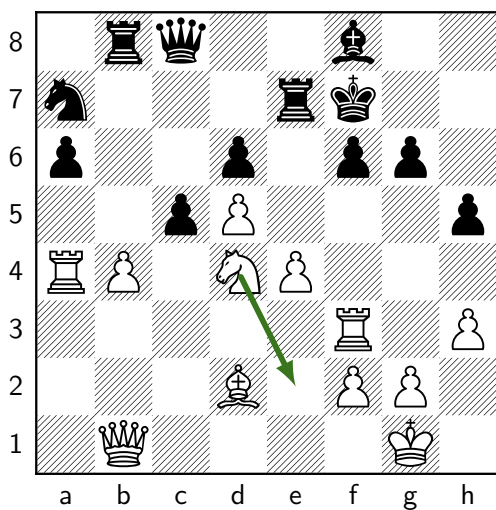


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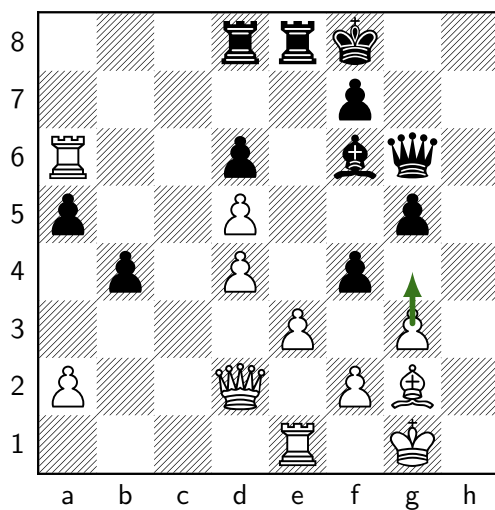


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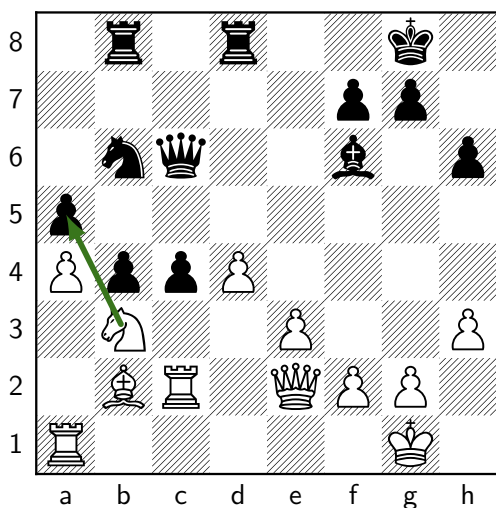
Figure 20: **Move 27, White: Rf3.** (a) Our position; Melanie plays Rf3 (e3f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1201` (ply 53) [structural distance $d=76$, QAP obj=81]: the master played Rdc6; under μ our Re3 plays the role of that game’s Rd3; cross-type roles: our Qb1 as their Rc8, our Nd4 as their Pa7; transferred move e3f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#523` (ply 58) [structural distance $d=74$, QAP obj=83]: the master played e4; under μ our Pe4 plays the role of that game’s Pe3; cross-type roles: our Qb1 as their Rb1, our Nd4 as their Qc1; transferred move e4e5. (d) Retrieved from `ccrl_top_p0003#1189` (ply 48) [structural distance $d=82$, QAP obj=58]: the master played Ra2; under μ our Re3 plays the role of that game’s Ra1; cross-type roles: our qc8 as their Rb1, our Bd2 as their Pf2; transferred move e3e2. Synthesis: 1 retrieved analog transferred to Rf3 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



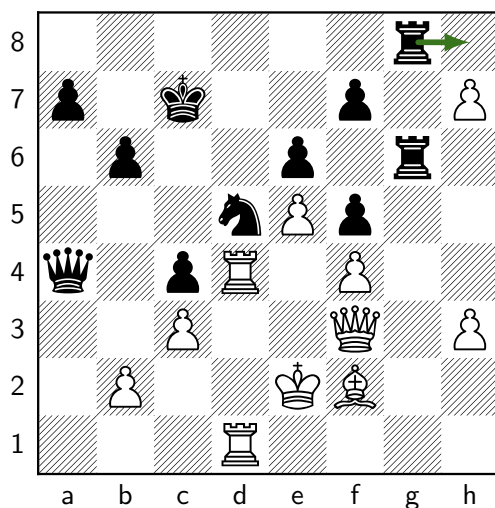
(a)



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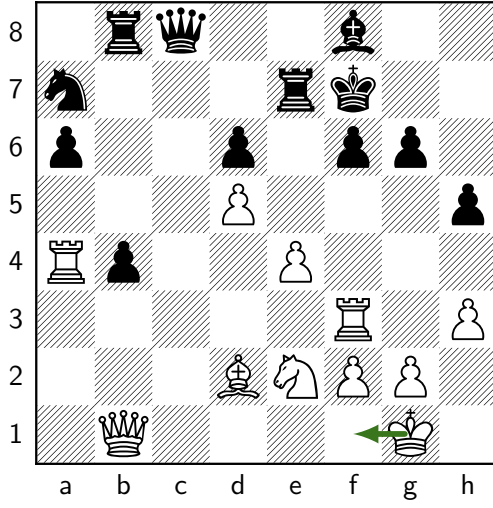


(c)

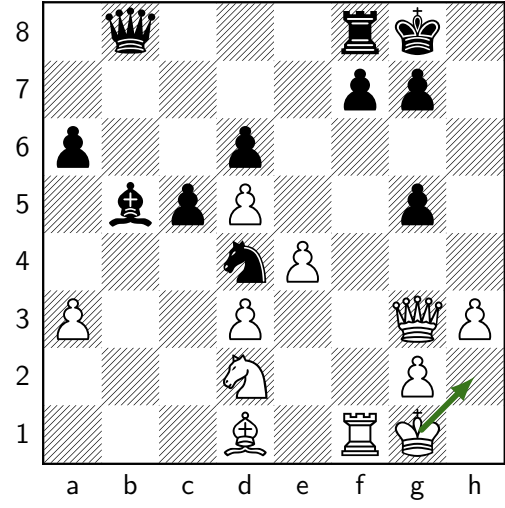


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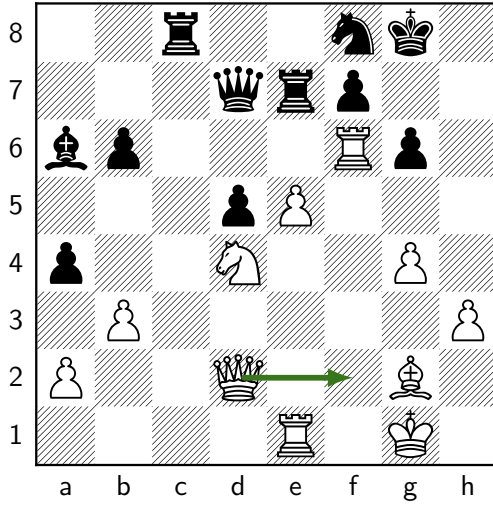
Figure 21: **Move 28, White: Ne2.** (a) Our position; Melanie plays Ne2 (d4e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#338` (ply 64) [structural distance $d=94$, QAP obj=74]: the master played g4; under μ our Nd4 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our re7 as their bf6, our na7 as their pf7; transferred move d4e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2417` (ply 56) [structural distance $d=93$, QAP obj=55]: the master played Nxa5; under μ our Nd4 plays the role of that game’s Nb3; cross-type roles: our Pd5 as their Bb2, our Pb4 as their Rc2; transferred move d4e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1975` (ply 69) [structural distance $d=90$, QAP obj=33]: the master played Rh8; under μ our Nd4 plays the role of that game’s Rg1 (a **cross-type** correspondence); cross-type roles: our qc8 as their Pe6, our pa6 as their Nd5; transferred move d4e2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ne2 (summed QAP objective 162, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



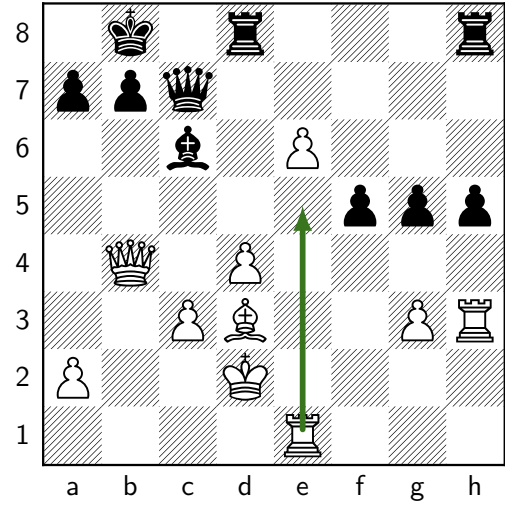
(a)



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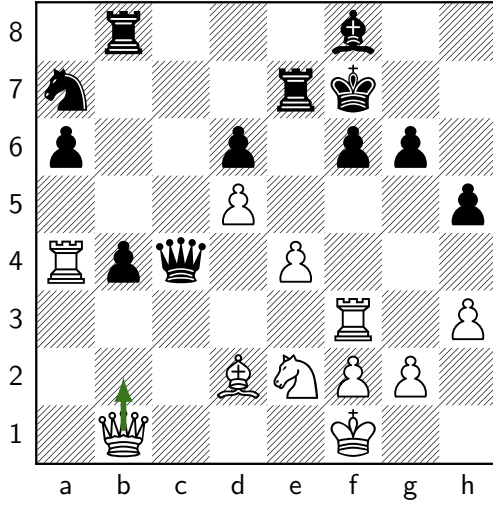


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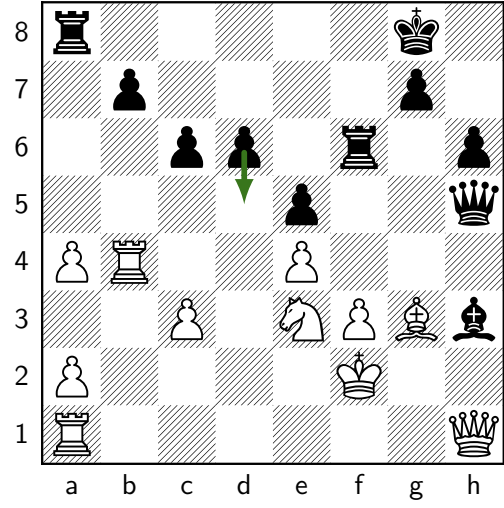


(d)

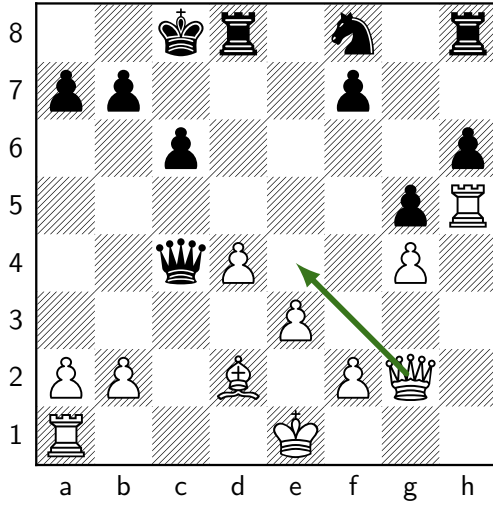
Figure 22: **Move 29, White: Kf1**. (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#134` (ply 48) [structural distance $d=91$, QAP obj=75]: the master played Kh2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pf2 as their Bd1, our bf8 as their Pg2; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1243` (ply 66) [structural distance $d=78$, QAP obj=96]: the master played Qf2; under μ our Pg2 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Re1, our Qb1 as their Pa2; transferred move g2g3. (d) Retrieved from `ccr1_top_p0002#1687` (ply 56) [structural distance $d=74$, QAP obj=84]: the master played Re5; under μ our Ne2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Qb1 as their Pa2, our Rf3 as their Qb4; transferred move e2d4. Synthesis: 1 retrieved analog transferred to Kf1 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



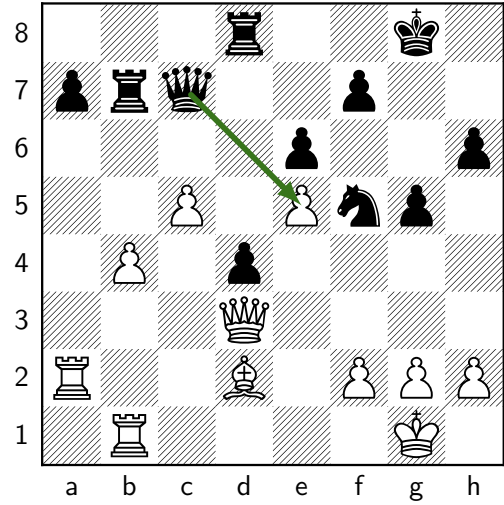
(a)



(b)

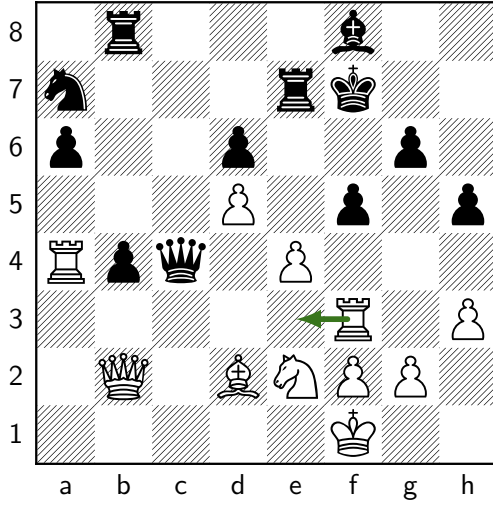


(c)

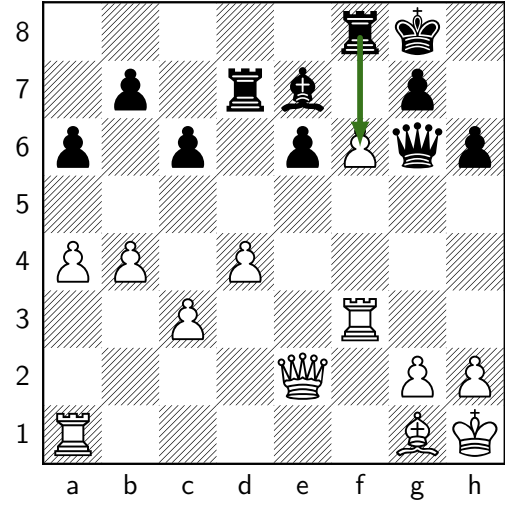


(d)

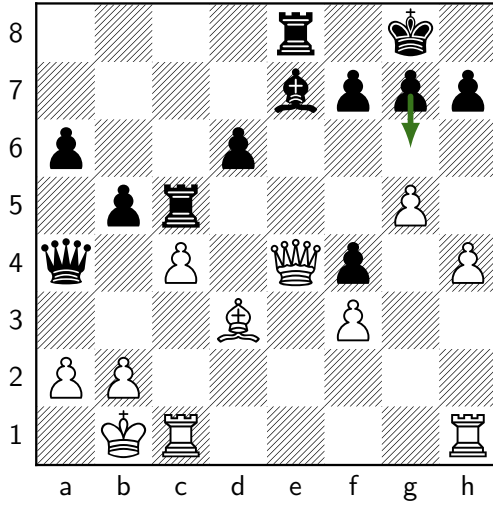
Figure 23: **Move 30, White: Qb2**. (a) Our position; Melanie plays Qb2 (b1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#718` (ply 65) [structural distance $d=87$, QAP obj=76]: the master played d5; under μ our *Ph3* plays the role of that game’s *Pd3*; cross-type roles: our *Pd5* as their *Ra8*, our *bf8* as their *Pg7*; transferred move *h3h4*. (c) Retrieved from `ccl1_top_p0002#2510` (ply 42) [structural distance $d=87$, QAP obj=67]: the master played Qe4; under μ our *Rf3* plays the role of that game’s *Qg2* (a **cross-type** correspondence); cross-type roles: our *bf8* as their *Pb2*, our *na7* as their *Bd2*; transferred move *f3e3*. (d) Retrieved from `ccl1_top_p0002#1911` (ply 63) [structural distance $d=95$, QAP obj=67]: the master played Qxe5; under μ our *Ra4* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qb1* as their *Rd8*, our *Ne2* as their *Pd4*; transferred move *a4a1*. Synthesis: no retrieved analog transferred to Qb2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



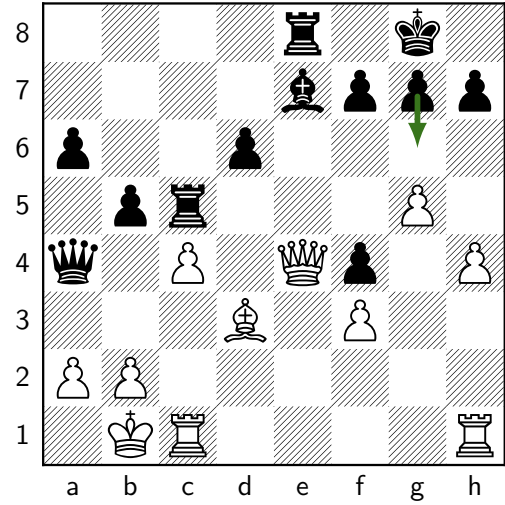
(a)



(b)

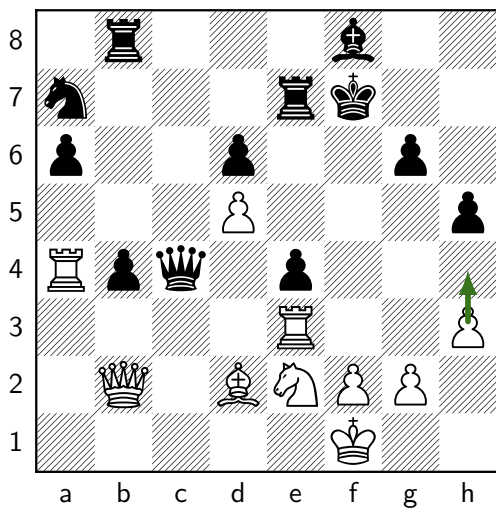


(c)

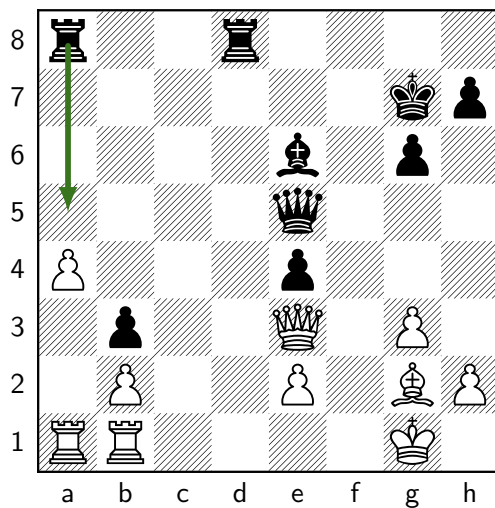


(d)

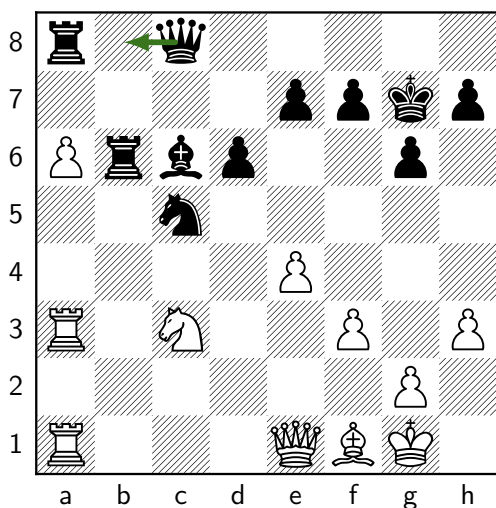
Figure 24: **Move 31, White: Re3.** (a) Our position; Melanie plays Re3 (f3e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#3135` (ply 45) [structural distance $d=85$, QAP obj=87]: the master played Rxf6; under μ our Rf3 plays the role of that game’s Rf1; cross-type roles: our Pe4 as their Be7, our Ne2 as their Pe6; transferred move f3e3 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#344` (ply 45) [structural distance $d=96$, QAP obj=194]: the master played g6; under μ our Ra4 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our ph5 as their Re8, our Ne2 as their Be7; transferred move a4a2. (d) Retrieved from `ccl1_top_p0000#2921` (ply 45) [structural distance $d=96$, QAP obj=194]: the master played g6; under μ our Ra4 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our ph5 as their Re8, our Ne2 as their Be7; transferred move a4a2. Synthesis: 1 retrieved analog transferred to Re3 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



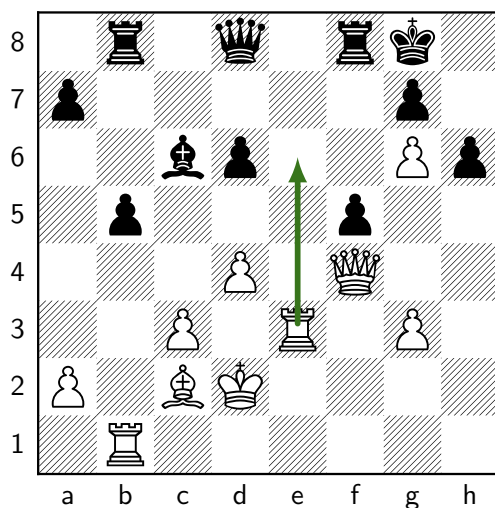
(a)



(b)

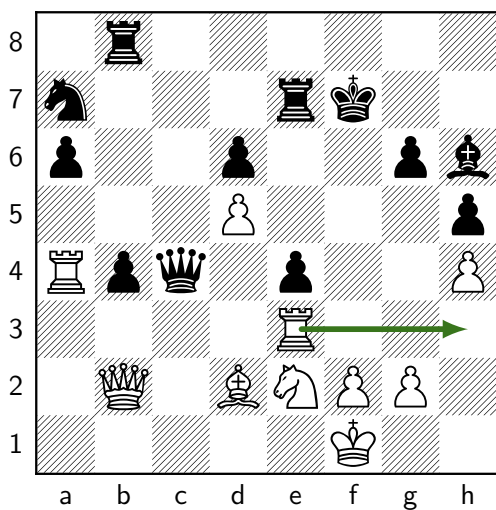


(c)

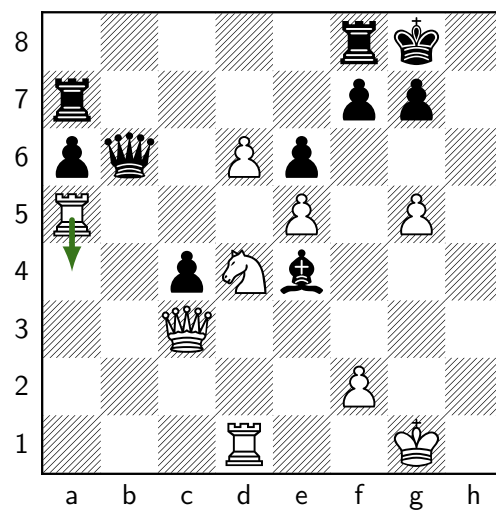


(d)

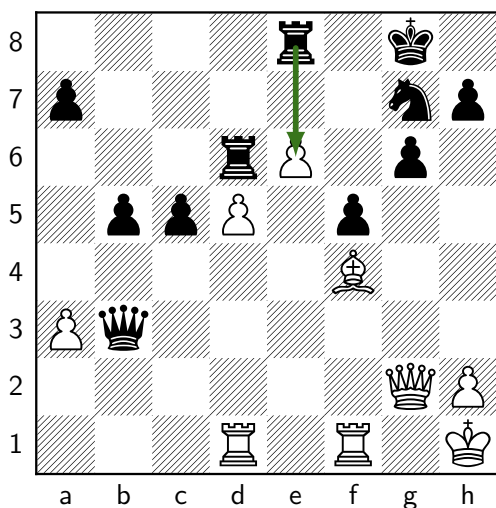
Figure 25: **Move 32, White: h4.** (a) Our position; Melanie plays h4 (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#593` (ply 61) [structural distance $d=93$, QAP obj=74]: the master played **Ra5**; under μ our *Ph3* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *na7* as their *Be6*, our *Ra4* as their *qe3*; transferred move **h3h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2320` (ply 57) [structural distance $d=94$, QAP obj=72]: the master played **Qb8**; under μ our *Qb2* plays the role of that game’s *Qc1*; cross-type roles: our *Qb2* as their *Rb6*, our *Qb2* as their *Bc6*; transferred move **b2a1**. (d) Retrieved from `ccr1_top_p0001#2435` (ply 56) [structural distance $d=95$, QAP obj=56]: the master played **Re6**; under μ our *Bd2* plays the role of that game’s *Re3* (a **cross-type** correspondence); cross-type roles: our *Ne2* as their *Pa2*, our *Qb2* as their *Bc2*; transferred move **d2c1**. Synthesis: 1 retrieved analog transferred to h4 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



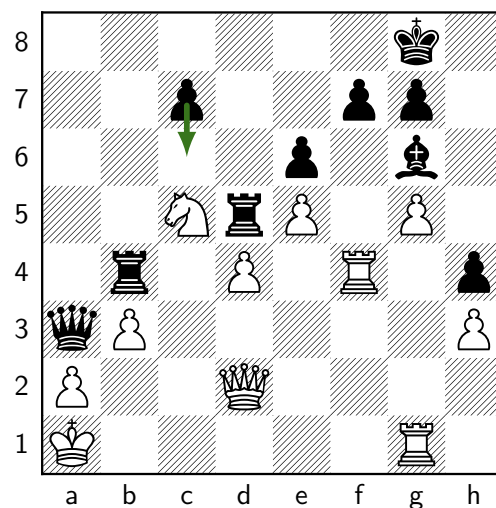
(a)



(b)

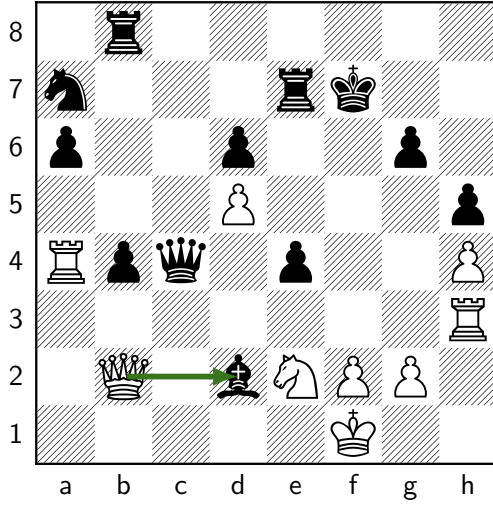


(c)

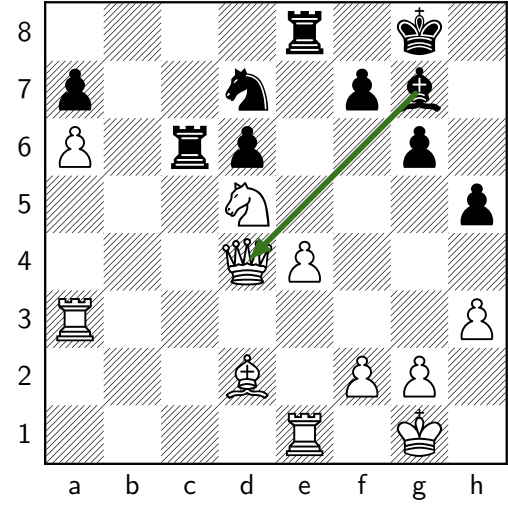


(d)

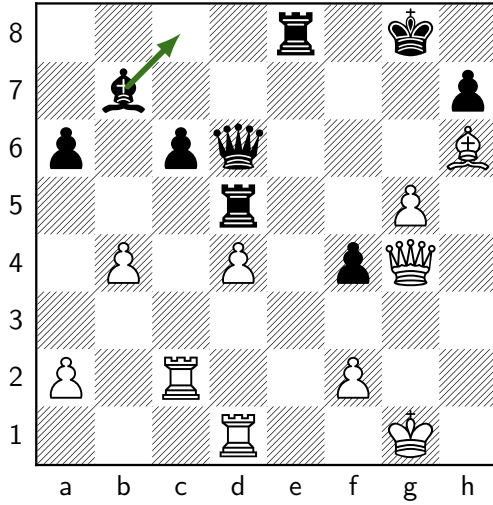
Figure 26: **Move 33, White: Rh3.** (a) Our position; Melanie plays Rh3 (e3h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3088` (ply 66) [structural distance $d=85$, QAP obj=44]: the master played Ra4; under μ our Re3 plays the role of that game’s Ra5; cross-type roles: our Qb2 as their Rd1, our Bd2 as their Qc3; transferred move e3h3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2117` (ply 79) [structural distance $d=105$, QAP obj=94]: the master played Rxe6; under μ our Qb2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Bd2 as their Pa7, our Ra4 as their Ng7; transferred move b2a3. (d) Retrieved from `ccrl_top_p0001#731` (ply 77) [structural distance $d=99$, QAP obj=87]: the master played c6; under μ our Pg2 plays the role of that game’s Pc2; cross-type roles: our Bd2 as their Pg7, our Qb2 as their Bg6; transferred move g2g4. Synthesis: 1 retrieved analog transferred to Rh3 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



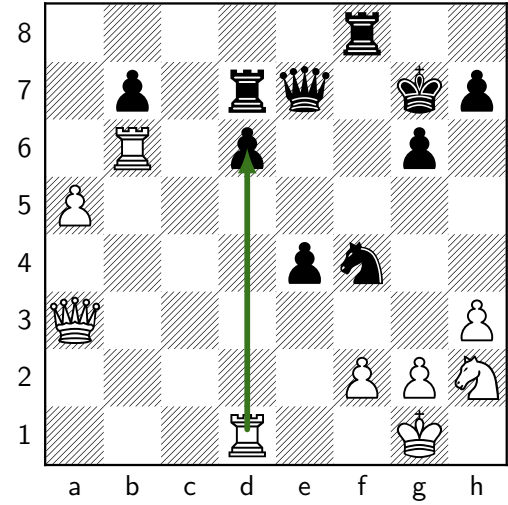
(a)



(b)

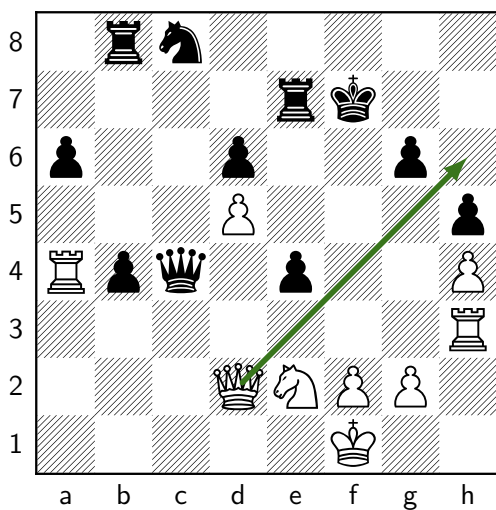


(c)

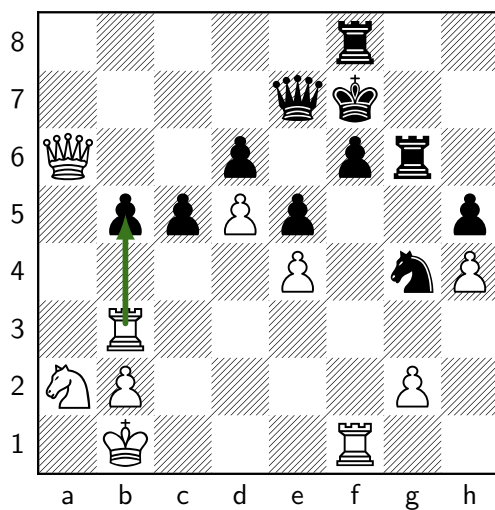


(d)

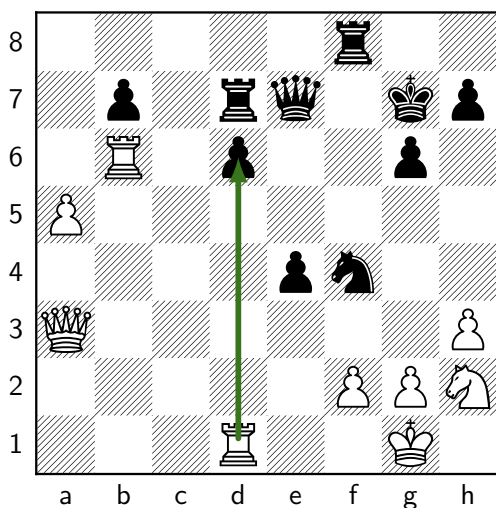
Figure 27: **Move 34, White: Qxd2.** (a) Our position; Melanie plays Qxd2 (b2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#549` (ply 63) [structural distance $d=90$, QAP obj=13]: the master played Bxd4; under μ our Qb2 plays the role of that game's Bg2 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Pa7, our Rh3 as their Nd7; transferred move b2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2606` (ply 61) [structural distance $d=86$, QAP obj=60]: the master played Bc8; under μ our Pg2 plays the role of that game's Bb2 (a **cross-type** correspondence); cross-type roles: our ph5 as their Re8, our na7 as their Pa6; transferred move g2g3. (d) Retrieved from `ccr1_top_p0001#2040` (ply 62) [structural distance $d=86$, QAP obj=56]: the master played Rdx6; under μ our Qb2 plays the role of that game's Rd1 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Pg2, our Pf2 as their Nh2; transferred move b2d4. Synthesis: 1 retrieved analog transferred to Qxd2 (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



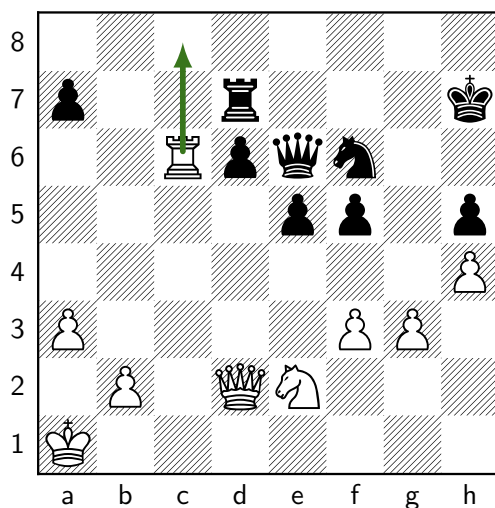
(a)



(b)

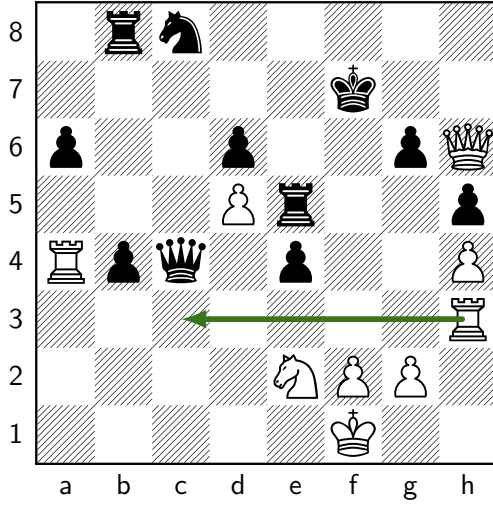


(c)

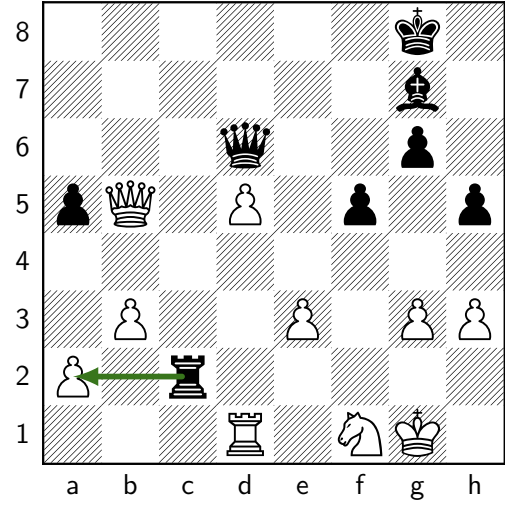


(d)

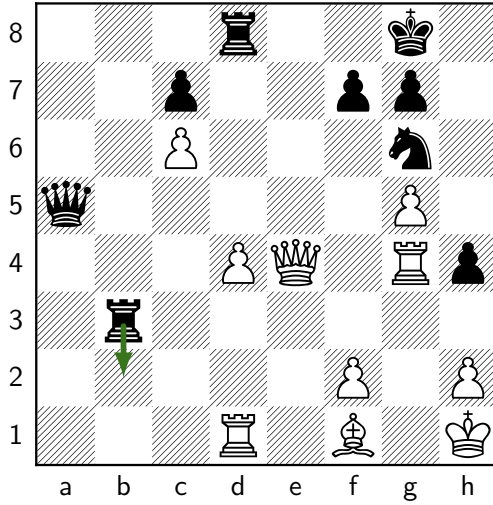
Figure 28: **Move 35, White: Qh6.** (a) Our position; Melanie plays Qh6 (d2h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#2290` (ply 68) [structural distance $d=70$, QAP obj=25]: the master played Rxb5; under μ our Qd2 plays the role of that game’s Rb3 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Na2, our nc8 as their Pb2; transferred move d2h6 (**endorses this move**). (c) Retrieved from `ccr1.top_p0001#2040` (ply 62) [structural distance $d=49$, QAP obj=41]: the master played Rdx6; under μ our Ra4 plays the role of that game’s Rd1; cross-type roles: our Ne2 as their Pg2, our Pg2 as their Nh2; transferred move a4b4. (d) Retrieved from `ccr1.top_p0001#758` (ply 58) [structural distance $d=80$, QAP obj=37]: the master played Rc8; under μ our Ra4 plays the role of that game’s Rc6; cross-type roles: our Rh3 as their Pg3, our re7 as their nf6; transferred move a4a6. Synthesis: 1 retrieved analog transferred to Qh6 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



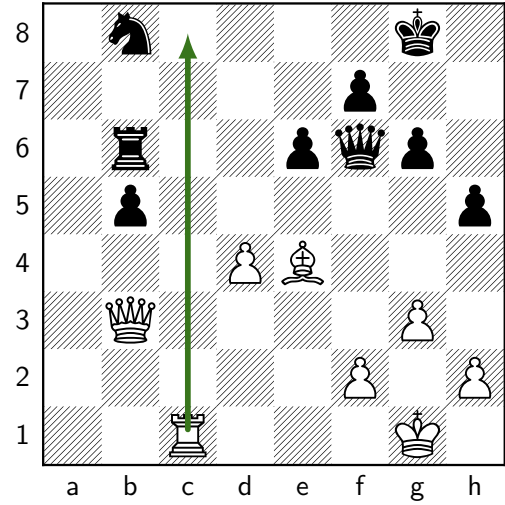
(a)



(b)

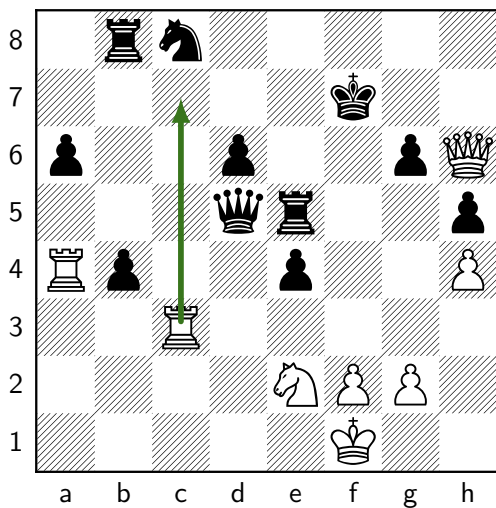


(c)

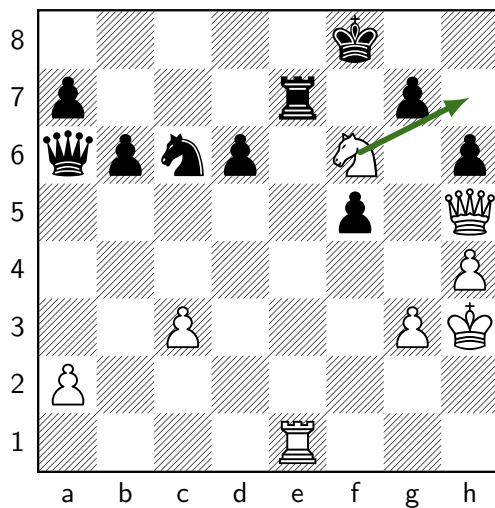


(d)

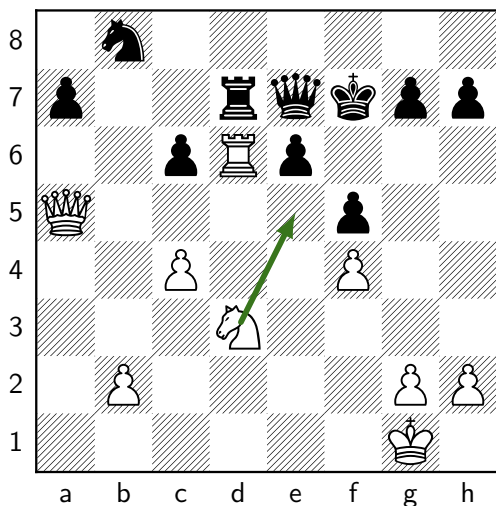
Figure 29: **Move 36, White: Rc3.** (a) Our position; Melanie plays Rc3 (h3c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2912` (ply 67) [structural distance $d=90$, QAP obj=55]: the master played Rxa2; under μ our Qh6 plays the role of that game’s Rc7 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Bg7, our Ra4 as their Qd6; transferred move h6h7. (c) Retrieved from `ccrl_top_p0003#458` (ply 87) [structural distance $d=84$, QAP obj=36]: the master played Rb2; under μ our Qh6 plays the role of that game’s Rb6 (a **cross-type** correspondence); cross-type roles: our Rh3 as their Qa5, our nc8 as their pf2; transferred move h6h7. (d) Retrieved from `ccrl_top_p0001#1795` (ply 54) [structural distance $d=91$, QAP obj=34]: the master played Rc8+; under μ our Rh3 plays the role of that game’s Rc1; cross-type roles: our Ne2 as their Pf2, our Ra4 as their Qb3; transferred move h3h2. Synthesis: no retrieved analog transferred to Rc3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



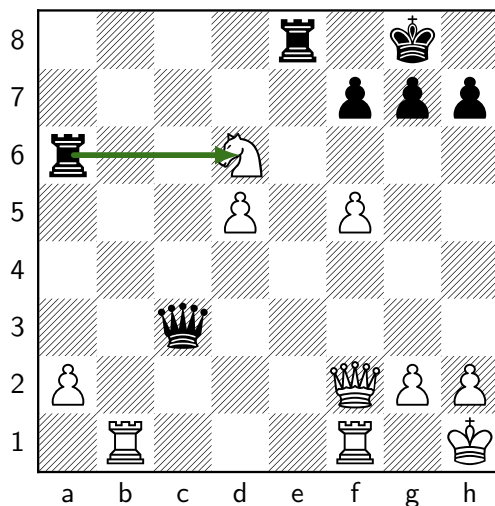
(a)



(b)

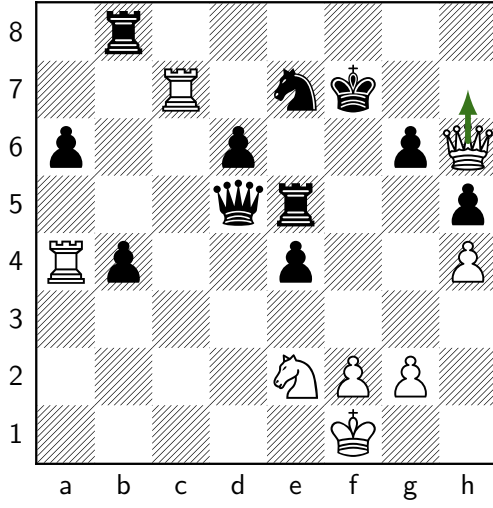


(c)

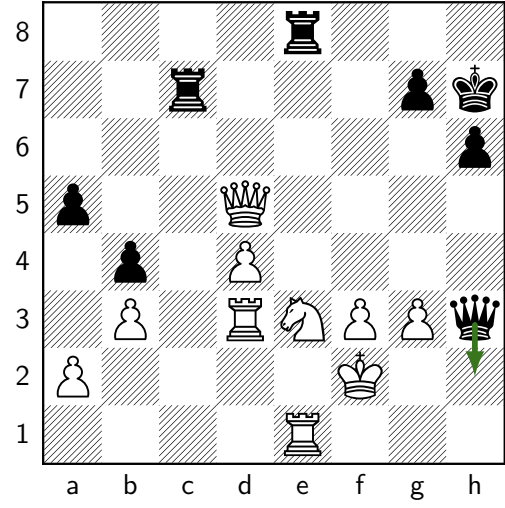


(d)

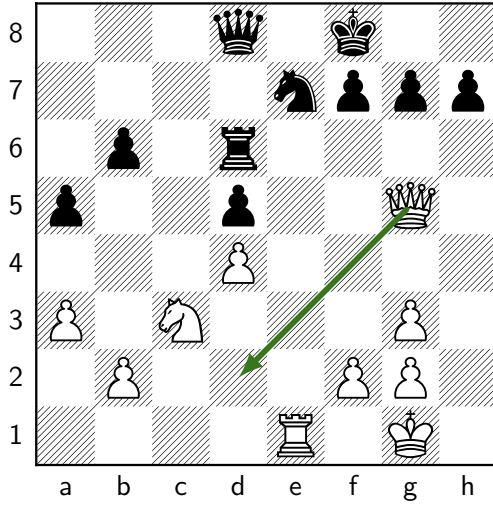
Figure 30: **Move 37, White: Rc7+.** (a) Our position; Melanie plays Rc7+ (c3c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2680` (ply 78) [structural distance $d=89$, QAP obj=14]: the master played Nh7+; under μ our Rc3 plays the role of that game’s Nf6 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Re1, our Ne2 as their Pa2; transferred move c3c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#335` (ply 60) [structural distance $d=88$, QAP obj=33]: the master played Ne5+; under μ our Qh6 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Pb2, our Ra4 as their Qa5; transferred move h6g5. (d) Retrieved from `ccr1_top_p0003#2898` (ply 65) [structural distance $d=87$, QAP obj=22]: the master played Rxd6; under μ our Ra4 plays the role of that game’s Ra3; cross-type roles: our Ne2 as their Ph7, our pb4 as their nd6; transferred move a4a6. Synthesis: 1 retrieved analog transferred to Rc7+ (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



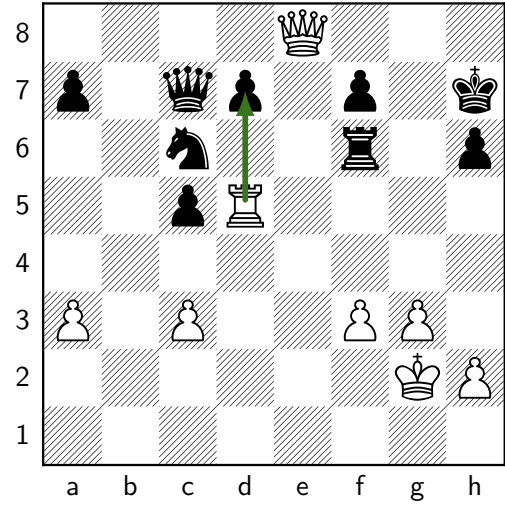
(a)



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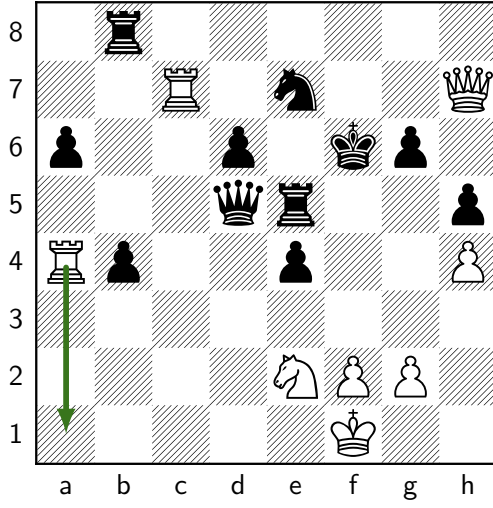


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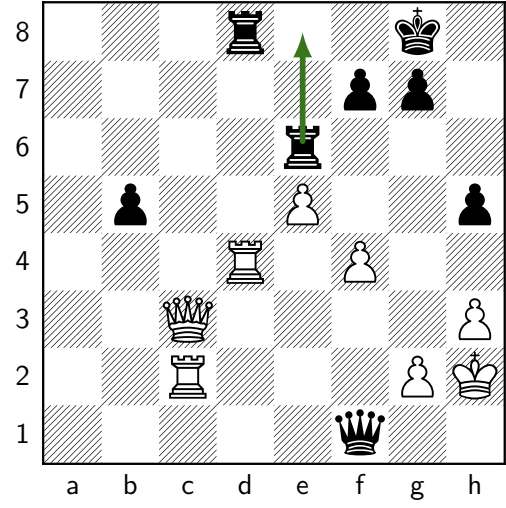


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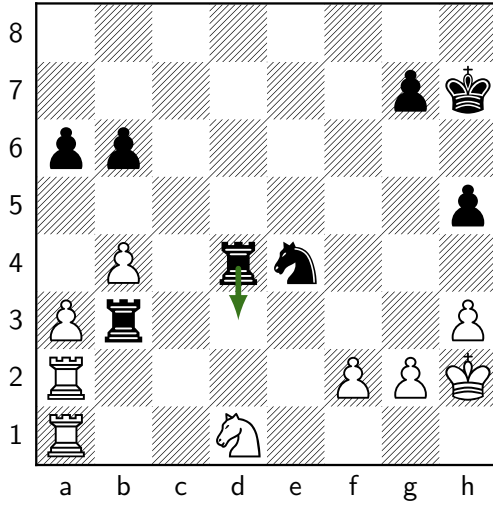
Figure 31: **Move 38, White: Qh7+.** (a) Our position; Melanie plays Qh7+ (h6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2304` (ply 85) [structural distance $d=85$, QAP obj=39]: the master played Qh2+; under μ our Qh6 plays the role of that game’s Qh6; cross-type roles: our qd5 as their Rc7, our Ra4 as their qd5; transferred move h6h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3337` (ply 62) [structural distance $d=93$, QAP obj=53]: the master played Qd2; under μ our Rc7 plays the role of that game’s Qg5 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Pf2, our Ra4 as their Pg2; transferred move c7c2. (d) Retrieved from `ccr1_top_p0000#1779` (ply 62) [structural distance $d=90$, QAP obj=53]: the master played Rxd7; under μ our Ra4 plays the role of that game’s Rd5; cross-type roles: our Ne2 as their Pa3, our qd5 as their nc6; transferred move a4a6. Synthesis: 1 retrieved analog transferred to Qh7+ (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



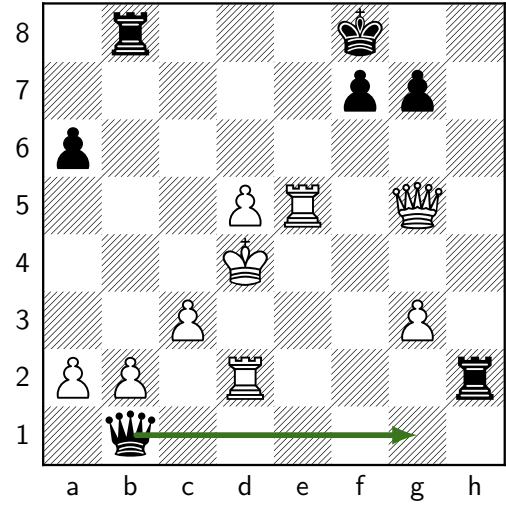
(a)



(b)

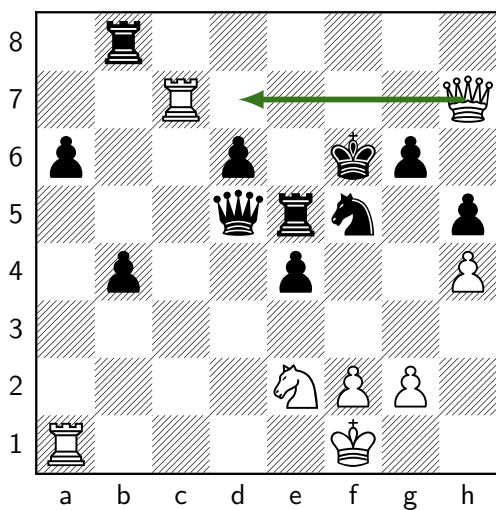


(c)

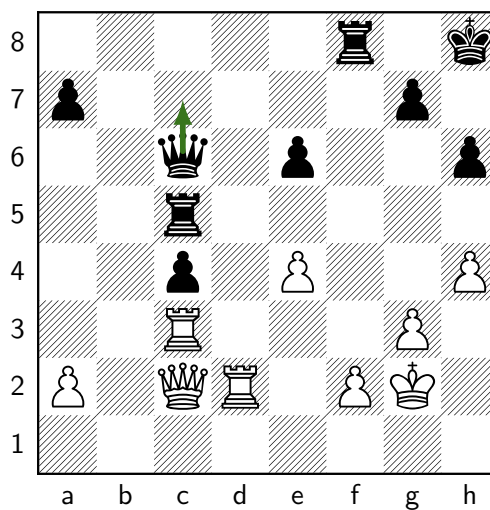


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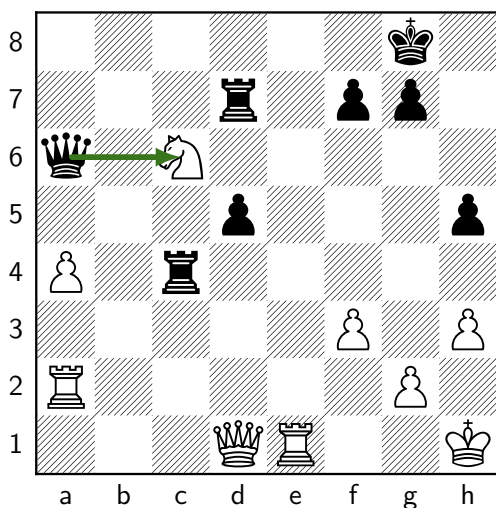
Figure 32: **Move 39, White: Ra1.** (a) Our position; Melanie plays **Ra1** (**a4a1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2444` (ply 71) [structural distance $d=79$, QAP obj=48]: the master played **Ree8**; under μ our **Ra4** plays the role of that game’s **Re3**; cross-type roles: our **Ne2** as their **Pg7**, our **ne7** as their **rd4**; transferred move **a4a1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2152` (ply 69) [structural distance $d=101$, QAP obj=51]: the master played **Rdd3**; under μ our **Rc7** plays the role of that game’s **Rd5**; cross-type roles: our **Ne2** as their **Pg7**, our **ne7** as their **pb4**; transferred move **c7a7**. (d) Retrieved from `ccr1_top_p0002#2330` (ply 69) [structural distance $d=87$, QAP obj=43]: the master played **Qg1+**; under μ our **Qh7** plays the role of that game’s **Qb8**; cross-type roles: our **ph5** as their **Rb8**, our **Ne2** as their **Pa6**; transferred move **h7h6**. Synthesis: 1 retrieved analog transferred to **Ra1** (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



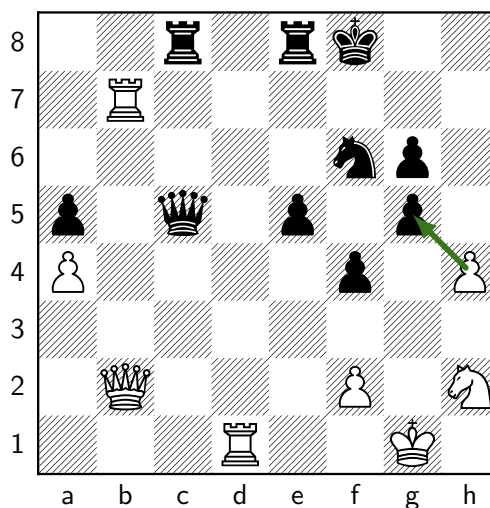
(a)



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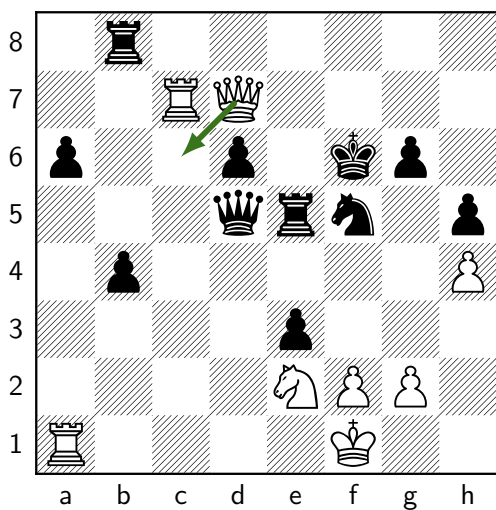


(c)

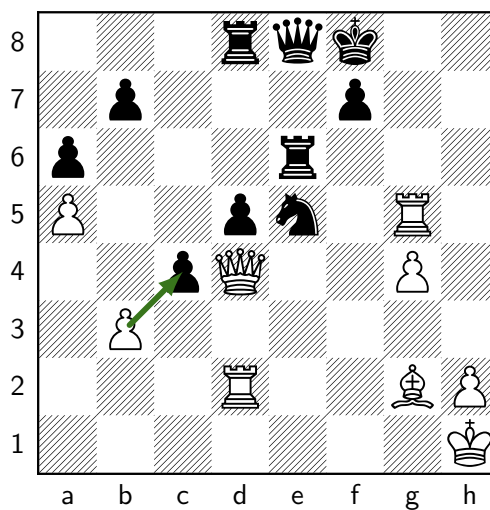


(d)

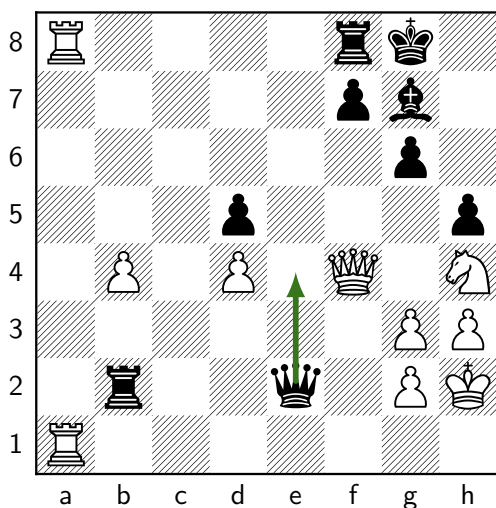
Figure 33: **Move 40, White: Qd7.** (a) Our position; Melanie plays Qd7 (h7d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1783` (ply 85) [structural distance $d=75$, QAP obj=54]: the master played Qc7; under μ our Qh7 plays the role of that game’s Qc3; cross-type roles: our Ne2 as their Pa7, our Qh7 as their rc3; transferred move h7d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1500` (ply 75) [structural distance $d=76$, QAP obj=45]: the master played Qxc6; under μ our Qh7 plays the role of that game’s Qa3; cross-type roles: our Ne2 as their Pg7, our ph5 as their nc6; transferred move h7d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#452` (ply 78) [structural distance $d=76$, QAP obj=35]: the master played hxg5; under μ our Qh7 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Rd1, our pb4 as their Nh2; transferred move h7d7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd7 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



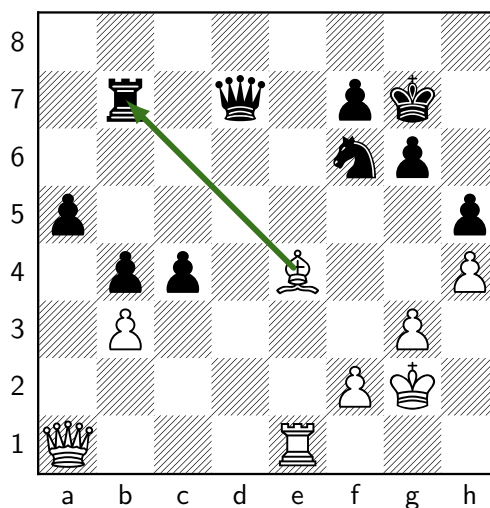
(a)



(b)

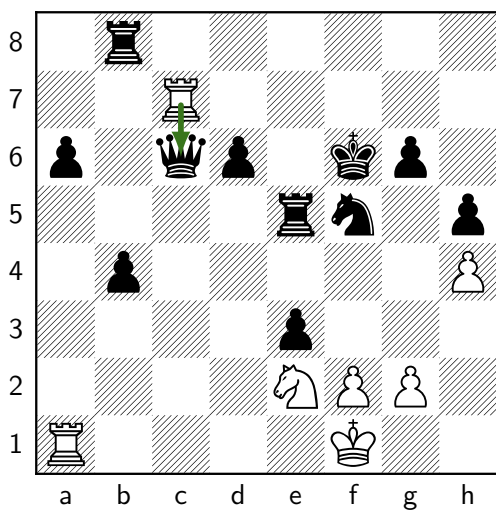


(c)

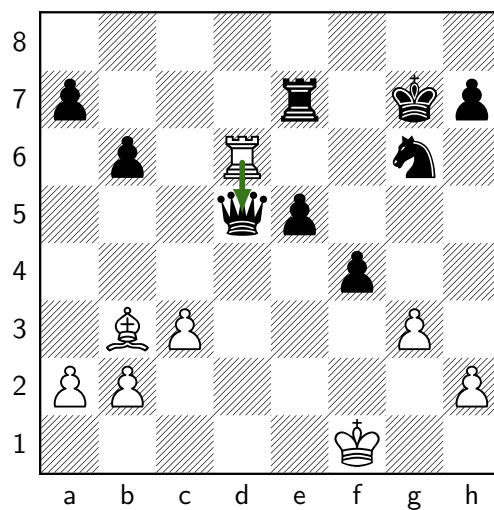


(d)

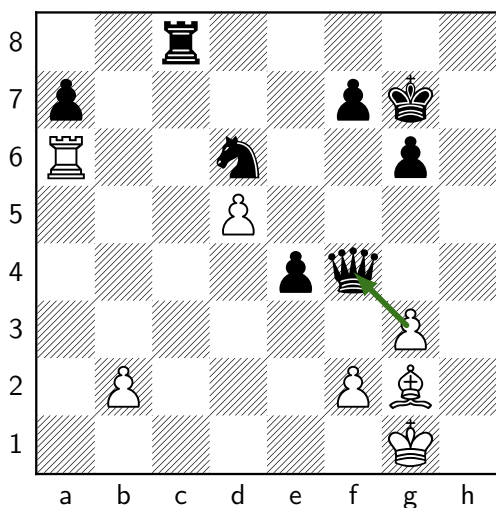
Figure 34: **Move 41, White: Qc6.** (a) Our position; Melanie plays Qc6 (d7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1124` (ply 70) [structural distance $d=77$, QAP obj=83]: the master played **bxc4**; under μ our Qd7 plays the role of that game's Pb3 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Bg2, our Ra1 as their Ph2; transferred move d7c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1230` (ply 75) [structural distance $d=88$, QAP obj=58]: the master played Qe4; under μ our Qd7 plays the role of that game's Qe7; cross-type roles: our Qd7 as their ra8, our pd6 as their Rf8; transferred move d7c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1338` (ply 70) [structural distance $d=87$, QAP obj=65]: the master played Bxb7; under μ our Qd7 plays the role of that game's Be4 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pg3, our re5 as their pb4; transferred move d7d8. Synthesis: 2 retrieved analogs transferred to Qc6 (summed QAP objective 141, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



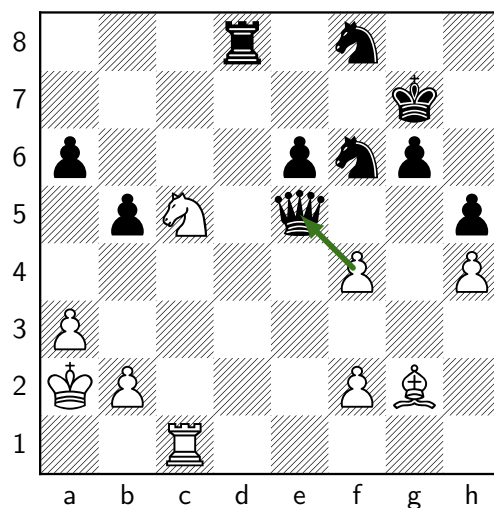
(a)



(b)

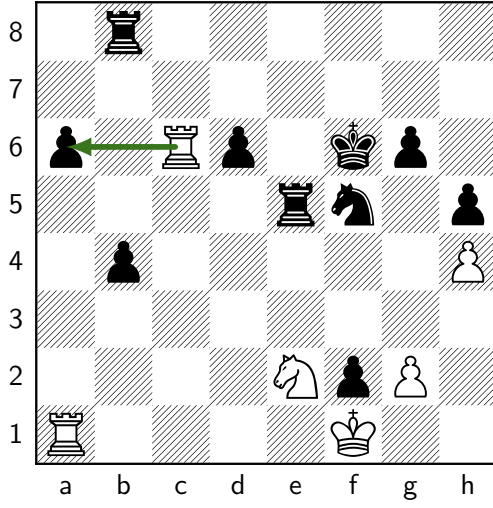


(c)

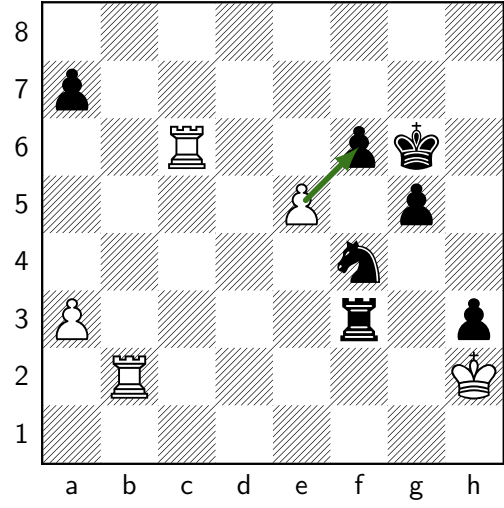


(d)

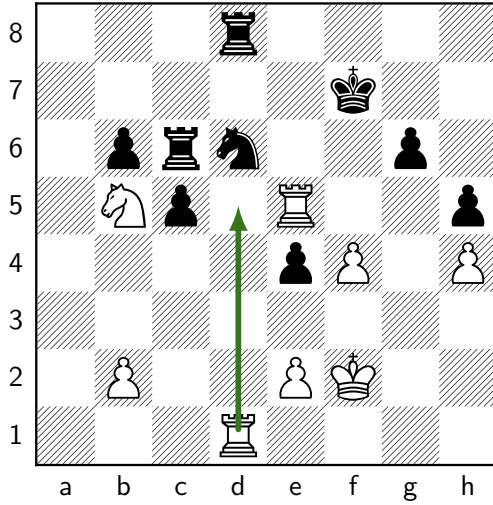
Figure 35: **Move 42, White: Rxc6**. (a) Our position; Melanie plays Rxc6 (c7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2437 (ply 72) [structural distance $d=83$, QAP obj=39]: the master played Rxd5; under μ our Rc7 plays the role of that game’s Rd6; cross-type roles: our Ne2 as their Pa2, our Rc7 as their Bb3; transferred move c7c6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3367 (ply 76) [structural distance $d=88$, QAP obj=22]: the master played gxf4; under μ our Rc7 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our pg6 as their Bg2, our pd6 as their nd6; transferred move c7c6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#1344 (ply 68) [structural distance $d=79$, QAP obj=22]: the master played fxe5; under μ our Rc7 plays the role of that game’s Pf4 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Pa3, our nf5 as their pb5; transferred move c7c6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rxc6 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



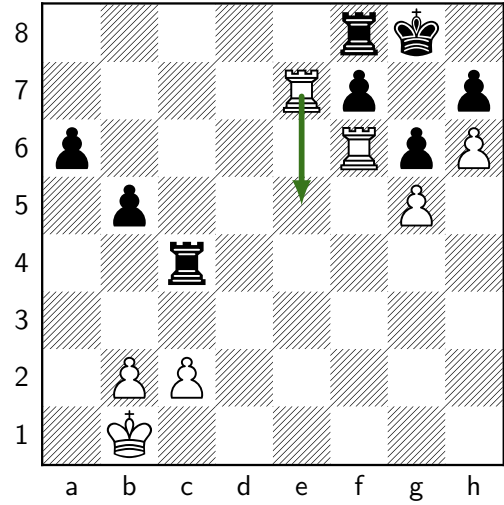
(a)



(b)



(c)



(d)

Figure 36: **Move 43, White: Rcxa6.** (a) Our position; Melanie plays Rcxa6 (c6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1492` (ply 94) [structural distance $d=66$, QAP obj=47]: the master played `exf6`; under μ our `Rc6` plays the role of that game’s `Pe5` (a **cross-type** correspondence); cross-type roles: our `Ne2` as their `Pa3`, our `Pg2` as their `rf3`; transferred move `c6a6` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3597` (ply 80) [structural distance $d=64$, QAP obj=43]: the master played `Rdd5`; under μ our `Rc6` plays the role of that game’s `Rd1`; cross-type roles: our `Ne2` as their `pe4`, our `Ra1` as their `Pf4`; transferred move `c6a6` (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2538` (ply 76) [structural distance $d=67$, QAP obj=79]: the master played `Re5`; under μ our `Rc6` plays the role of that game’s `Re7`; cross-type roles: our `Ne2` as their `Pc2`, our `Ra1` as their `pb5`; transferred move `c6c4`. Synthesis: 2 retrieved analogs transferred to Rcxa6 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

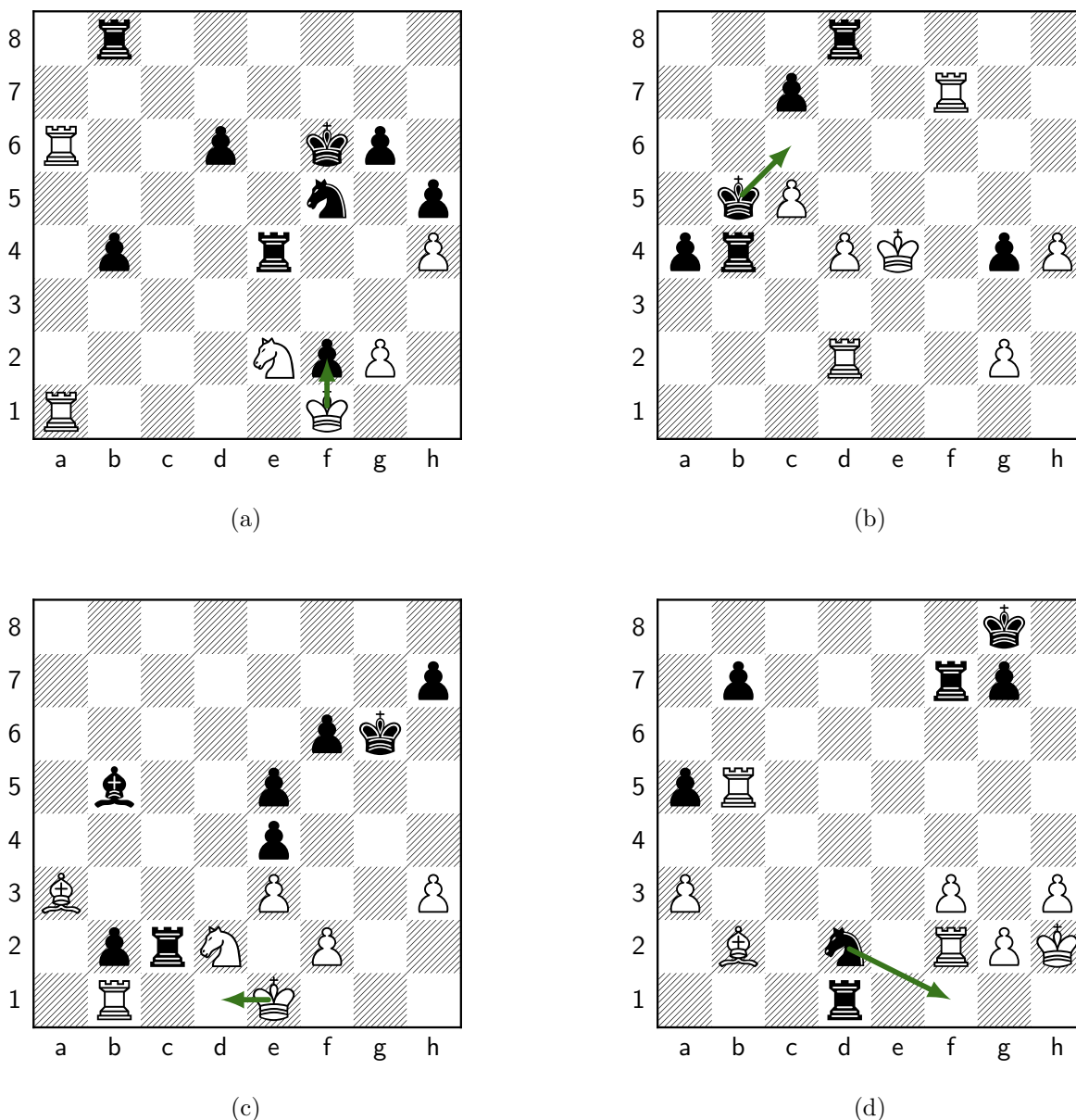
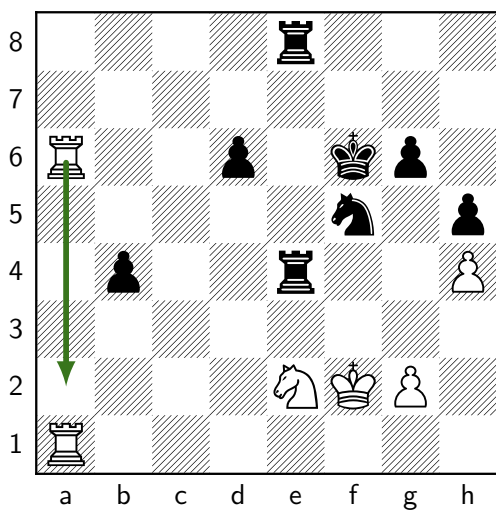
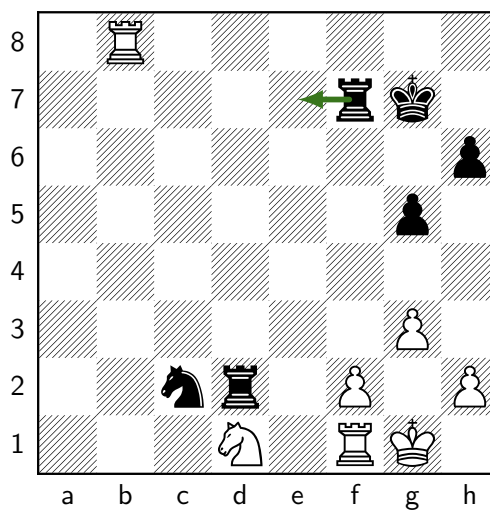


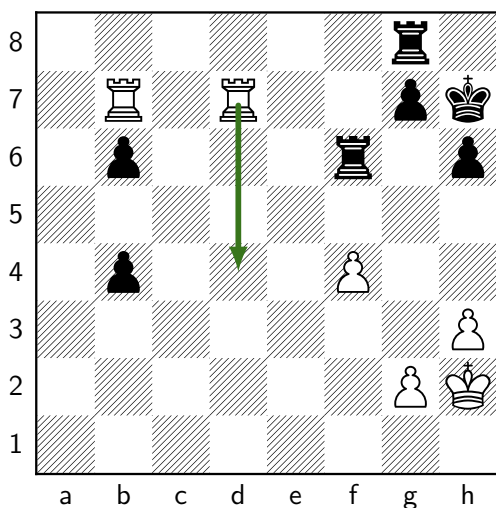
Figure 37: **Move 44, White: Kxf2.** (a) Our position; Melanie plays Kxf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2817` (ply 99) [structural distance $d=50$, QAP obj=42]: the master played Kc6; under μ our Kf1 plays the role of that game’s Kb4; cross-type roles: our Ne2 as their Pc7, our Ra1 as their ph4; transferred move f1f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#488` (ply 86) [structural distance $d=49$, QAP obj=18]: the master played Kd1; under μ our Kf1 plays the role of that game’s Ke1; cross-type roles: our Pg2 as their Ba3, our nf5 as their bb5; transferred move f1f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2178` (ply 85) [structural distance $d=51$, QAP obj=44]: the master played Nf1+; under μ our Ne2 plays the role of that game’s Nd7; cross-type roles: our Ra1 as their Pg7, our Kf1 as their rb5; transferred move e2c3. Synthesis: 2 retrieved analogs transferred to Kxf2 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



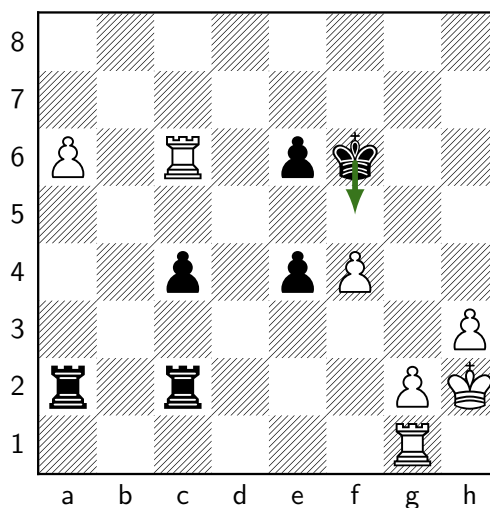
(a)



(b)

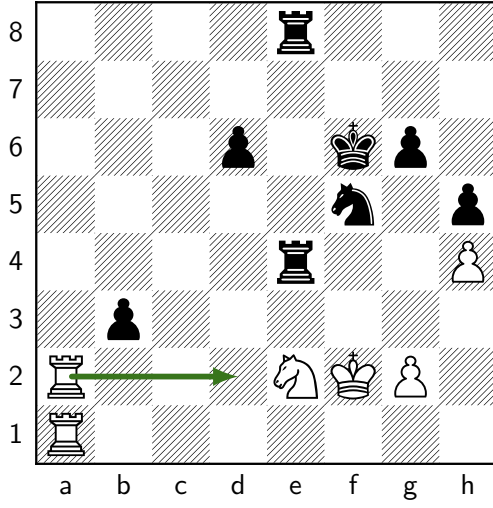


(c)

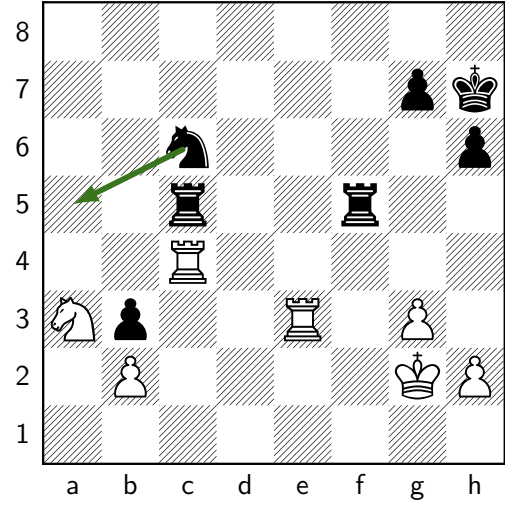


(d)

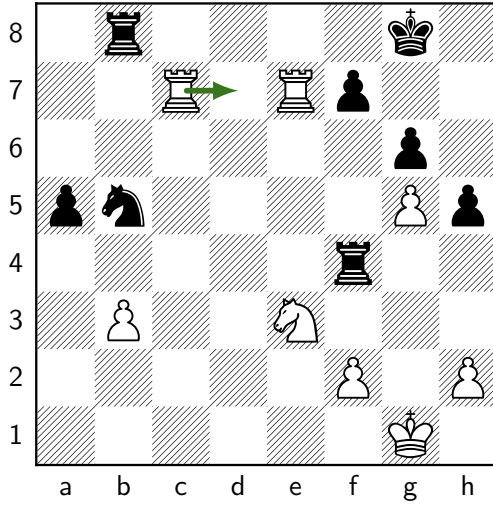
Figure 38: **Move 45, White: R6a2.** (a) Our position; Melanie plays R6a2 (a6a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2721` (ply 73) [structural distance $d=42$, QAP obj=25]: the master played Re7 ; under μ our Ra6 plays the role of that game’s Rf2 ; cross-type roles: our pb4 as their Nc2 , our Ra1 as their ph2 ; transferred move a6a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2345` (ply 82) [structural distance $d=42$, QAP obj=44]: the master played Rd4 ; under μ our Ra6 plays the role of that game’s Rd7 ; cross-type roles: our re8 as their Ph3 , our Ra1 as their pb4 ; transferred move a6c6 . (d) Retrieved from `ccr1_top_p0001#3368` (ply 95) [structural distance $d=44$, QAP obj=37]: the master played Kf5 ; under μ our Kf2 plays the role of that game’s Kf3 ; cross-type roles: our re4 as their pf4 , our nf5 as their rg1 ; transferred move f2f3 . Synthesis: 1 retrieved analog transferred to R6a2 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



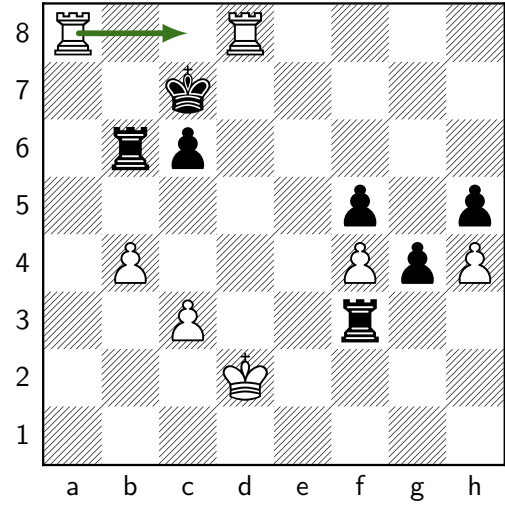
(a)



(b)



(c)



(d)

Figure 39: **Move 46, White: Rd2.** (a) Our position; Melanie plays Rd2 (a2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3063` (ply 79) [structural distance $d=29$, QAP obj=22]: the master played Na5; under μ our Ne2 plays the role of that game’s Nc3; cross-type roles: our pb3 as their rc4, our re4 as their na3; transferred move e2c1. (c) Retrieved from `ccr1_top_p0002#3780` (ply 78) [structural distance $d=47$, QAP obj=12]: the master played Rcd7; under μ our Ra2 plays the role of that game’s Rc7; cross-type roles: our Ne2 as their Pb3, our Ph4 as their Ne3; transferred move a2a7. (d) Retrieved from `ccr1_top_p0001#2616` (ply 106) [structural distance $d=42$, QAP obj=3]: the master played Rac8+; under μ our Pg2 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our Ne2 as their Pc3, our Ra1 as their Pf4; transferred move g2g4. Synthesis: no retrieved analog transferred to Rd2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

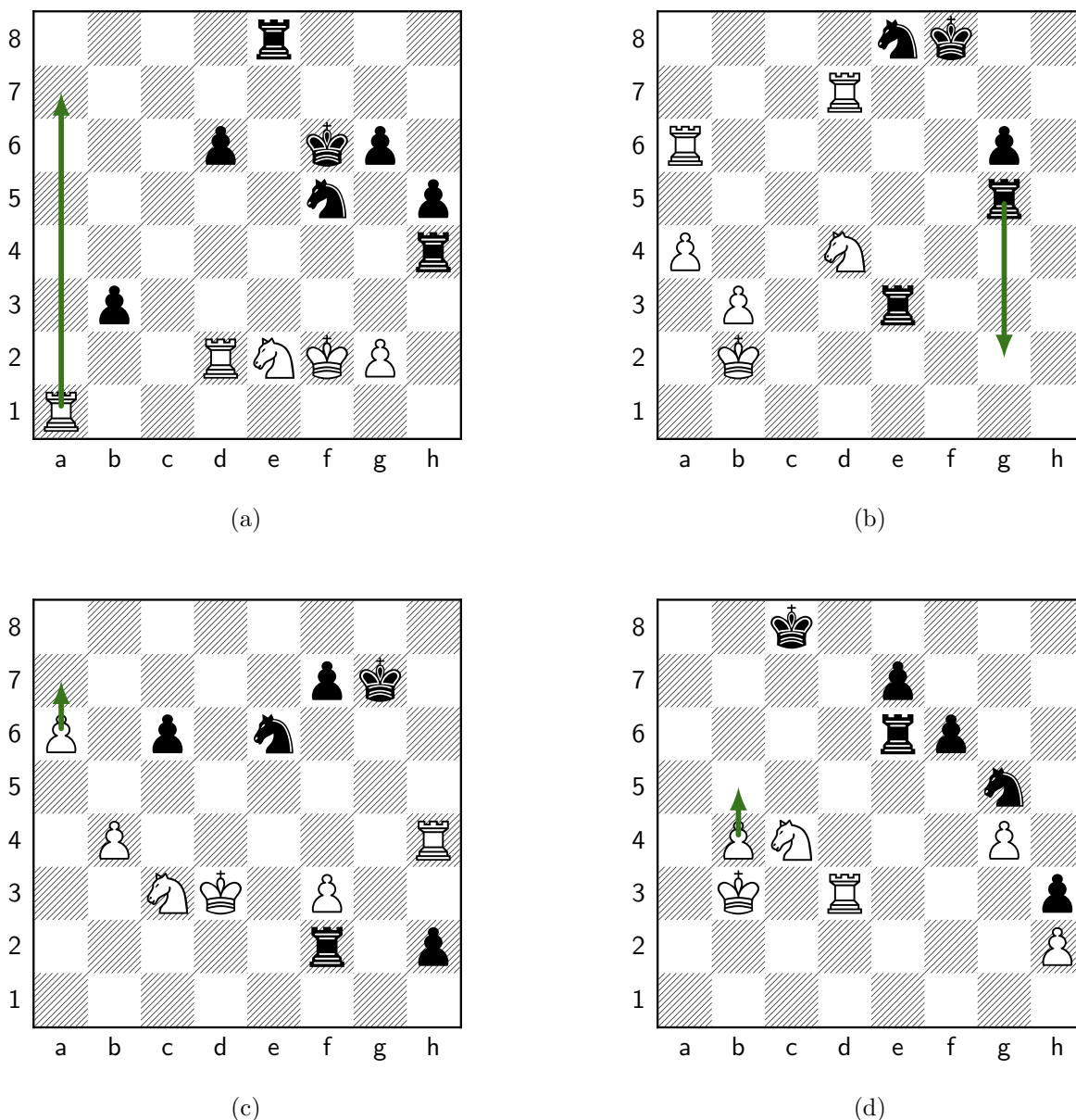
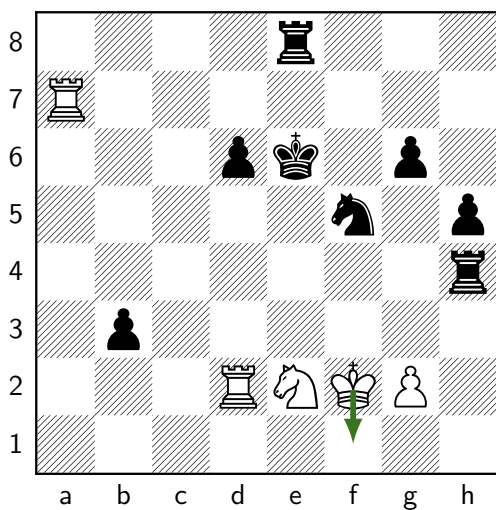
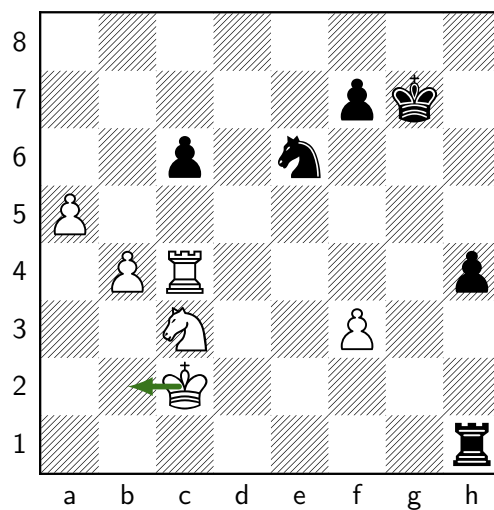


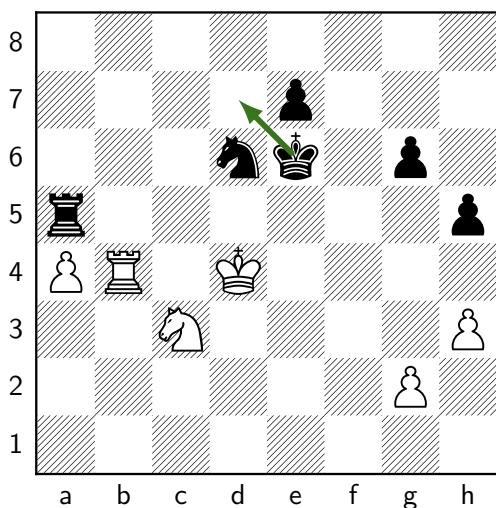
Figure 40: **Move 47, White: Ra7.** (a) Our position; Melanie plays Ra7 (a1a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2783` (ply 95) [structural distance $d=38$, QAP obj=19]: the master played Rg2+; under μ our Ra1 plays the role of that game’s Rg4; cross-type roles: our Pg2 as their Ne8, our nf5 as their rd7; transferred move a1a7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2859` (ply 90) [structural distance $d=40$, QAP obj=16]: the master played a7; under μ our Ra1 plays the role of that game’s Pa6 (a **cross-type** correspondence); cross-type roles: our Ne2 as their ph2, our rh4 as their ne6; transferred move a1a7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3336` (ply 104) [structural distance $d=33$, QAP obj=11]: the master played b5; under μ our Ra1 plays the role of that game’s Pb4 (a **cross-type** correspondence); cross-type roles: our rh4 as their Nc4, our Ne2 as their re6; transferred move a1a7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ra7 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



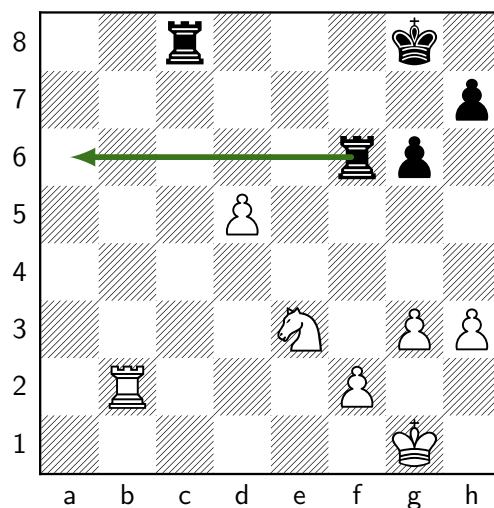
(a)



(b)

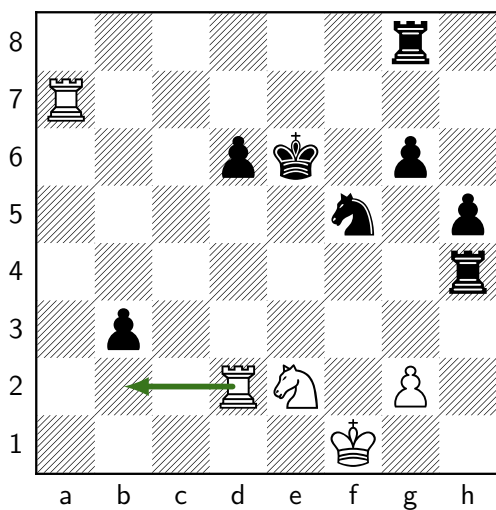


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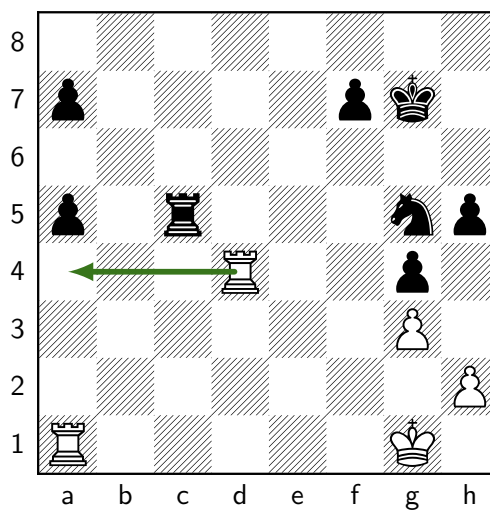


(d)

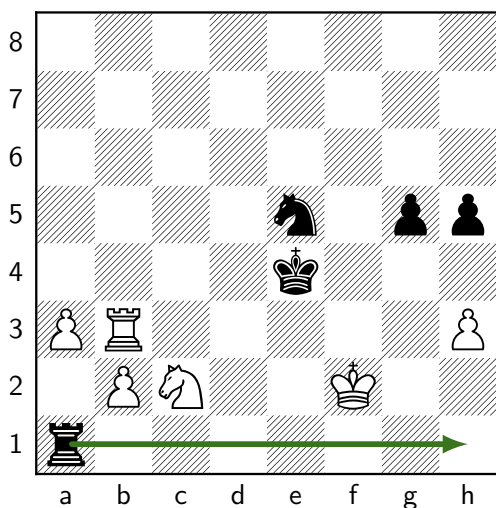
Figure 41: **Move 48, White: Kf1 (f2f1).** (a) Our position; Melanie plays Kf1 (f2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2859` (ply 78) [structural distance $d=47$, QAP obj=17]: the master played Kb2; under μ our Kf2 plays the role of that game’s Kc2; cross-type roles: our nf5 as their rh1, our rh4 as their Pa5; transferred move f2f1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2895` (ply 99) [structural distance $d=40$, QAP obj=8]: the master played Kd7; under μ our Kf2 plays the role of that game’s Ke3; cross-type roles: our Ra7 as their Nd6, our Ne2 as their Ph5; transferred move f2f1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3398` (ply 87) [structural distance $d=46$, QAP obj=22]: the master played Ra6; under μ our Rd2 plays the role of that game’s Rf3; cross-type roles: our Ne2 as their Pg6, our re8 as their pg3; transferred move d2b2. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



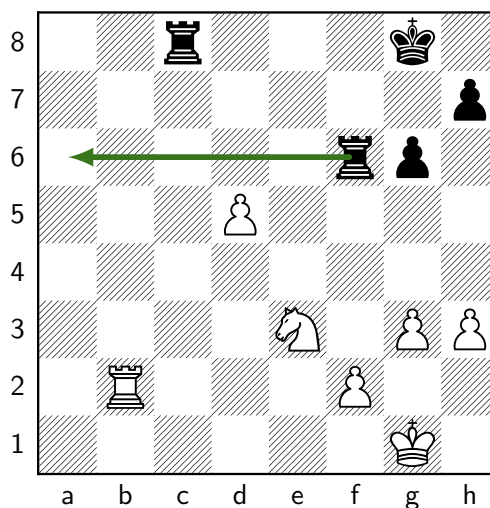
(a)



(b)



(c)



(d)

Figure 42: **Move 49, White: Rb2.** (a) Our position; Melanie plays Rb2 (d2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2725` (ply 94) [structural distance $d=47$, QAP obj=20]: the master played Rda4; under μ our Rd2 plays the role of that game’s Rd4; cross-type roles: our Ra7 as their Pg3, our nf5 as their rc5; transferred move d2b2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1459` (ply 101) [structural distance $d=45$, QAP obj=4]: the master played Rh1; under μ our Rd2 plays the role of that game’s Ra8; cross-type roles: our Ra7 as their ph3; transferred move d2b2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3398` (ply 87) [structural distance $d=45$, QAP obj=23]: the master played Ra6; under μ our Rd2 plays the role of that game’s Rf3; cross-type roles: our Ne2 as their Pg6, our rh4 as their ne3; transferred move d2d1. Synthesis: 2 retrieved analogs transferred to Rb2 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

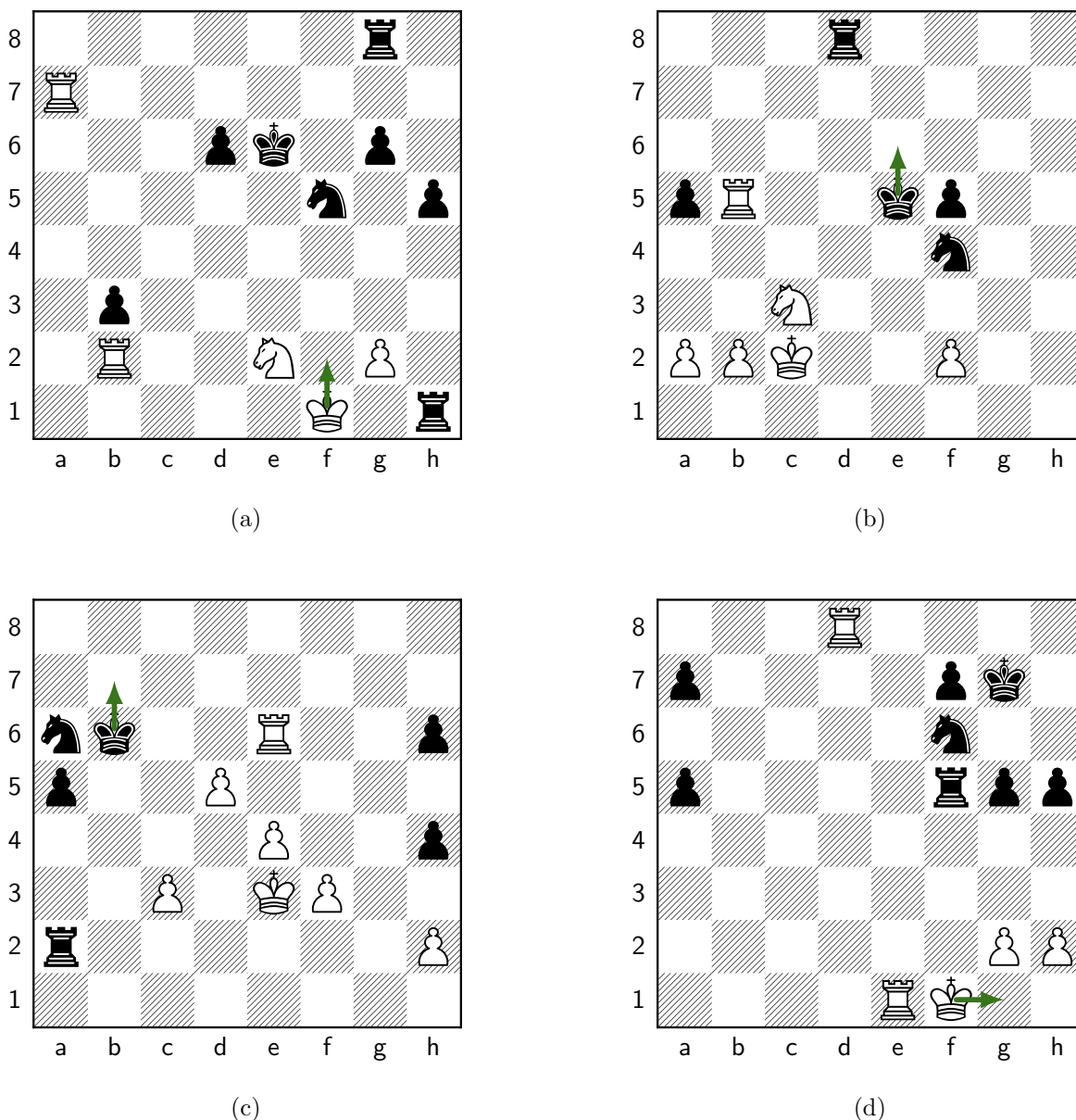


Figure 43: **Move 50, White: Kf2**. (a) Our position; Melanie plays Kf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2508 (ply 83) [structural distance $d=36$, QAP obj=54]: the master played Ke6; under μ our Kf1 plays the role of that game's Ke4; cross-type roles: our Rb2 as their pa2; transferred move f1f2 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2514 (ply 91) [structural distance $d=37$, QAP obj=52]: the master played Kb7; under μ our Kf1 plays the role of that game's Kb3; cross-type roles: our Rb2 as their Ph6, our rg8 as their pd5; transferred move f1f2 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#2725 (ply 86) [structural distance $d=39$, QAP obj=36]: the master played Kg1; under μ our Kf1 plays the role of that game's Kf1; cross-type roles: our Ne2 as their Re1, our rg8 as their Ph2; transferred move f1f2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf2 (summed QAP objective 279, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

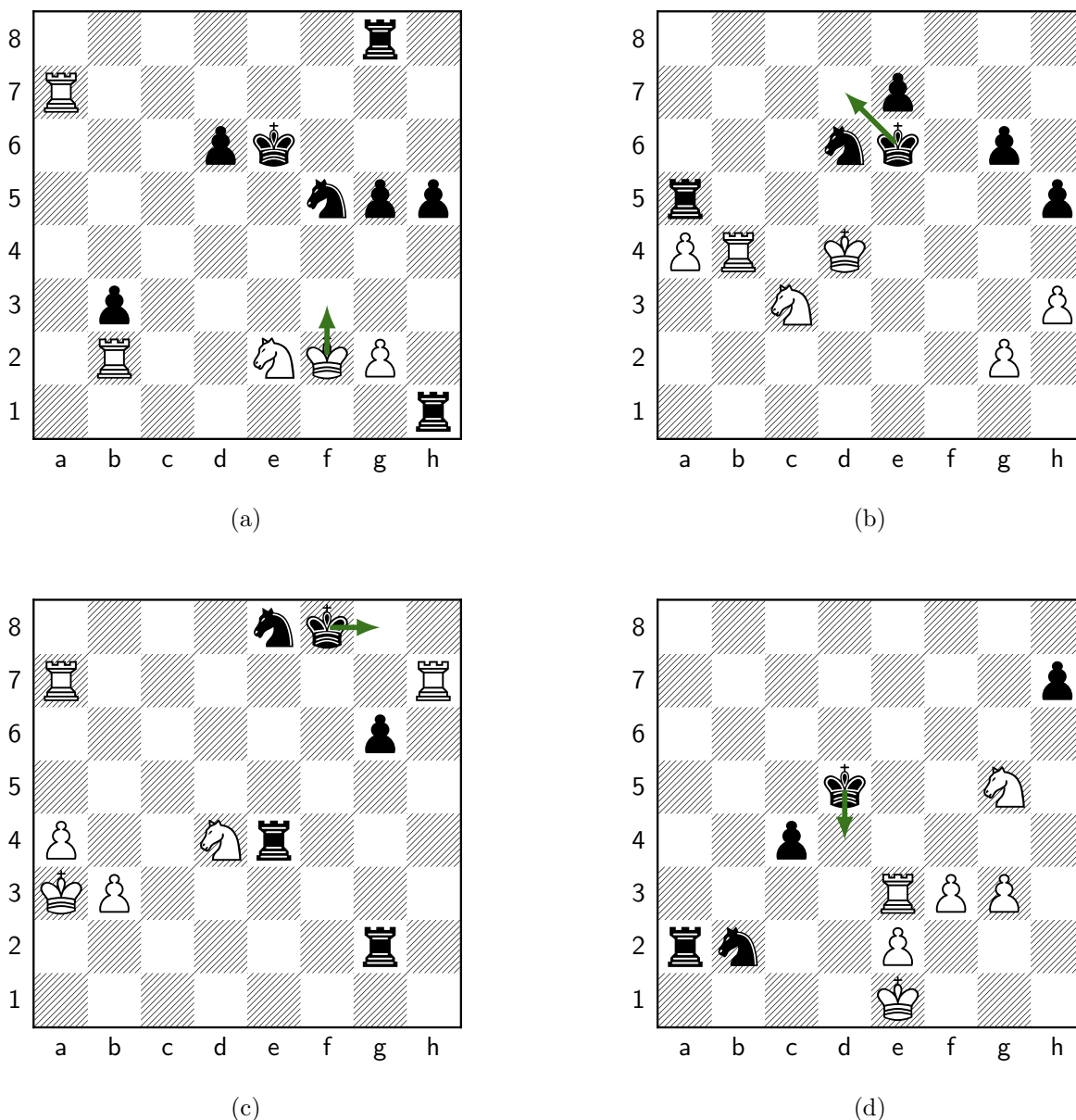


Figure 44: **Move 51, White: Kf3.** (a) Our position; Melanie plays Kf3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2895` (ply 99) [structural distance $d=35$, QAP obj=20]: the master played Kd7; under μ our Kf2 plays the role of that game’s Ke3; cross-type roles: our rg8 as their Pe7, our Ra7 as their Nd6; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2783` (ply 101) [structural distance $d=35$, QAP obj=12]: the master played Kg8; under μ our Kf2 plays the role of that game’s Kf1; cross-type roles: our pb3 as their nd4, our nf5 as their pb3; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#653` (ply 91) [structural distance $d=37$, QAP obj=11]: the master played Kd4; under μ our Kf2 plays the role of that game’s Kd4; cross-type roles: our Rb2 as their ng5, our Ne2 as their pf3; transferred move f2f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf3 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

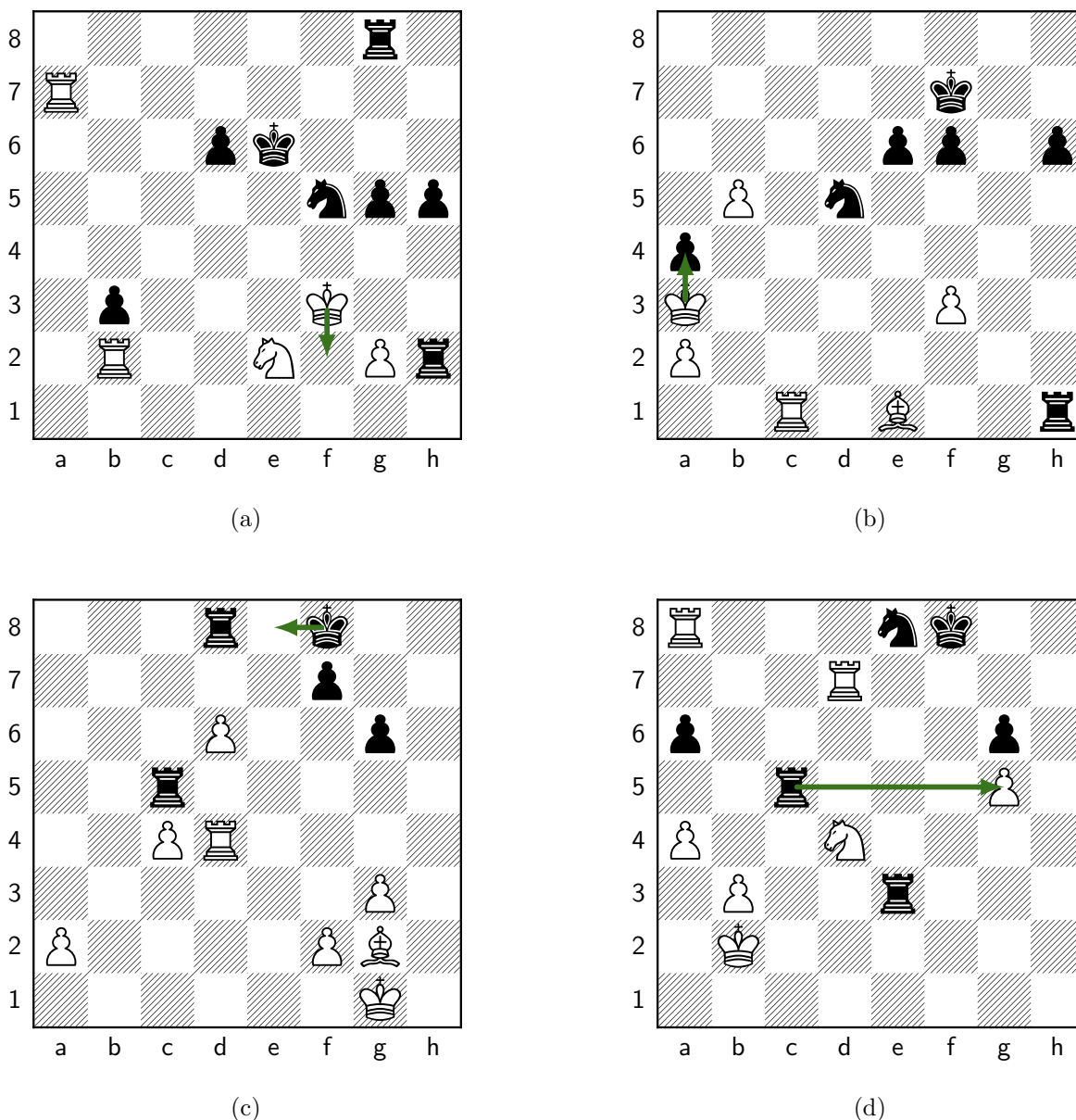


Figure 45: **Move 52, White: Kf2.** (a) Our position; Melanie plays Kf2 (f3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#540` (ply 92) [structural distance $d=44$, QAP obj=40]: the master played Kxa4; under μ our Kf3 plays the role of that game’s Ka3; cross-type roles: our Pg2 as their Be1, our Ne2 as their Pa2; transferred move f3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1292` (ply 89) [structural distance $d=43$, QAP obj=39]: the master played Ke8; under μ our Kf3 plays the role of that game’s Kf1; cross-type roles: our Ne2 as their Pf7, our nf5 as their Pg6; transferred move f3f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2783` (ply 93) [structural distance $d=44$, QAP obj=41]: the master played Rxc5; under μ our Ra7 plays the role of that game’s Rc4; cross-type roles: our Pg2 as their Ne8, our ph5 as their rd7; transferred move a7a4. Synthesis: 2 retrieved analogs transferred to Kf2 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

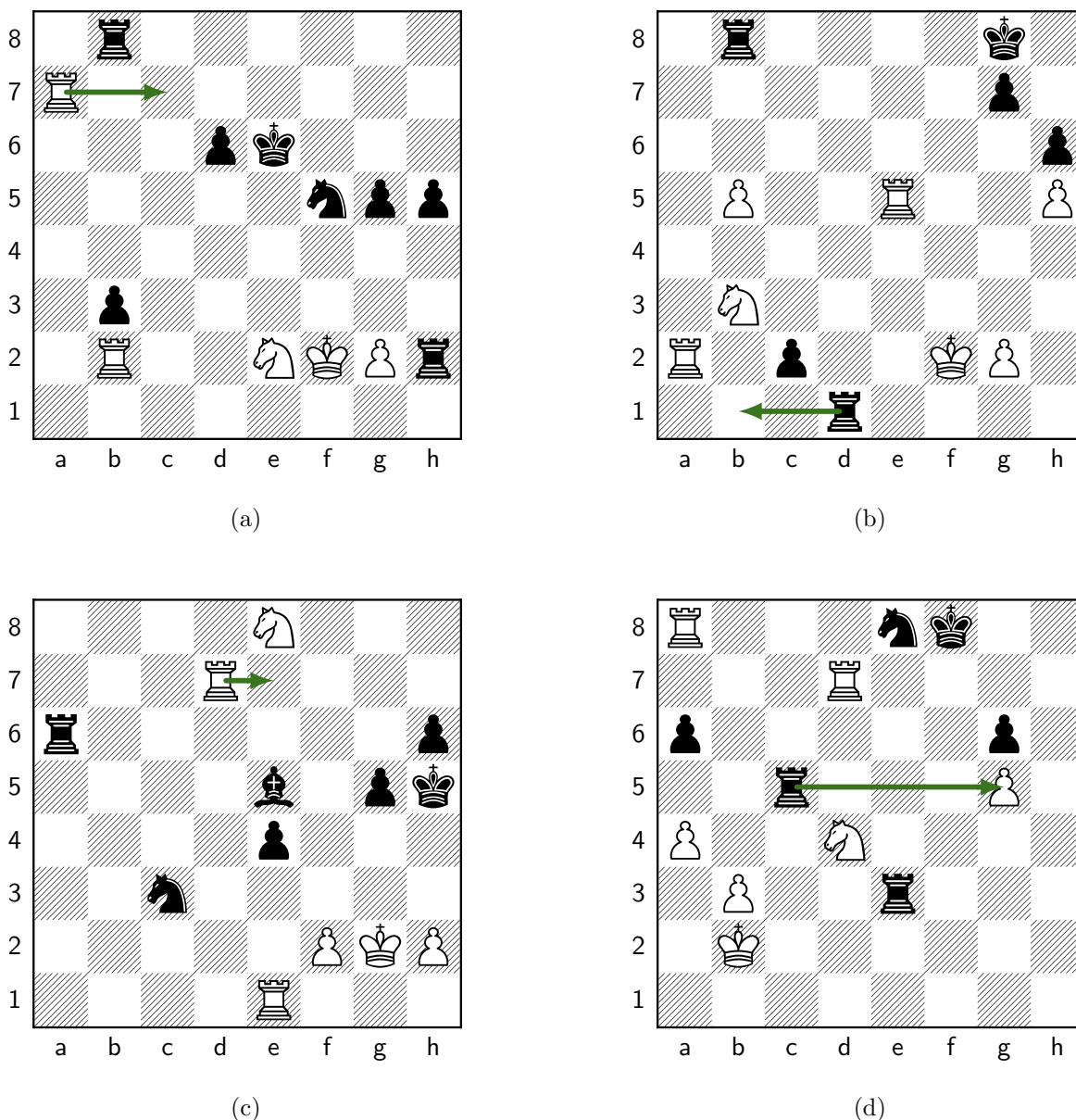
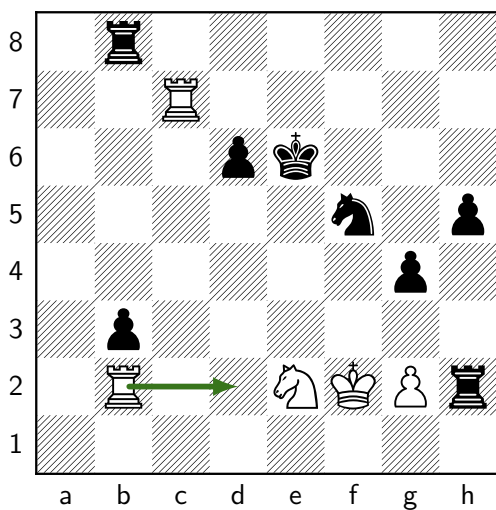
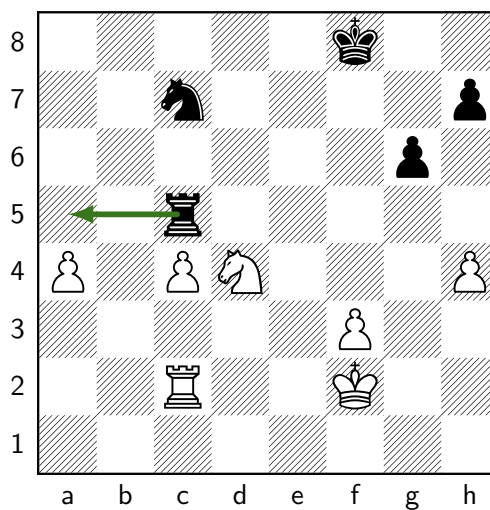


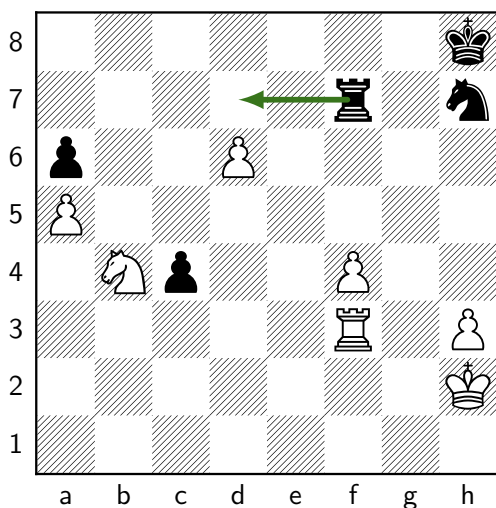
Figure 46: **Move 53, White: Rc7 (a7c7)**. (a) Our position; Melanie plays Rc7 (a7c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#118` (ply 121) [structural distance $d=43$, QAP obj=24]: the master played Rb1; under μ our Ra7 plays the role of that game’s Rd8; cross-type roles: our Ne2 as their Pg7; transferred move a7c7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1632` (ply 98) [structural distance $d=41$, QAP obj=23]: the master played Re7; under μ our Ra7 plays the role of that game’s Rd7; cross-type roles: our Ne2 as their Pf2, our rb8 as their nc3; transferred move a7c7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2783` (ply 93) [structural distance $d=42$, QAP obj=42]: the master played Rxc5; under μ our Ra7 plays the role of that game’s Rc4; cross-type roles: our Pg2 as their Ne8, our nf5 as their rd7; transferred move a7a4. Synthesis: 2 retrieved analogs transferred to Rc7 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



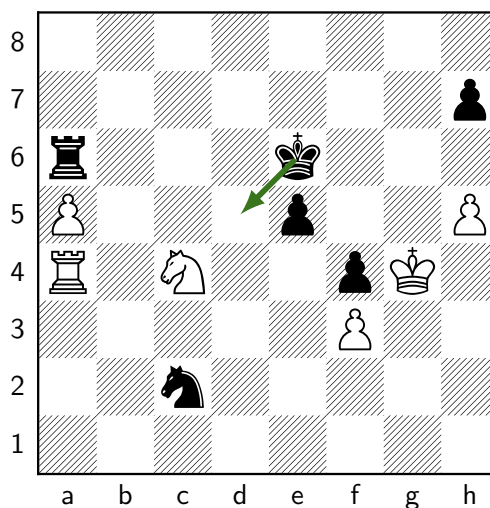
(a)



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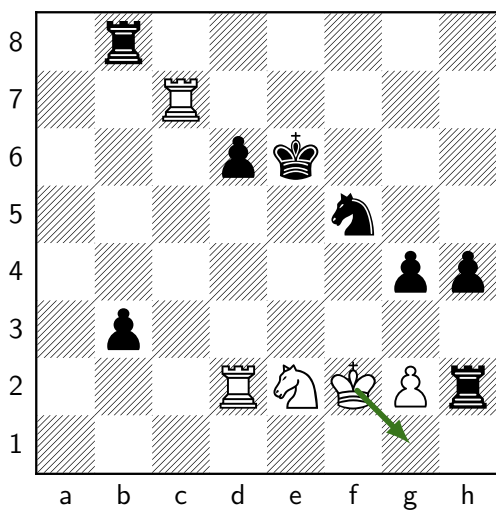


(c)

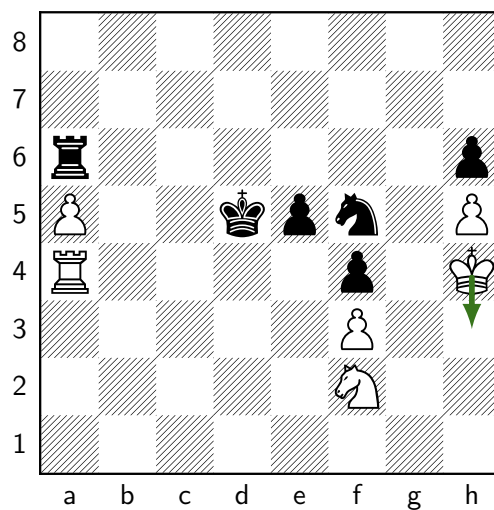


(d)

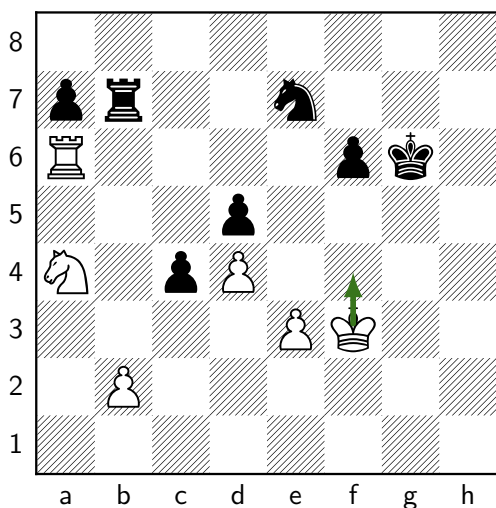
Figure 47: **Move 54, White: Rd2.** (a) Our position; Melanie plays Rd2 (b2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3595` (ply 107) [structural distance $d=39$, QAP obj=44]: the master played Ra5; under μ our Rb2 plays the role of that game’s Rc4; cross-type roles: our rh2 as their ph4; transferred move b2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3553` (ply 121) [structural distance $d=32$, QAP obj=31]: the master played Rd7; under μ our Rb2 plays the role of that game’s Rf2; cross-type roles: our rh2 as their nb4, our nf5 as their Pc4; transferred move b2d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1072` (ply 99) [structural distance $d=43$, QAP obj=50]: the master played Kd5; under μ our Kf2 plays the role of that game’s Ke3; cross-type roles: our nf5 as their Ph7, our Ne2 as their ph5; transferred move f2g1. Synthesis: 2 retrieved analogs transferred to Rd2 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



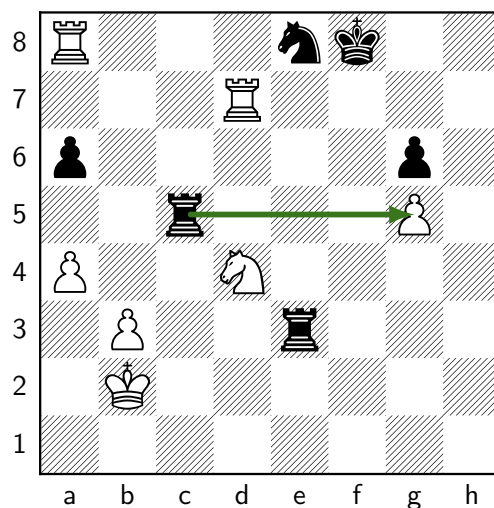
(a)



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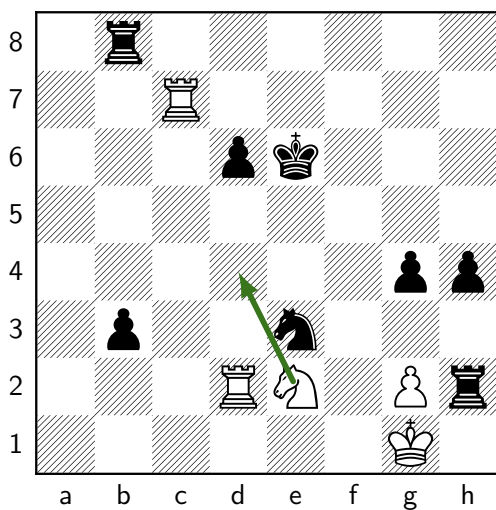


(c)

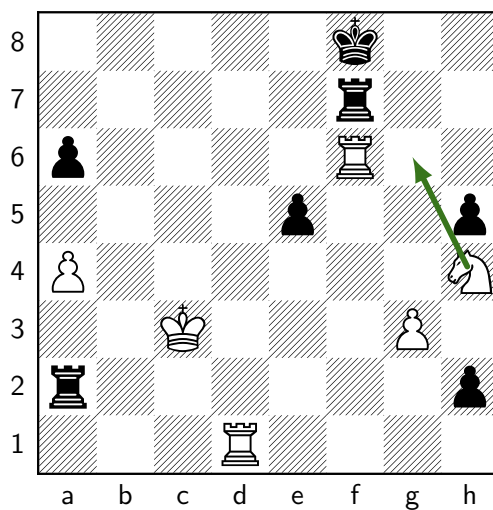


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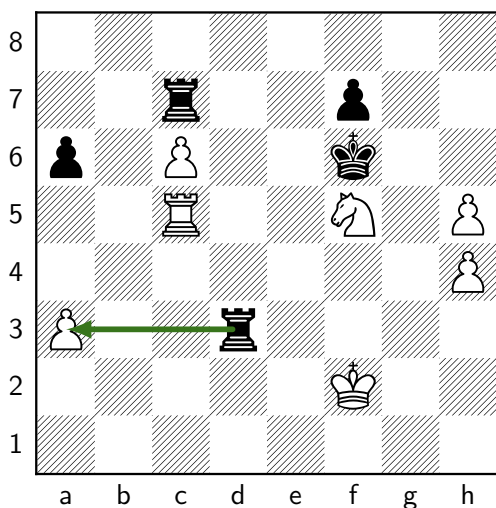
Figure 48: **Move 55, White: Kg1.** (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1072` (ply 110) [structural distance $d=50$, QAP obj=24]: the master played Kh3; under μ our Kf2 plays the role of that game's Kh4; cross-type roles: our pb3 as their Nf2, our Rc7 as their Pf3; transferred move f2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#109` (ply 120) [structural distance $d=51$, QAP obj=21]: the master played Kf4; under μ our Kf2 plays the role of that game's Kf3; cross-type roles: our ph4 as their Na4, our nf5 as their Pd4; transferred move f2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2783` (ply 93) [structural distance $d=45$, QAP obj=43]: the master played Rrg5; under μ our Rc7 plays the role of that game's Rc4; cross-type roles: our Pg2 as their Ne8, our pg4 as their rd7; transferred move c7a7. Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



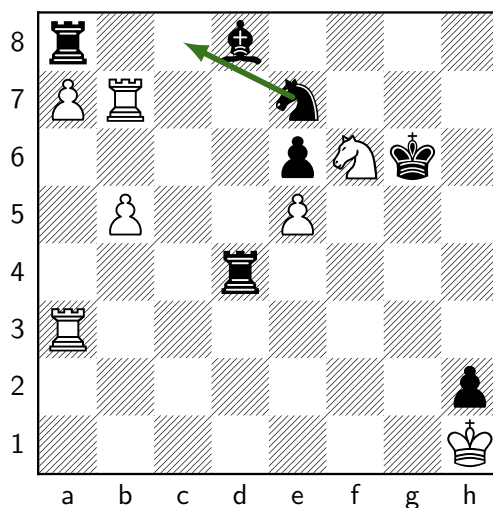
(a)



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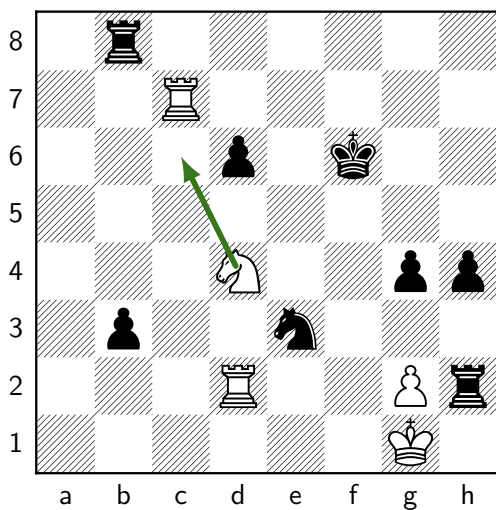


(c)

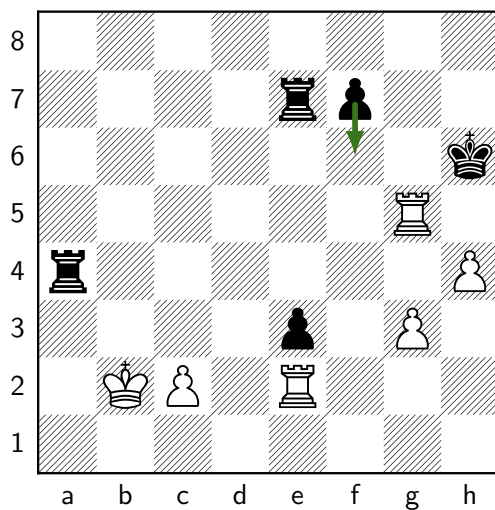


(d)

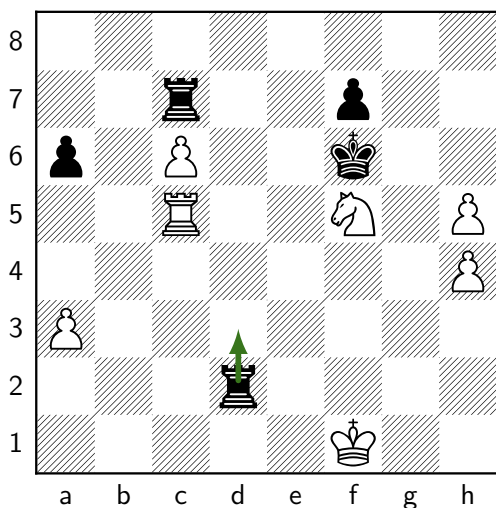
Figure 49: **Move 56, White: Nd4+.** (a) Our position; Melanie plays Nd4+ (e2d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1627` (ply 104) [structural distance $d=45$, QAP obj=15]: the master played Ng6+; under μ our Ne2 plays the role of that game’s Nh4; cross-type roles: our ne3 as their ra2, our Rc7 as their Pg3; transferred move e2d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2398` (ply 101) [structural distance $d=45$, QAP obj=30]: the master played Rxa3; under μ our Rc7 plays the role of that game’s Rd6; cross-type roles: our Ne2 as their Pf7, our rh2 as their nf5; transferred move c7a7. (d) Retrieved from `ccr1_top_p0001#3459` (ply 105) [structural distance $d=44$, QAP obj=30]: the master played Nc8; under μ our Ne2 plays the role of that game’s Ne2; cross-type roles: our pd6 as their Bd8, our Rd2 as their Pe6; transferred move e2c1. Synthesis: 1 retrieved analog transferred to Nd4+ (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



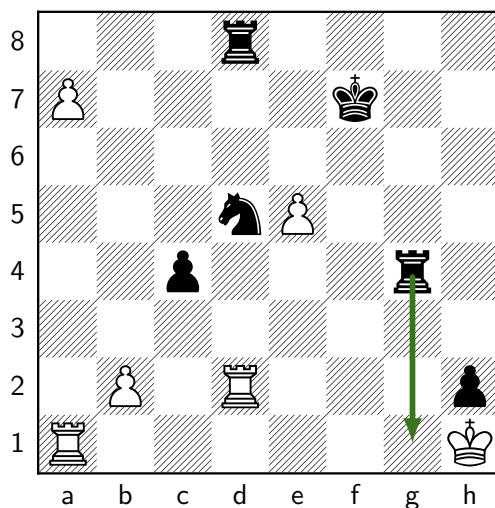
(a)



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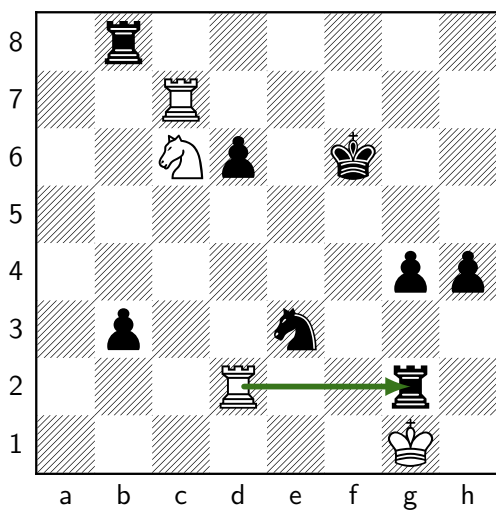


(c)

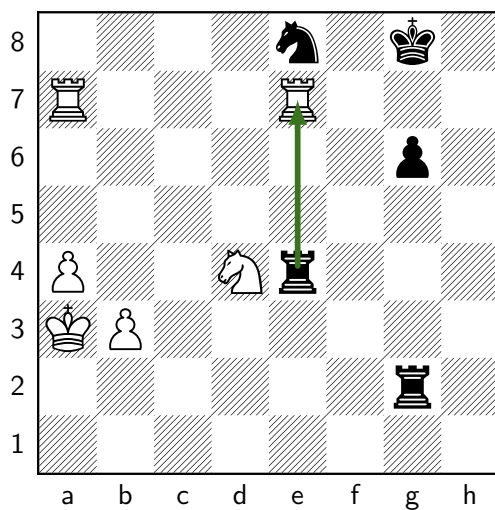


(d)

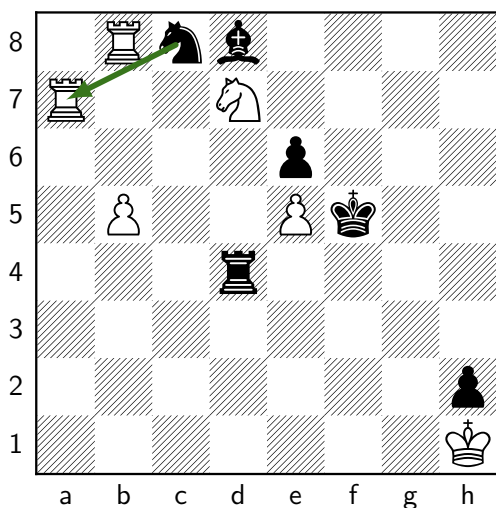
Figure 50: **Move 57, White: Nc6**. (a) Our position; Melanie plays Nc6 (d4c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1719` (ply 105) [structural distance $d=42$, QAP obj=39]: the master played f6; under μ our Nd4 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our ne3 as their ph4; transferred move d4e2. (c) Retrieved from `ccr1_top_p0001#2398` (ply 99) [structural distance $d=43$, QAP obj=30]: the master played Rd3; under μ our Rc7 plays the role of that game’s Rd7; cross-type roles: our rh2 as their nf5, our ne3 as their ph5; transferred move c7c6. (d) Retrieved from `ccr1_top_p0000#1858` (ply 97) [structural distance $d=36$, QAP obj=24]: the master played Rg1+; under μ our Rc7 plays the role of that game’s Rg5; cross-type roles: our Pg2 as their Nd5, our ne3 as their pe5; transferred move c7f7. Synthesis: no retrieved analog transferred to Nc6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



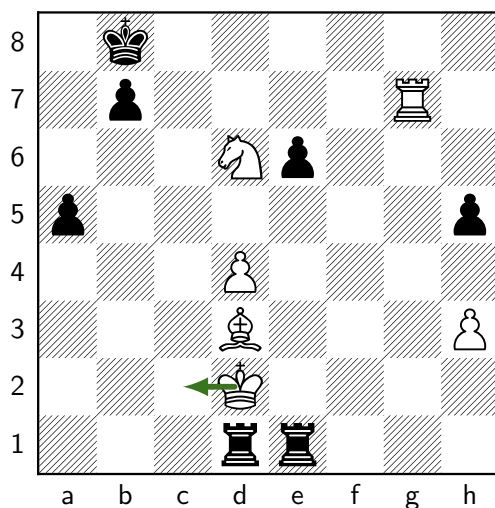
(a)



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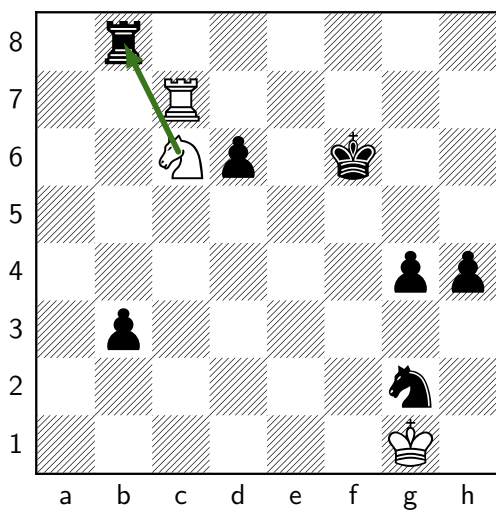


(c)

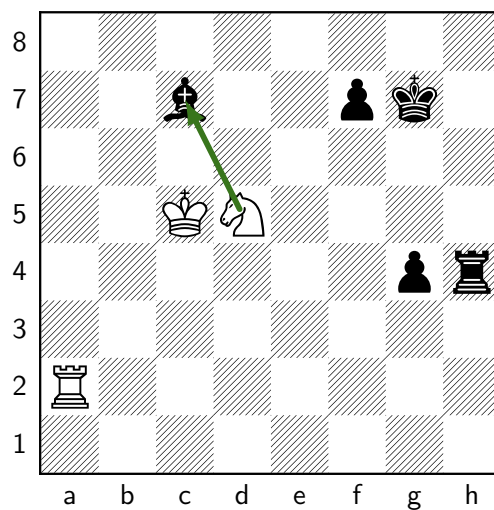


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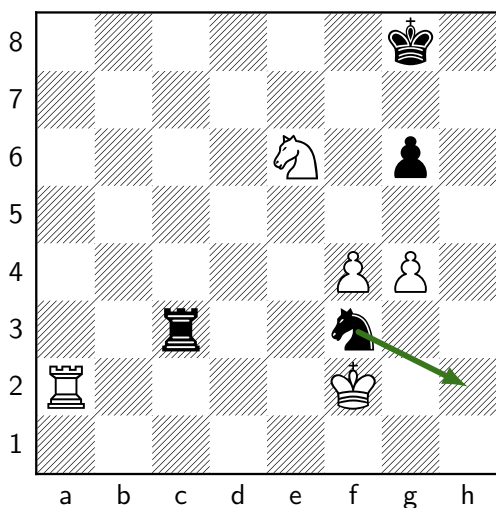
Figure 51: **Move 58, White: Rxc2**. (a) Our position; Melanie plays Rxc2 (d2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2783` (ply 103) [structural distance $d=49$, QAP obj=3]: the master played Rxe7; under μ our Rd2 plays the role of that game's Re5; cross-type roles: our ph4 as their Ne8, our ne3 as their ra7; transferred move d2g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3459` (ply 111) [structural distance $d=55$, QAP obj=3]: the master played Nxa7; under μ our Rd2 plays the role of that game's Nc1 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Bd8, our ph4 as their Rd4; transferred move d2g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1082` (ply 110) [structural distance $d=53$, QAP obj=69]: the master played Kc2; under μ our Kg1 plays the role of that game's Kd2; cross-type roles: our ne3 as their re1, our Rd2 as their Bd3; transferred move g1h1. Synthesis: 2 retrieved analogs transferred to Rxc2 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



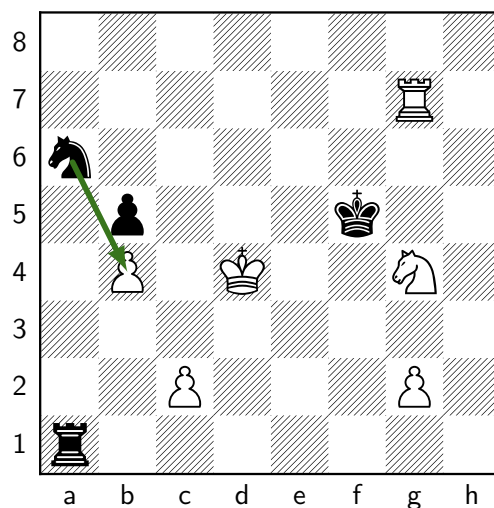
(a)



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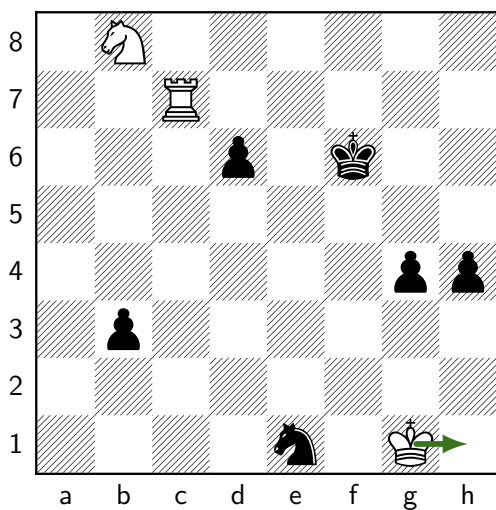


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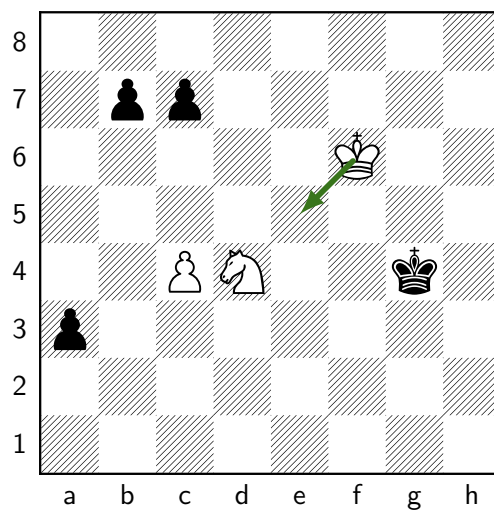


(d)

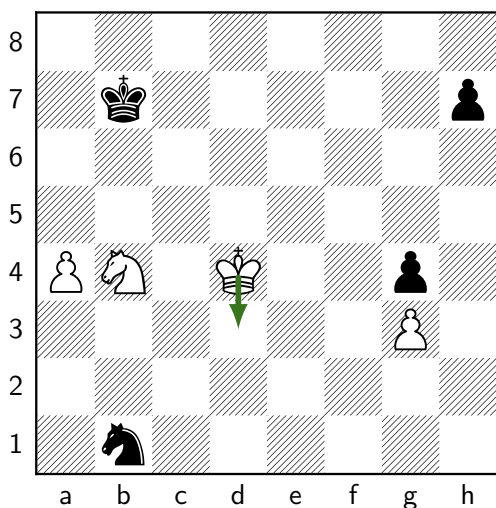
Figure 52: **Move 59, White: Nxb8**. (a) Our position; Melanie plays Nxb8 (c6b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top-p0002#602 (ply 102) [structural distance $d=25$, QAP obj=11]: the master played Nxc7; under μ our Nc6 plays the role of that game's Nd5; cross-type roles: our ng2 as their pg4, our pb3 as their rh4; transferred move c6b8 (**endorses this move**). (c) Retrieved from ccr1_top-p0002#308 (ply 111) [structural distance $d=26$, QAP obj=2]: the master played Nh2; under μ our Nc6 plays the role of that game's Nf6; cross-type roles: our rb8 as their ne6, our ng2 as their Pg6; transferred move c6b8 (**endorses this move**). (d) Retrieved from ccr1_top-p0003#310 (ply 105) [structural distance $d=23$, QAP obj=20]: the master played Nxb4; under μ our Nc6 plays the role of that game's Na3; cross-type roles: our pd6 as their rg7, our Rc7 as their Pb5; transferred move c6a5. Synthesis: 2 retrieved analogs transferred to Nxb8 (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



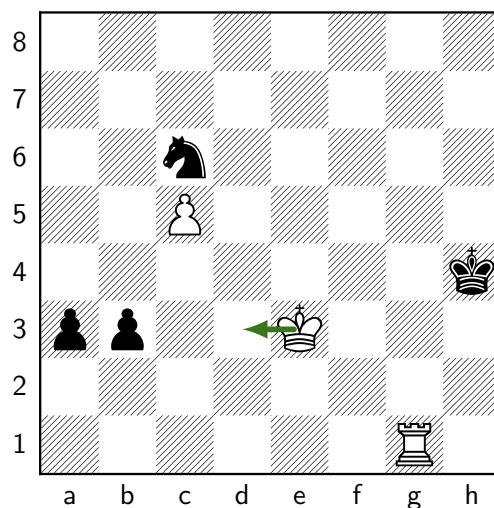
(a)



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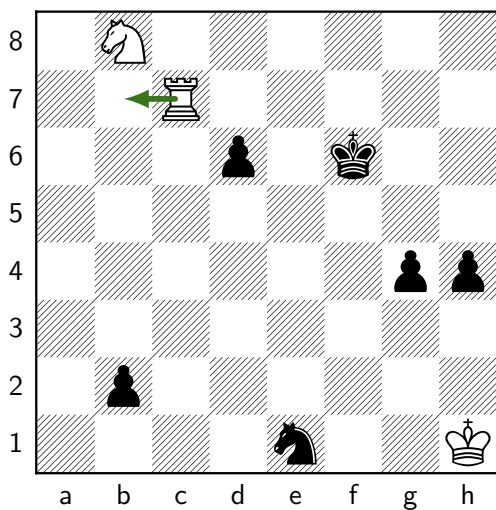


(c)

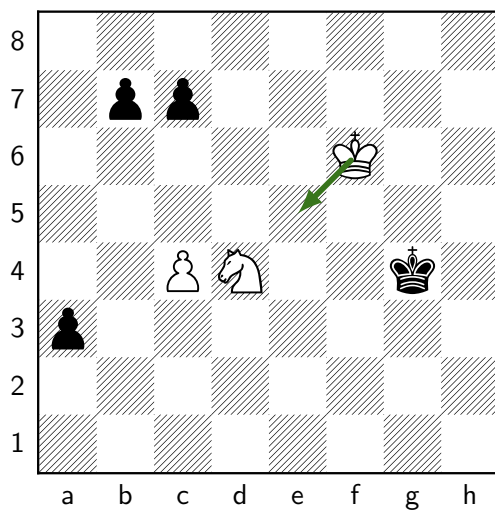


(d)

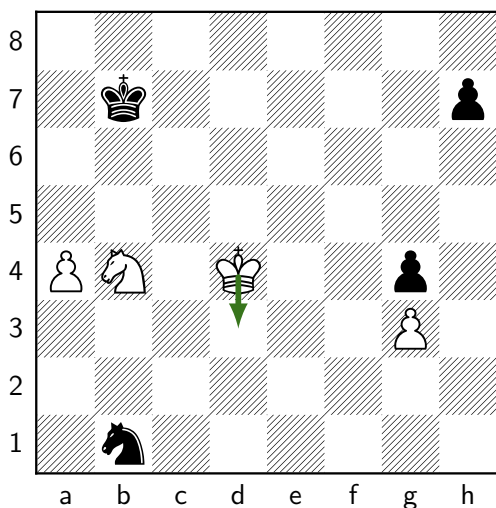
Figure 53: **Move 60, White: Kh1**. (a) Our position; Melanie plays Kh1 (**g1h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3690` (ply 122) [structural distance $d=11$, QAP obj=1]: the master played Ke5; under μ our *Kg1* plays the role of that game’s *Kf6*; cross-type roles: our *ne1* as their *pa3*, our *Rc7* as their *Nd4*; transferred move **g1h1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#1133` (ply 126) [structural distance $d=12$, QAP obj=1]: the master played Kd3; under μ our *Kg1* plays the role of that game’s *Kd4*; cross-type roles: our *Rc7* as their *Nb4*; transferred move **g1h1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#1901` (ply 124) [structural distance $d=11$, QAP obj=0]: the master played Kd3; under μ our *Kg1* plays the role of that game’s *Ke3*; cross-type roles: our *ne1* as their *pb3*, our *pb3* as their *nc6*; transferred move **g1h1 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to Kh1 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



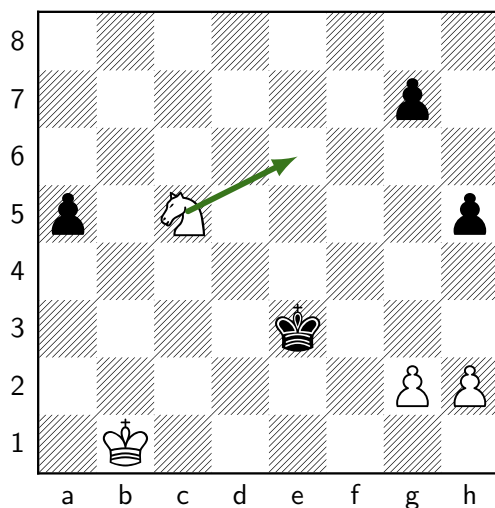
(a)



(b)

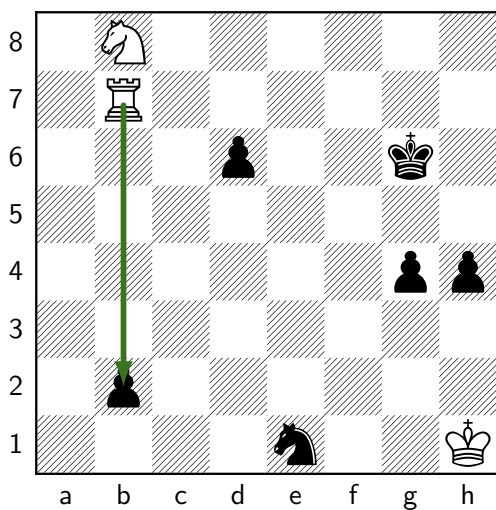


(c)

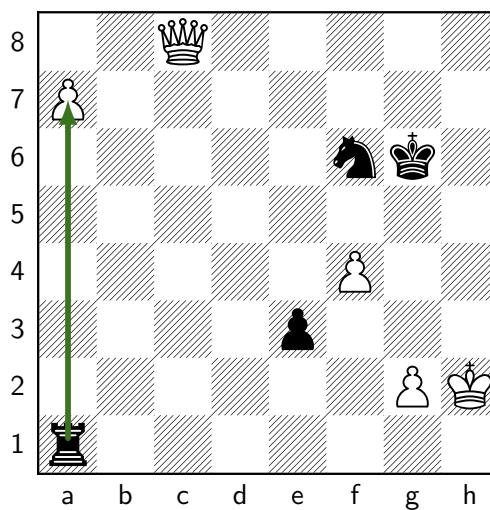


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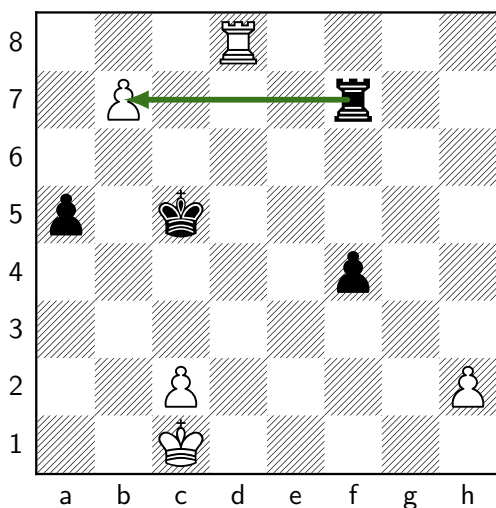
Figure 54: **Move 61, White: Rb7.** (a) Our position; Melanie plays Rb7 (c7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3690` (ply 122) [structural distance $d=11$, QAP obj=1]: the master played Ke5; under μ our *Kh1* plays the role of that game's *Kf6*; cross-type roles: our *ne1* as their *pa3*, our *Rc7* as their *Nd4*; transferred move h1g1. (c) Retrieved from `ccr1_top_p0000#1133` (ply 126) [structural distance $d=13$, QAP obj=1]: the master played Kd3; under μ our *Kh1* plays the role of that game's *Kd4*; cross-type roles: our *Rc7* as their *Nb4*; transferred move h1g1. (d) Retrieved from `ccr1_top_p0003#1588` (ply 110) [structural distance $d=11$, QAP obj=1]: the master played Ne6; under μ our *Rc7* plays the role of that game's *Nc5* (a **cross-type** correspondence); cross-type roles: our *ne1* as their *Pg2*; transferred move c7h7. Synthesis: no retrieved analog transferred to Rb7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



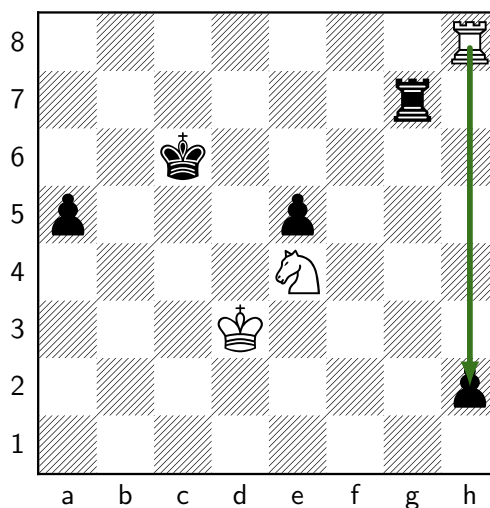
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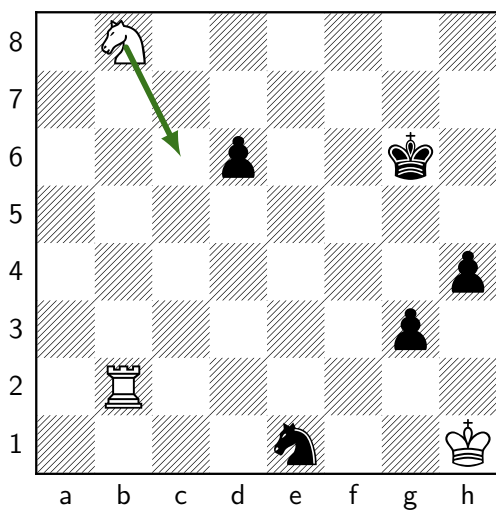


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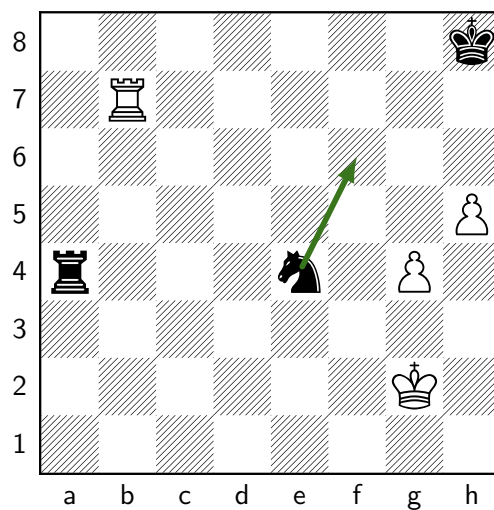


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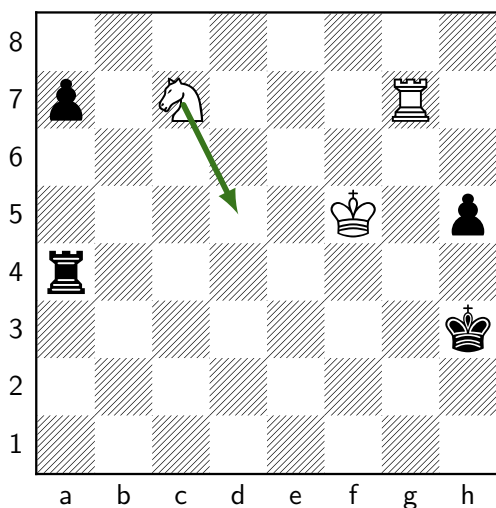
Figure 55: **Move 62, White: Rxb2**. (a) Our position; Melanie plays Rxb2 (b7b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3674` (ply 121) [structural distance $d=14$, QAP obj=16]: the master played Rxa7; under μ our Rb7 plays the role of that game’s Ra8; cross-type roles: our pg4 as their qc8, our ph4 as their Nf6; transferred move b7b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2511` (ply 119) [structural distance $d=18$, QAP obj=16]: the master played Rxb7; under μ our Rb7 plays the role of that game’s Rf2; cross-type roles: our ne1 as their rd8, our Nb8 as their Pf4; transferred move b7b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1657` (ply 114) [structural distance $d=11$, QAP obj=15]: the master played Rxh2; under μ our Rb7 plays the role of that game’s Rh8; cross-type roles: our pd6 as their Ne4, our ne1 as their pe5; transferred move b7b2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxb2 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



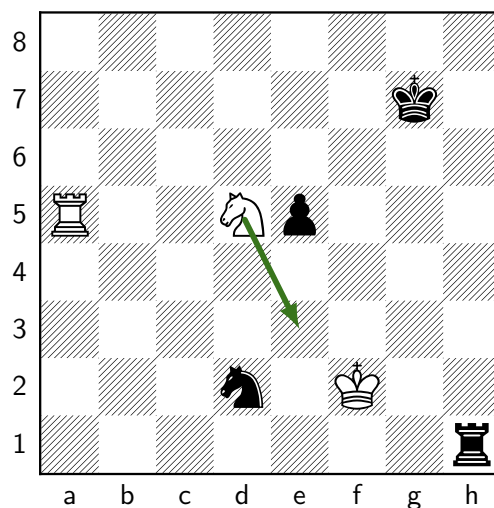
(a)



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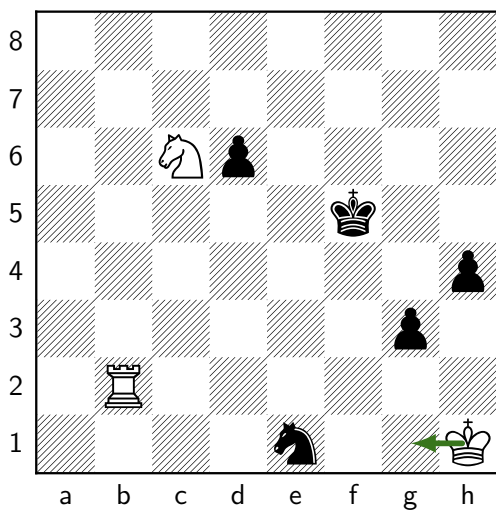


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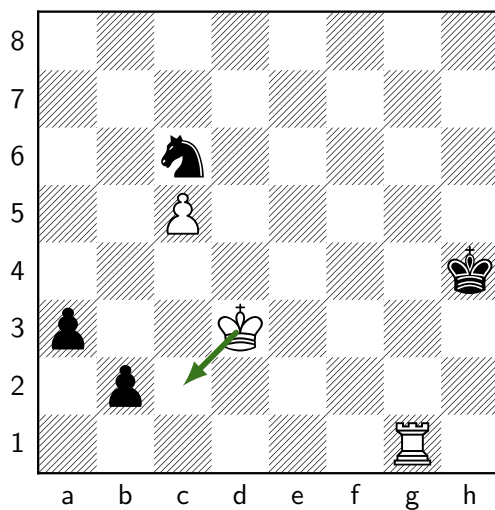


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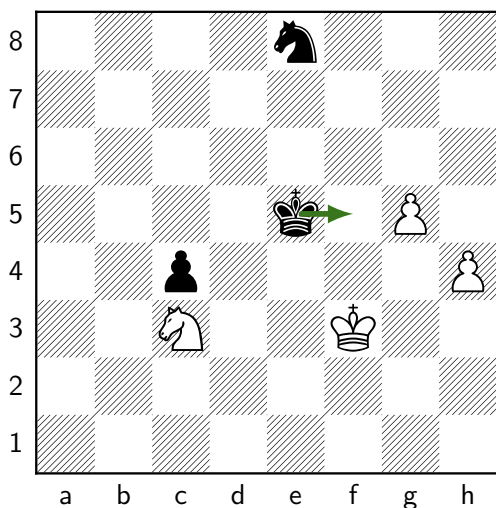
Figure 56: **Move 63, White: Nc6**. (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1087` (ply 113) [structural distance $d=11$, QAP obj=2]: the master played Nf6; under μ our Nb8 plays the role of that game’s Ne5; cross-type roles: our ne1 as their rb7; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1778` (ply 114) [structural distance $d=12$, QAP obj=2]: the master played Nd5; under μ our Nb8 plays the role of that game’s Nc7; cross-type roles: our ne1 as their ra4; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2829` (ply 116) [structural distance $d=15$, QAP obj=1]: the master played Ne3; under μ our Nb8 plays the role of that game’s Nd5; cross-type roles: our ne1 as their rh1, our pg3 as their nd2; transferred move b8c6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nc6 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



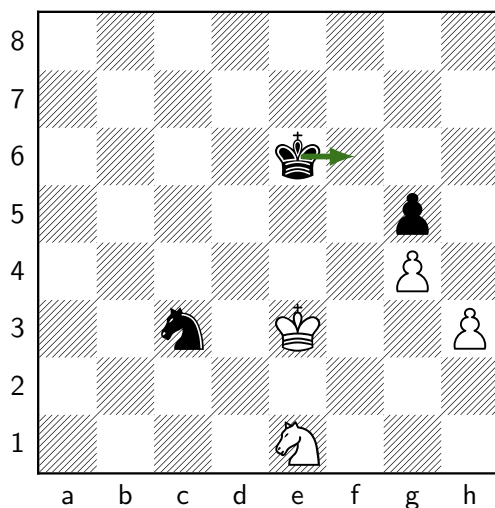
(a)



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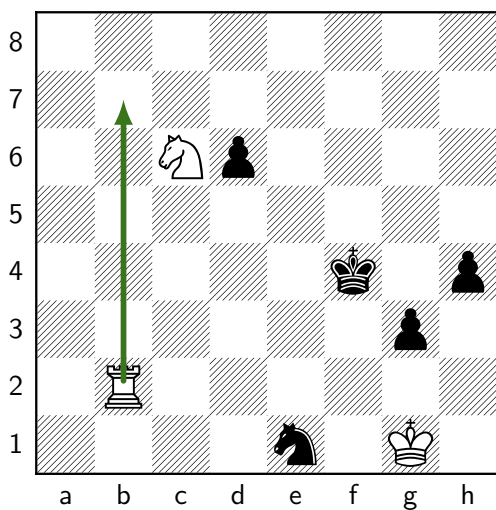


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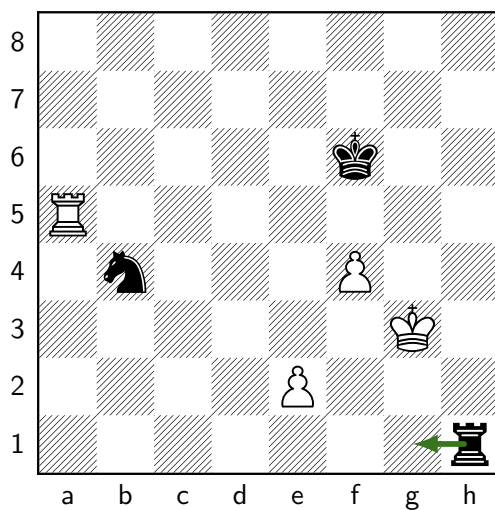


(d)

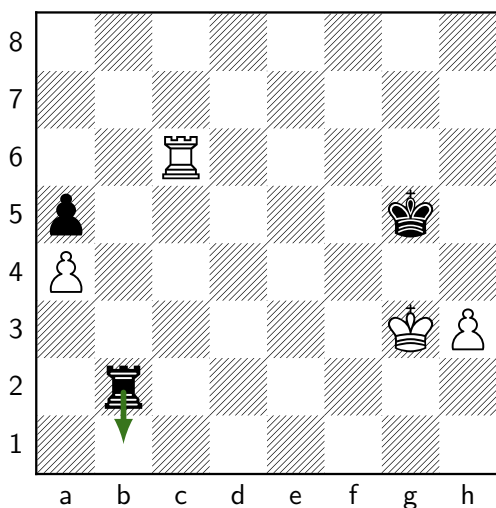
Figure 57: **Move 64, White: Kg1**. (a) Our position; Melanie plays Kg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1901 (ply 126) [structural distance $d=7$, QAP obj=2]: the master played Kc2; under μ our *Kh1* plays the role of that game’s *Kd3*; transferred move h1g1 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2681 (ply 129) [structural distance $d=9$, QAP obj=2]: the master played Kf5; under μ our *Kh1* plays the role of that game’s *Ke4*; cross-type roles: our *Rb2* as their *Ne8*, our *Nc6* as their *Pc4*; transferred move h1g1 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3462 (ply 131) [structural distance $d=11$, QAP obj=1]: the master played Kf6; under μ our *Kh1* plays the role of that game’s *Ke3*; cross-type roles: our *Rb2* as their *Pg5*; transferred move h1g1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg1 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



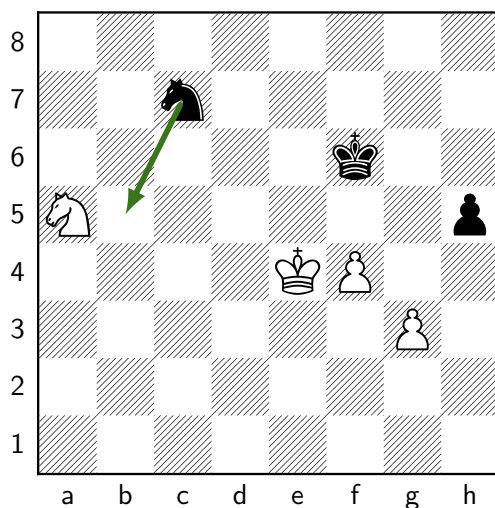
(a)



(b)

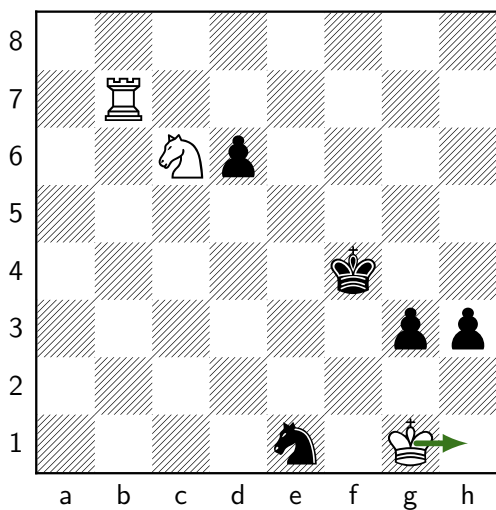


(c)

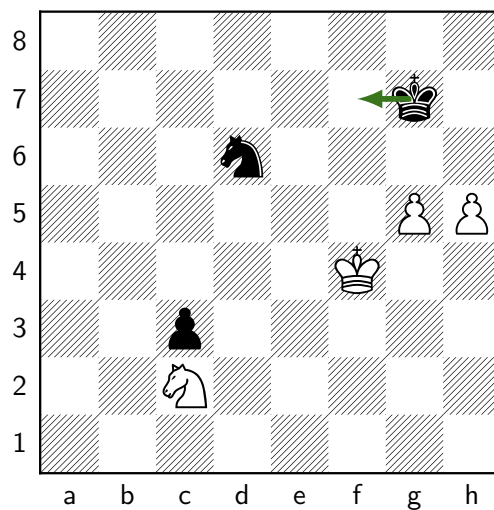


(d)

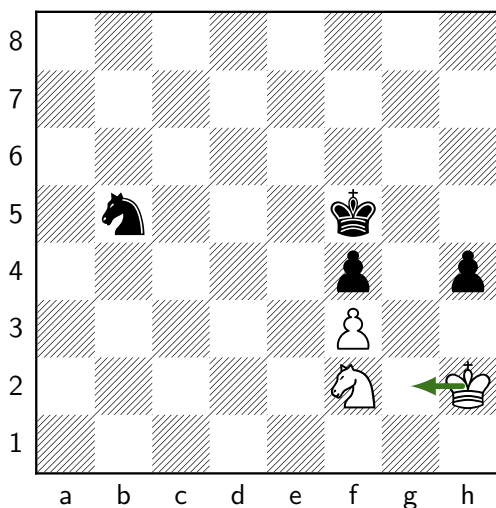
Figure 58: **Move 65, White: Rb7.** (a) Our position; Melanie plays Rb7 (b2b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1517` (ply 139) [structural distance $d=12$, QAP obj=3]: the master played Rg1+; under μ our Rb2 plays the role of that game’s Rh8; cross-type roles: our ne1 as their ra5; transferred move b2b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2643` (ply 141) [structural distance $d=12$, QAP obj=3]: the master played Rb1; under μ our Rb2 plays the role of that game’s Rb7; cross-type roles: our ne1 as their rc6, our Nc6 as their pa4; transferred move b2b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3097` (ply 131) [structural distance $d=9$, QAP obj=4]: the master played Nb5; under μ our Nc6 plays the role of that game’s Nc2; cross-type roles: our Rb2 as their Ph5; transferred move c6a5. Synthesis: 2 retrieved analogs transferred to Rb7 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



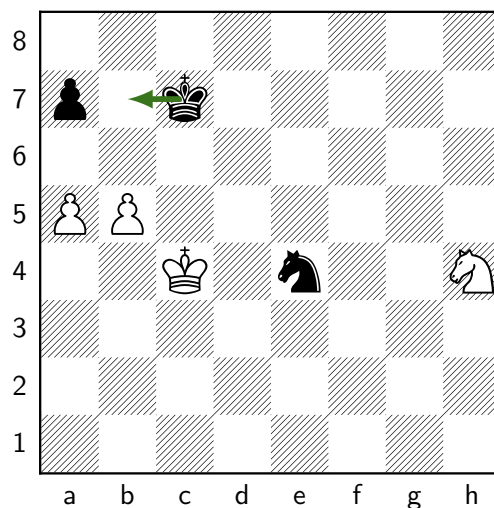
(a)



(b)

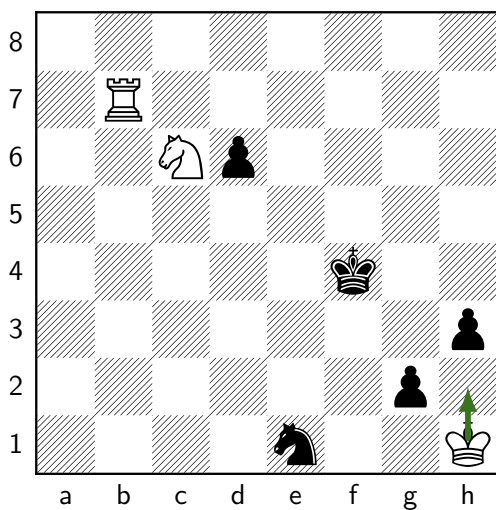


(c)

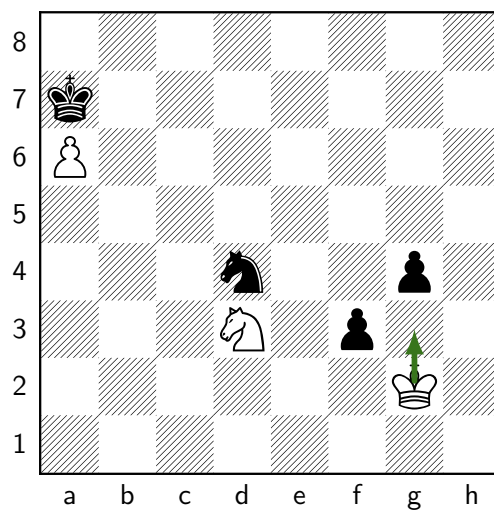


(d)

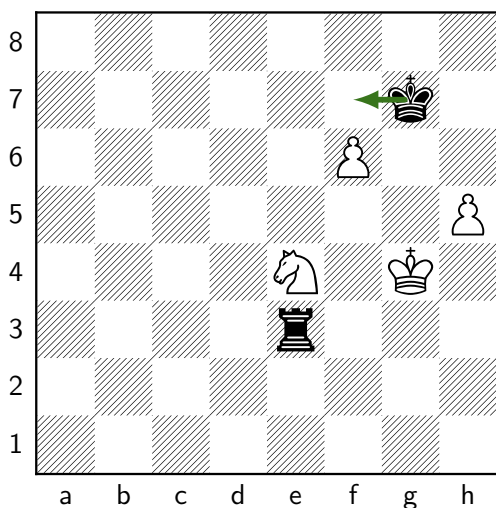
Figure 59: **Move 66, White: Kh1.** (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2681` (ply 141) [structural distance $d=5$, QAP obj=3]: the master played Kf7; under μ our *Kg1* plays the role of that game’s *Kg2*; transferred move g1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#438` (ply 146) [structural distance $d=9$, QAP obj=3]: the master played Kg2; under μ our *Kg1* plays the role of that game’s *Kh2*; cross-type roles: our *Nc6* as their *Pf3*, our *pd6* as their *nb5*; transferred move g1h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1187` (ply 123) [structural distance $d=8$, QAP obj=2]: the master played Kb7; under μ our *Kg1* plays the role of that game’s *Kc2*; transferred move g1h1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh1 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



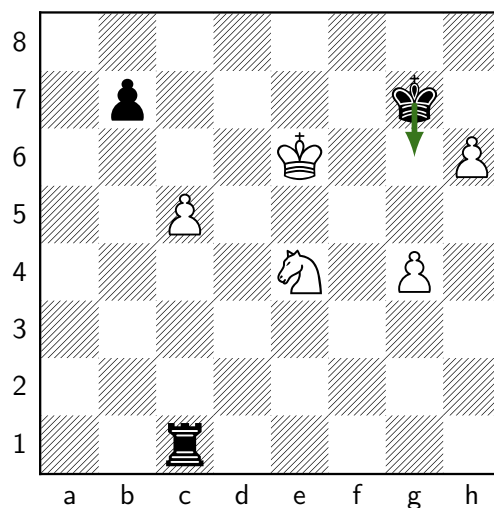
(a)



(b)

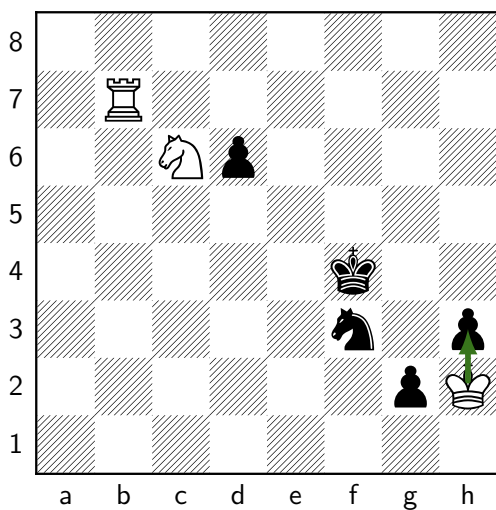


(c)

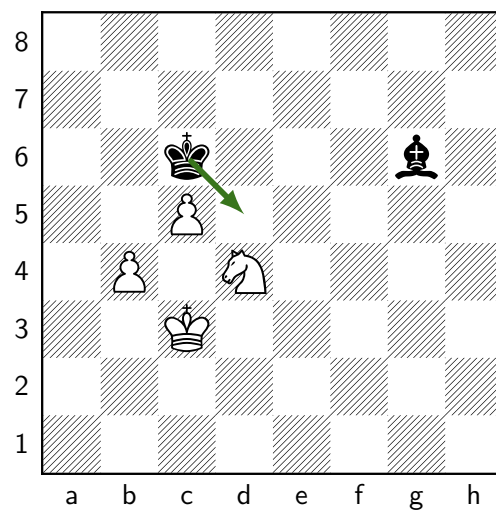


(d)

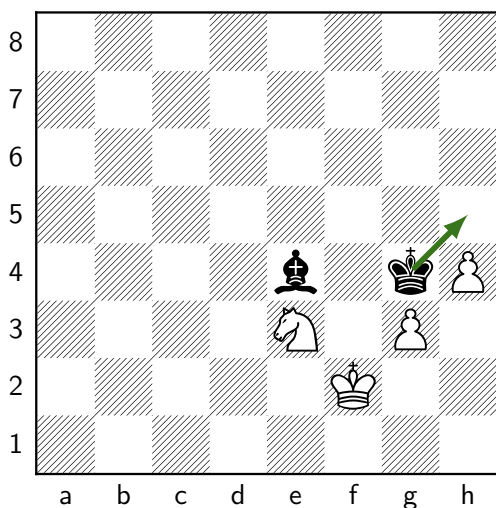
Figure 60: **Move 67, White: Kh2.** (a) Our position; Melanie plays Kh2 (**h1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3894` (ply 124) [structural distance $d=10$, QAP obj=72]: the master played Kg3; under μ our Kh1 plays the role of that game’s Kg2; cross-type roles: our pd6 as their Nd3, our Nc6 as their Pa6; transferred move h1h2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2289` (ply 121) [structural distance $d=16$, QAP obj=70]: the master played Kf7; under μ our Kh1 plays the role of that game’s Kg2; cross-type roles: our Nc6 as their Re3; transferred move h1h2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1656` (ply 121) [structural distance $d=10$, QAP obj=66]: the master played Kg6; under μ our Kh1 plays the role of that game’s Kg2; cross-type roles: our Nc6 as their Pb7; transferred move h1h2 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh2 (summed QAP objective 253, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



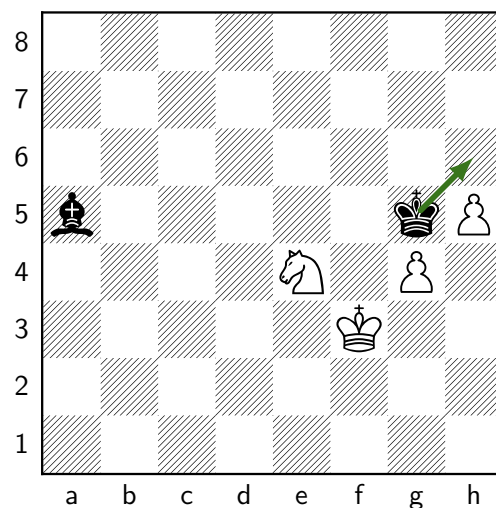
(a)



(b)

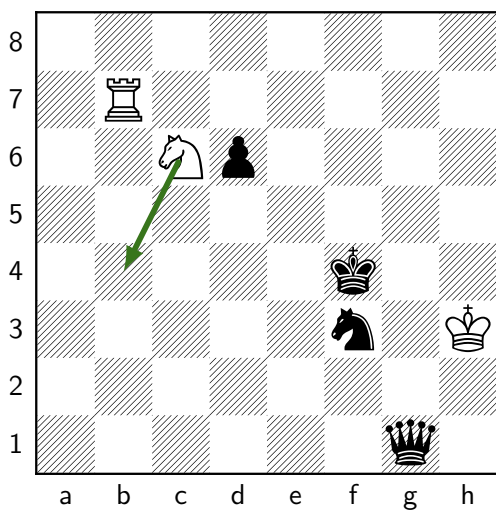


(c)

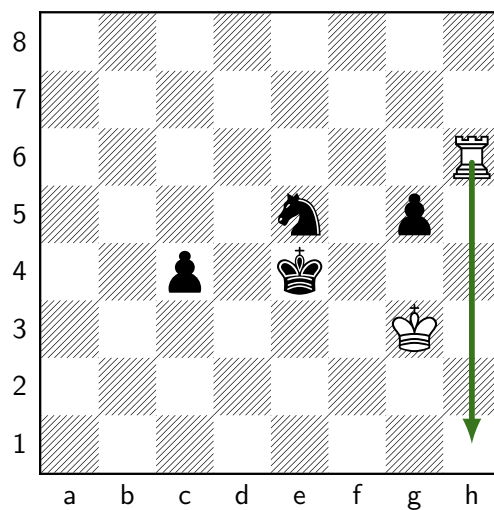


(d)

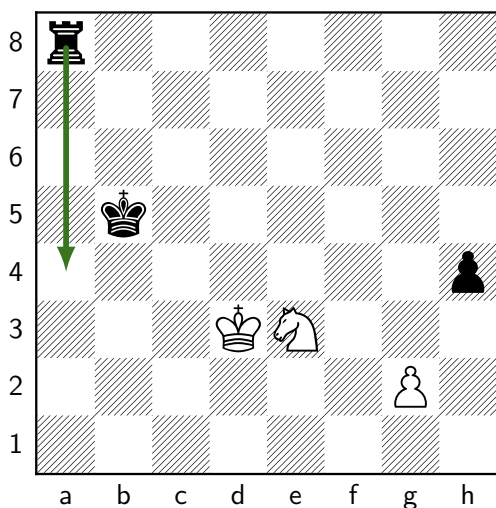
Figure 61: **Move 68, White: Kxh3**. (a) Our position; Melanie plays Kxh3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2855` (ply 145) [structural distance $d=17$, QAP obj=87]: the master played Kd5; under μ our Kh2 plays the role of that game’s Kc3; cross-type roles: our Nc6 as their Bg6; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1758` (ply 121) [structural distance $d=12$, QAP obj=86]: the master played Kh5; under μ our Kh2 plays the role of that game’s Kg5; cross-type roles: our Nc6 as their Be4; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#743` (ply 141) [structural distance $d=11$, QAP obj=86]: the master played Kh6; under μ our Kh2 plays the role of that game’s Kg4; cross-type roles: our Nc6 as their Ba5; transferred move h2h3 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kxh3 (summed QAP objective 1049, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



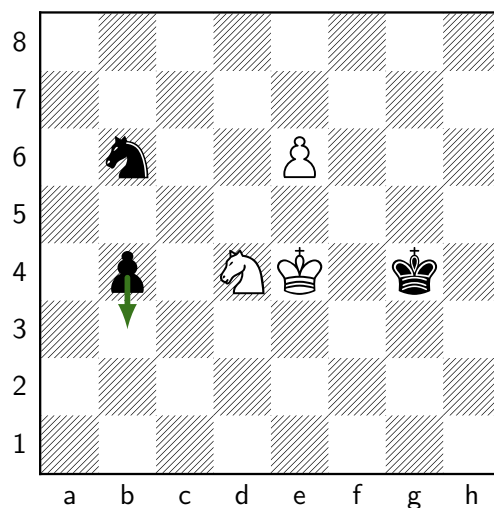
(a)



(b)



(c)



(d)

Figure 62: **Move 69, White: Nb4.** (a) Our position; Melanie plays Nb4 (c6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1337 (ply 128) [structural distance $d=14$, QAP obj=2]: the master played Rh1; under μ our Nc6 plays the role of that game’s Rh6 (a **cross-type** correspondence); cross-type roles: our qg1 as their pc4; transferred move c6b4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3602 (ply 125) [structural distance $d=15$, QAP obj=2]: the master played Ra4; under μ our Nc6 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rb7 as their Ph4, our qg1 as their pg2; transferred move c6b4 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#444 (ply 123) [structural distance $d=12$, QAP obj=2]: the master played b3; under μ our Nc6 plays the role of that game’s Pb5 (a **cross-type** correspondence); cross-type roles: our qg1 as their Nb6, our Rb7 as their pe6; transferred move c6a5. Synthesis: 3 retrieved analogs transferred to Nb4 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.